


University of Verona

Department of Economics

Master's Degree in Economics and Data Analysis

# Investigating the Financial Effects of Twitter Sentiment Through Machine Learning and Transformers: Evidence from Elon Musk's Tweets

**Supervisor**:

Professor Claudia Di Caterina

**Examiner**:

Professor Federico Perali

**Student:** Helena Oprandi

**ID number:** VR501746

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*"La cultura è un'arma potente, può cambiare il mondo."*

*"Culture is a powerful weapon, it can change the world."*

— *L'amica Geniale, Elena Ferrante*

# Acknowledgements

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To my family, thank you for your support, even from afar and in your own unique way. I know you are proud, and I am happy to share this achievement with you. A special mention goes to my grandmothers, whose resilience has always inspired me, and to my little nephews, with the hope that they may one day see in this journey an example of dedication and perseverance.

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# Abstract

This thesis explores the relationship between Elon Musk's tweets and the stock price movements of Tesla, employing machine learning, deep learning and econometric methods. Tweets are pre-processed to extract sentiment features using transformer models such as BERT, LSTM, and RoBERTa. The impact of sentiment on Tesla stock returns is then analyzed using GARCH and GARCH-X models. The study investigates the influence of social media on financial markets and contributes to the growing field of sentiment-based forecasting.

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# Nomenclature

ACF Autocorrelation Function

AI Artificial Intelligence

BERT Bidirectional Encoder Representations from Transformers

BoW Bag-of-Words

CBOW Continuous Bag-of-Words

CST Combined Sentiment-Topic

MLM Masked Language Modeling

NLP Natural Language Processing

NSP Next Sentence Prediction

OHLC Open-High-Low-Close

PACF Partial Autocorrelation Function

POS Part-Of-Speech

RoBERTa Robustly Optimized BERT Pretraining Approach

TF-IDF Term Frequency-Inverse Document Frequency

# Chapter 1

# Introduction

## 1.1 Motivation and Research Relevance

In recent years, the increasing availability of user-generated content on social media has opened new opportunities to analyze public sentiment in real time. Among the many influential figures on these platforms, Elon Musk stands out for the significant attention his tweets receive from the public, media, and financial markets. His posts have often been linked to immediate and sharp reactions in the price of Tesla stock, prompting both academic and professional interest in whether such sentiment-driven effects can be measured and predicted (Nofsinger, 2005; Kraaijeveld and De Smedt, 2020).

Sentiment analysis, a growing field within Natural Language Processing (NLP), enables researchers to extract emotional cues from text data and use them as inputs for forecasting or explanatory models. While much of the existing research has focused on traditional financial news articles (Tetlock, 2007), more recent contributions show that social media content can offer predictive signals for market behavior, including stock returns and volatility (Zhang et al., 2011; Tetlock et al., 2008).

Moreover, studies have demonstrated that machine learning methods outperform traditional dictionary-based techniques when it comes to interpreting informal and

context-dependent expressions in platforms like Twitter[1] (Joulin et al., 2016; Mikolov et al., 2013a).

Given this background, this thesis aims to explore the relationship between sentiment extracted from Elon Musk's tweets and Tesla's stock performance. By combining machine learning models for sentiment analysis in Python with econometric tools in R, the study investigates whether sentiment signals can help explain or anticipate financial outcomes in a data-driven and interdisciplinary way.

## 1.2 Objectives of the Study

The goal of this thesis is to investigate the financial relevance of social media sentiment, with a specific focus on Elon Musk's tweets and their potential influence on Tesla's stock performance. The study is grounded in the growing body of literature that explores the intersection between sentiment analysis and financial market prediction, which has demonstrated that textual sentiment extracted from social media and news sources can carry significant explanatory and even predictive power with respect to asset price movements and volatility (Bollen et al., 2011b; Zhang et al., 2011; Shapiro et al., 2022).

The first objective is to employ a machine learning-based framework to analyze the emotional tone of Elon Musk's tweets, addressing the limitations of traditional lexicon-based approaches. While lexicons often fail to capture informal, sarcastic, or context-dependent language common on Twitter (Wankhade et al., 2022), our methodology leverages RoBERTa as both a high-accuracy benchmark and a pseudo-labeling tool. This dual role capitalizes on RoBERTa's state-of-the-art performance in contextual sentiment analysis (Liu et al., 2019), while its pseudo-labels enable supervised training of comparative models, logistic regression, random forests, and

[1]Although the platform is now officially rebranded as X, the name "Twitter" is used throughout this thesis for consistency and alignment with the existing academic literature.

LSTMs, without reliance on manual annotation.

Unlike hybrid lexicon-machine learning approaches (Kolchyna et al., 2015), our framework prioritizes scalability and consistency. RoBERTa, fine-tuned for tweet sentiment analysis, outperforms static embeddings (e.g., Word2Vec) and lexicons in detecting nuanced sentiment (Barbieri et al., 2020). The pseudo-labeling process uses RoBERTa-generated labels to train lightweight models, balancing computational efficiency with transformer-level accuracy. These trained models are then evaluated against RoBERTa's predictions to assess trade-offs in speed, interpretability, and performance for financial applications. This approach ensures robustness against ambiguity while maintaining reproducibility, a critical advantage for analyzing large-scale social media data.

The second objective involves aligning the extracted sentiment information with Tesla's stock market data, creating an integrated dataset that merges financial time series with aggregated sentiment indicators. This methodological step follows earlier work by Price et al. (2012), who emphasized the importance of synchronizing textual tone with financial event timing to capture its predictive effect on asset prices.

The third objective is to evaluate whether sentiment derived from social media can explain or predict financial market movements. This analysis is conducted in R, focusing on econometric modeling techniques suited to financial time series data. Specifically, GARCH and GARCH-X models are employed to examine volatility dynamics and the potential impact of sentiment signals on Tesla's stock returns and volatility. This approach aligns with best practices in financial econometrics for modeling time-varying volatility and incorporating external regressors such as sentiment measures (Bollerslev, 1986; Engle, 1982).

By addressing these three objectives, this thesis aims to provide an integrated, cross-disciplinary framework that combines NLP in Python with econometric modeling in R. It seeks not only to explore a high-profile case study, like Elon Musk and Tesla, but also to contribute more broadly to the literature on alternative data

in finance, showing how unstructured textual information can be systematically transformed into meaningful financial insight (Baker and Wurgler, 2007; Das and Chen, 2007).

## 1.3 Research Questions

This thesis seeks to answer whether public sentiment, as expressed in Elon Musk's tweets, has a measurable and potentially predictive impact on the financial performance of Tesla's stock (Pagolu et al., 2016; Jiang, 2021). The study is guided by a central question: *Can machine learning-based sentiment analysis of Elon Musk's tweets explain or predict short-term movements in Tesla's stock price or volatility?*

To support this primary inquiry, the thesis also addresses several secondary questions. First, which machine learning techniques are most effective in capturing sentiment from short, informal, and often ambiguous texts like tweets? This concern mirrors earlier challenges highlighted in financial text analysis, where researchers noted that even structured corporate language, such as earnings call transcripts, requires sensitive handling of tone and context for predictive insight (Price et al., 2012).

Second, how does sentiment derived from these tweets correlate with Tesla stock metrics such as returns, volatility, and trading volume? Prior research using financial news sentiment shows that negative tone is particularly informative during periods of low quantitative visibility, reinforcing the importance of such textual data (Tetlock et al., 2008).

Finally, is there any evidence of a causal relationship between sentiment and market movement, as suggested in previous works on Twitter and financial markets (Bollen et al., 2011b; Zhang et al., 2011)? Similar to findings on the informativeness of tone in financial disclosures and Q&A sessions, where market reactions closely followed sentiment changes, this study attempts to identify whether Musk's tweets

produce parallel market effects (Price et al., 2012).

Such questions aim to bridge the gap between computational text analysis and empirical financial modeling by combining tools from both domains.

## 1.4 Methodological Overview

The research follows a structured two-phase approach, integrating NLP with financial econometrics, as illustrated in Figure 1.1.

In Phase 1, the sentiment analysis pipeline begins with data processing where collected tweets undergo cleaning (removal of URLs, mentions) and preprocessing such as lemmatization and stopword removal. Feature extraction is performed using BoW, TF-IDF (Qaiser and Ali, 2018) and word embeddings (Word2Vec and BERT )[2] for sparse lexical representations (Qaiser and Ali, 2018) and word embeddings such as Word2Vec and BERT for semantic encoding (Mikolov et al., 2013a; Devlin et al., 2019). RoBERTa is employed in two roles: first as a benchmark classifier providing state-of-the-art sentiment predictions, and second as a pseudo-labeler generating training labels for comparative models including logistic regression, random forest, and LSTM (Barbieri et al., 2020). The models are evaluated based on accuracy and F1-score to assess performance, training and inference time to measure computational efficiency, and interpretability.

Phase 2 focuses on financial impact analysis by integrating daily sentiment scores, aggregated from tweet-level predictions, with Tesla's stock returns and volatility metrics. Econometric modeling uses GARCH to model stock volatility clustering and GARCH-X to extend this by incorporating sentiment scores as exogenous variables.

[2]TF-IDF vectors represent word importance based on term frequency and document rarity; word embeddings are vector representations of words that encode semantic similarity (Qaiser and Ali, 2018; Mikolov et al., 2013a).

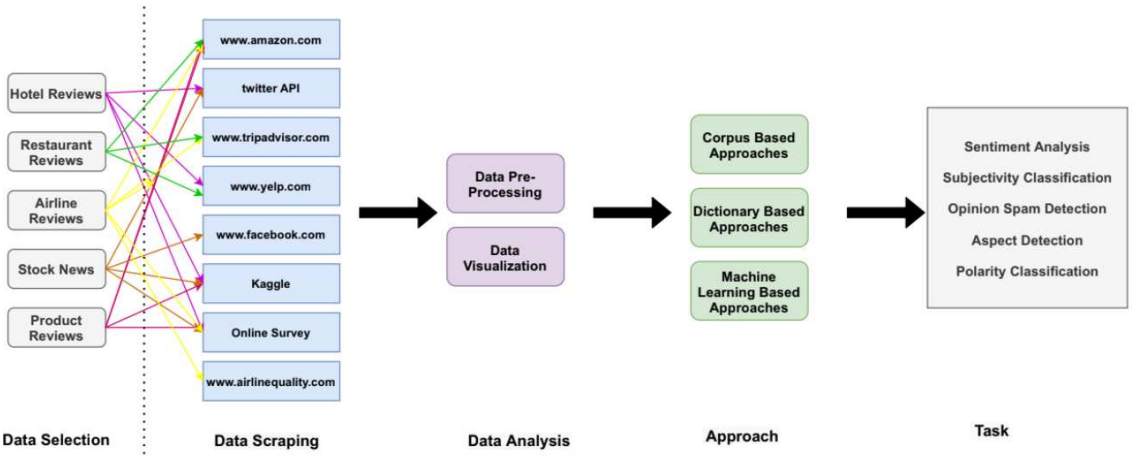



**Figure 1.1:** Sentiment analysis pipeline from data selection to task-specific modelin. Source: Wankhade et al. (2022).

## 1.5 Structure of the Thesis

The thesis is organized into seven chapters. Chapter 1 introduces the topic, presents the motivation, and outlines the key research questions and methodological design. Chapter 2 provides a theoretical background on sentiment analysis, its applications, and the intersection between social media and financial markets. Chapter 3 offers a critical review of the literature, focusing on prior research in sentiment-based forecasting and the role of influential social media activity in financial contexts.

Chapter 4 details the datasets used and the pre-processing steps applied to prepare both textual and financial data for analysis. Chapter 5 presents the implementation and evaluation of sentiment classification models developed in Python. Chapter 6 shifts the focus to the econometric analysis conducted in R, where the relationship between tweet sentiment and Tesla's stock performance is examined. Finally, Chapter 7 discusses the results in the context of the literature, addresses the study's limitations, and outlines potential directions for future research. Appendices A-C include relevant code, technical details, and additional results.

The structure reflects a sequential and integrated analysis, moving from raw textual input to empirical financial interpretation. A visual summary of the overall

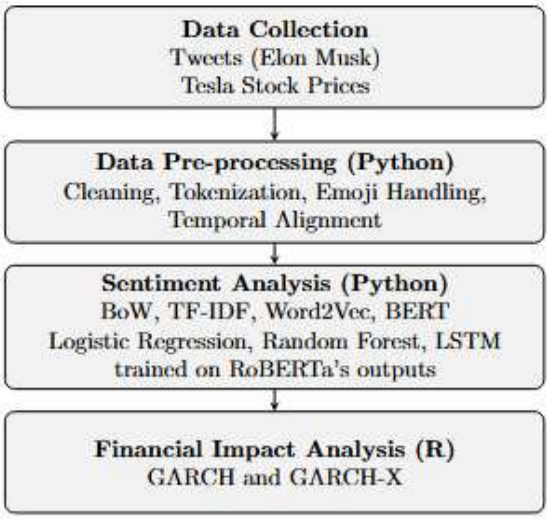



**Figure 1.2:** Overview of the analytical pipeline used in the thesis.

research pipeline is provided in Figure 1.2. The flowchart illustrates the sequential structure of the thesis, beginning with data collection and pre-processing, followed by sentiment analysis using machine learning and deep learning models in Python, and concluding with financial impact analysis and econometric testing in R.

This diagram reflects the interdisciplinary nature of the study, which integrates NLP with quantitative financial modeling to investigate the influence of Elon Musk's tweets on Tesla's stock performance. Each chapter of the thesis corresponds to a specific phase of the pipeline, ensuring coherence between the methodological flow and the structure of the written work.

# Chapter 2

# Theoretical Background

## 2.1 Introduction to Sentiment Analysis

Sentiment analysis, also known as opinion mining, is a subfield of NLP focused on detecting and categorizing opinions, emotions, and attitudes expressed in text. Sentiments can be classified into binary categories, such as positive, negative, and neutral, or mapped onto a multi-point scale.

In this context, sentiment analysis can be viewed as a classification task, where each sentiment category represents a distinct class label (Prabowo and Thelwall, 2009). Although this seems simple, it is actually a complex task, because human emotions can be expressed subtly or indirectly. As Prabowo and Thelwall (2009) explain, sentiment may be directly stated through evaluative words or implied through context, which has led to the development of various approaches, such as lexicon-based, statistical, and hybrid methods. Zhang and Liu (2017) provides a foundational taxonomy of these techniques, distinguishing between document-level, sentence-level, and aspect-level sentiment analysis, each presenting its own set of challenges in accurately identifying subjective information. Their work emphasizes the importance of opinion targets and opinion holders[1], which must be carefully

[1]The opinion target is the entity being evaluated, while the opinion holder is the source of the

extracted for sentiment analysis to be meaningful in real-world applications (Zhang and Liu, 2017).

With the rise of user-generated content on social media and review platforms, sentiment analysis has become increasingly important for businesses, researchers, and institutions. According to Wankhade et al. (2022), it plays a key role in understanding public opinion across different languages and sectors. Hussein (2018) reviews the main challenges in the field, dividing the methods into rule-based systems that use predefined lexicons and learning-based models that rely on machine learning to analyze large amounts of text.

Theoretical support for sentiment analysis also comes from emotion psychology. Ekman (1992) identified a set of basic emotions, like joy, anger, and sadness, that are universal and biologically rooted. These emotions can be used as a foundation for detecting sentiment in text. Early work in sentiment analysis includes the General Inquirer by Stone et al. (1966), which used dictionaries to classify words by emotional and psychological meaning. This early system influenced many of the techniques still used in sentiment analysis today.

More recent work emphasizes the emotional complexity and contextual nature of text, particularly in dynamic online environments such as Twitter. Sailunaz and Alhajj (2019) highlight that while other forms of online communication such as images and videos are increasingly popular, text remains the dominant mode for expressing opinions and emotions in social networks. Their research underscores the nuanced distinction between sentiment and emotion, with sentiment referring to the polarity (positive, negative, or neutral) and emotion reflecting specific affective states such as anger, joy, or disgust. They further demonstrate how integrating both tweets and their replies can lead to more accurate sentiment and emotion detection. This

opinion. For example, in "*The iPhone's call quality is good, but its battery life is short*," the targets are "call quality" (positive) and "battery life" (negative). In "*The mayor is loved by the people in the city, but he has been criticized by the state government*," the holders are "people in the city" and "state government" (Zhang and Liu, 2017).

work illustrates the growing sophistication in sentiment analysis methods and the importance of incorporating emotional dynamics in social media contexts (Sailunaz and Alhajj, 2019).

## 2.2 Levels of Sentiment Analysis

Sentiment analysis can be performed at different levels of granularity, specifically, at the document, sentence, phrase, and aspect levels, depending on the objectives of the study and the complexity of the textual data. Each level corresponds to a different linguistic unit and offers a distinct perspective on the sentiment being expressed (Zhang and Liu, 2017; Wankhade et al., 2022).

These levels are illustrated in Figure 2.1, which shows a nested structure where the document level encompasses broader text units, while more fine-grained emotional cues can be captured progressively at the sentence, phrase, and aspect levels. This hierarchical representation reflects the increasing specificity and analytical depth required to detect sentiment with greater precision (Zhang and Liu, 2017; Wankhade et al., 2022).

A multi-level approach is particularly valuable in domains such as finance, where sentiment can vary across multiple aspects of a message, such as entities, events, or subtopics, and influence economic behavior in complex ways. Such approaches enable finer-grained sentiment detection by aligning sentiment with specific topics or aspects within a document, rather than treating sentiment as a single, undifferentiated signal (Xianghua et al., 2013; Lu et al., 2011).

Moreover, the layered structure of emotional expression, ranging from broad statements to targeted opinions, finds theoretical grounding in emotion science. Ekman (1992) argues that emotions are rapid and involuntary reactions that can manifest at various textual levels, reinforcing the importance of analyzing sentiment through multiple layers.

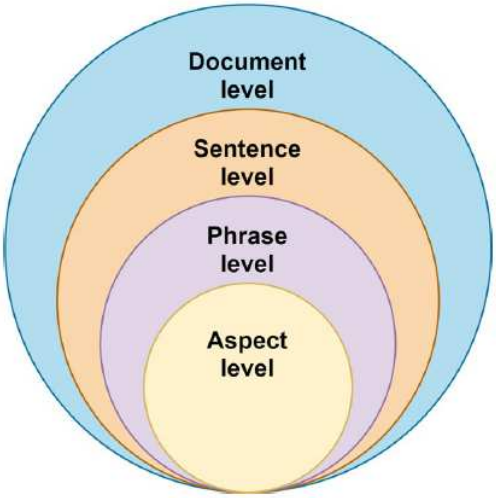



**Figure 2.1:** Different levels of sentiment analysis. Source: Wankhade et al. (2022).

The next subsections will explore each level in more detail, highlighting their methodological implications[2] and relevance for sentiment analysis in economic and financial texts.

### 2.2.1 Document-Level

Document-level sentiment analysis focuses on identifying the overall sentiment expressed in an entire text, such as a review, news article, or social media post. This level assumes that the document talks about one main topic or entity and expresses a consistent opinion throughout. The goal is to classify the whole document as either positive, negative, or neutral, depending on the language and context (Zhang and Liu, 2017; Wankhade et al., 2022).

Earlier methods for this task often used sentiment lexicons or supervised machine learning models trained on labeled data. These approaches work by analyzing individual words or sentence-level sentiments and then combining this information to predict the sentiment of the whole document. More recently, deep learning models

[2]The main computational approaches used at each level, such as lexicon-based methods, machine learning, and deep learning, are discussed in more detail in Chapter 3.

like neural networks and transformers have improved performance by better capturing meaning and context across the document (Mishev et al., 2020; Jiang, 2021).

One common technique used in document-level sentiment analysis is the TF–IDF method, which identify the most informative terms in a document by combining how often a word appears with how rare it is across other documents (Qaiser and Ali, 2018). Aizawa (2003) provided an information-theoretical interpretation of TF–IDF, showing how it helps highlight important words for sentiment classification, especially in specific fields like finance or economics.

However, traditional models like TF–IDF and early neural approaches sometimes struggle to capture complex sentiment patterns to adapt well to different domains. This problem is especially common when applying a model trained on one type of text (e.g., movie reviews) to another (e.g., financial reports). To address this, models like the combined sentiment-topic (CST) model have been developed. CST can identify both topics and their associated sentiments at the same time, making it more flexible for analyzing documents with multiple themes or ambiguous content (Usha and Devi, 2013).

In line with these developments, Behdenna et al. (2018) provide a comprehensive survey of document-level sentiment analysis methods, highlighting the differences between machine learning, lexicon-based, and hybrid approaches. Their findings show that SVM and Naïve Bayes are among the most widely used classifiers, and that deep learning models such as LSTM (Hochreiter and Schmidhuber, 1997) and convolutional neural network (Wang et al., 2016) are increasingly preferred due to their higher accuracy and better handling of linguistic complexity. However, they also note that bag-of-words (BoW) representations still limit many models' ability to capture sentiment subtleties (Behdenna et al., 2018).

Recently, models like RoBERTa (Liu et al., 2019) have been introduced to improve sentiment classification by using contextual embeddings and attention

mechanisms[3](Liu et al., 2019). While these models are often used for more detailed tasks, their ability to understand word relationships across a whole document also benefits document-level sentiment analysis. This is especially important in financial texts, where even short statements can significantly influence market perception (Liao et al., 2021).

In conclusion, document-level sentiment analysis is a foundational approach in NLP. While earlier methods provided the base for this field, recent advances in neural and transformer-based models continue to enhance the ability to interpret sentiment across different types of documents and domains (Aizawa, 2003; Usha and Devi, 2013; Behdenna et al., 2018; Liao et al., 2021).

### 2.2.2 Sentence-Level

Sentence-level sentiment analysis offers a more detailed view by focusing on the sentiment expressed in each individual sentence. This is especially useful when a document contains mixed opinions, with some parts being positive and others negative. Analyzing sentences one by one helps capture these differences and gives a more accurate picture of the overall sentiment (Zhang and Liu, 2017; Wankhade et al., 2022).

In this approach, each sentence is treated as a separate unit and classified as positive, negative, or neutral. The process usually involves identifying sentences that contain opinions and evaluating them. However, this can be challenging because some sentences are ambiguous or rely on surrounding context to understand the sentiment, especially in cases of sarcasm or implied meaning (Zhang and Liu, 2017).

Recent models like BERT (Devlin et al., 2019) have made major improvements in sentence-level sentiment analysis. Unlike older models that read text in one direction, BERT looks at both the left and right context of each word at the same time. This

[3]Attention mechanism is a technique that allows a model to focus on the most relevant parts of the input when making predictions (Liao et al., 2021).

allows it to better understand the full meaning of a sentence, which is especially useful for short or complex texts (Devlin et al., 2019).

BERT is trained using tasks like Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), helping it learn relationships between words and sentences. These features make it very effective in financial texts, where short statements like a CEO's quote or a market update can strongly influence investor sentiment (Devlin et al., 2019).

Other models, like LSTM, have also been applied to sentence-level analysis (Hochreiter and Schmidhuber, 1997). Rao et al. (2018) showed that using sentence representations in an LSTM framework can improve classification performance by preserving important context, even in longer documents. This helps overcome issues like loss of meaning and limited memory in earlier models.

Studies have also shown that sentence-level sentiment analysis works well for financial documents, especially forward-looking statements. For example, classifiers like naïve bayes and BERT variants can detect subtle changes in tone, which are often more important to investors than the overall tone of the document (Kearney and Liu, 2014).

In conclusion, sentence-level analysis fills the gap between full-document sentiment and more detailed methods. It is particularly useful for analyzing financial news, reports, or tweets where mixed opinions are common. By examining each sentence separately, it provides a clearer view of how sentiment is distributed throughout the text (Kearney and Liu, 2014; Rao et al., 2018; Devlin et al., 2019).

### 2.2.3 Phrase-Level

Phrase-level sentiment analysis looks at smaller parts of a sentence, such as specific phrases or words that express opinions. This level is helpful when sentiment is expressed in short segments of text, especially in documents with mixed or subtle emotional tones. Unlike sentence-level analysis, which assumes a single sentiment

per sentence, phrase-level analysis can capture shifts in tone that might otherwise be missed (Zhang and Liu, 2017; Wankhade et al., 2022).

To do this, systems often rely on tools like sentiment lexicons and syntactic parsing, which involves analyzing the grammatical structure of a sentence to understand how opinion words relate to the aspects they describe. These help identify words or phrases that carry emotion or judgment. For example, techniques like part-of-speech (POS) tagging, which identifies the grammatical role of each word (such as nouns, verbs, or adjectives), and dependency parsing, which analyzes how words in a sentence are syntactically connected, can highlight important terms such as adjectives or noun phrases. In financial news, a phrase such as "unexpectedly weak earnings" clearly suggests negative sentiment, even if the full sentence seems neutral (Zhang and Liu, 2017; Wankhade et al., 2022).

This level of analysis has improved with the use of word embeddings, mathematical models that represent words in a continuous vector space. Mikolov et al. (2013a) introduced Word2Vec, which learns these representations based on word context. With this approach, phrases that share similar meanings are mapped close to each other in the space. It also helps detect the overall sentiment of a phrase by analyzing the words it contains. This is particularly useful in finance, where certain phrases like "profit warning" or "rate hike" carry clear emotional weight (Mikolov et al., 2013a).

Research in financial text analysis supports the value of phrase-level sentiment. Schumaker and Chen (2009) showed that using noun phrases in an SVM model led to better predictions of stock price movements. Their AZFinText system used noun phrase extraction and named entity recognition to focus on impactful financial expressions. The system achieved better prediction accuracy and trading simulation results compared to models using simpler features, showing the power of phrase-level analysis in financial applications (Schumaker and Chen, 2009).

In short, phrase-level sentiment analysis provides a very detailed view of sentiment within texts. By combining linguistic techniques and modern embeddings, it helps

detect emotional signals that may be missed at higher levels. This is especially important in financial texts, where short, meaningful phrases can strongly influence how information is perceived (Mikolov et al., 2013a; Schumaker and Chen, 2009).

### 2.2.4 Aspect-Level

Aspect-level sentiment analysis is the most detailed type of sentiment analysis. Instead of classifying the sentiment of a whole document or sentence, it focuses on identifying what specific parts or features of a subject people are talking about—and what sentiment they express toward each one. This is useful in contexts like product or service reviews, where users often mention multiple features such as price, quality, or design, and give different opinions on each (Schouten and Frasincar, 2015; Zhang and Liu, 2017; Wankhade et al., 2022).

At this level, the process usually involves three steps: first, finding the aspects or targets mentioned in the text; second, assigning a sentiment to each one; third, aggregating for each aspect to provide a concise overview. For example, a user might say a Tesla car is innovative (positive sentiment) but also mention that its customer support is poor (negative sentiment). This can be challenging, especially when the aspect is implied rather than clearly stated. For this reason, more advanced models have been developed to detect both explicit and hidden aspects and link them to the correct sentiment (Schouten and Frasincar, 2015).

This method is becoming more popular because it helps companies and researchers to better understand detailed user feedback. In finance, for instance, investors may react differently to sentiment about stock performance, leadership, or future plans. Instead of just knowing whether a report is positive or negative, aspect-level analysis reveals exactly what part of the message people feel strongly about (Lu et al., 2011).

Some studies use topic modeling techniques like LDA to automatically find relevant aspects and their associated sentiments. Other methods rely on small sets of seed words or weak supervision to reduce the need for labeled data, making the

process more scalable (Lu et al., 2011).

Aspect-level sentiment analysis has also been applied in online discussions, such as movie review forums. Thet et al. (2010) proposed a method that combines NLP and machine learning to extract features (like acting, story, or soundtrack) and analyze the sentiment linked to each one. Their study showed that this type of analysis helps detect how different aspects are perceived, especially when people express mixed or complex opinions.

This level of analysis is also important in financial and social media texts, where a single message might contain both praise and criticism. Schouten and Frasincar (2015) point out that both the aspect and its sentiment can be either obvious or hidden, and detecting them correctly often requires domain-specific knowledge. This adds complexity, but it also increases the value of the insights produced[4].

Overall, aspect-level sentiment analysis provides a much clearer and more detailed picture of how people feel about different parts of a subject. It is especially helpful in fields like marketing, finance or politics, where understanding the focus of an opinion can make a big difference (Thet et al., 2010; Lu et al., 2011; Schouten and Frasincar, 2015).

## 2.3 Applications of Sentiment Analysis

Sentiment analysis has evolved into a versatile tool with wide-ranging applications across numerous fields. By enabling the extraction of subjective information from textual data, it supports decision-making processes in both commercial and public sectors. This section explores how sentiment analysis is being applied in various domains, including finance, marketing, healthcare, political science, and public

[4]Both sentiment and target can be expressed explicitly or remain implicit, independent of each other. When explicitly mentioned, a sentiment or target is literally present in the text. In contrast, implicit expressions of sentiment or target must be inferred, often requiring additional context or domain knowledge (Schouten and Frasincar, 2015).

administration, highlighting its role in generating actionable insights, improving user engagement, and informing policy and strategy (Birjali et al., 2021).

In financial contexts, crucial to this thesis, sentiment analysis has evolved into a powerful tool for market prediction and risk assessment. Deep learning models like BERT, when fine-tuned on financial texts, have shown remarkable proficiency in capturing subtle investor sentiments and market cues (Mishev et al., 2020). For instance, sentiment extracted from financial news articles and social media platforms is now routinely incorporated into quantitative trading strategies and predictive analytics models. Such integration enables more accurate forecasting of asset price movements and volatility patterns by incorporating the psychological dimensions of investor behavior (Valencia et al., 2019; Jiang, 2021, 2024). A notable example is provided by Bhardwaj et al. (2015), who demonstrated that sentiment analysis applied to Indian financial news could effectively predict trends in major stock indices such as the Sensex and Nifty[5], thereby illustrating the potential of these tools in emerging market contexts where traditional financial indicators may lack transparency or timeliness. Nevertheless, as shown by Aït-Sahalia et al. (2024), many high-frequency price jumps remain unexplained by identifiable public news, despite the availability of machine-readable news feeds such as Thomson Reuters News Analytics[6]. This suggests persistent inefficiencies in the market and reinforces the value of sentiment-based models capable of capturing more nuanced or latent informational signals beyond formal news disclosures.

During times of crisis, sentiment analysis assumes an even more central role due to its ability to provide real-time insights into public emotion and behavioral response.

[5]Sensex (Sensitive Index) is the benchmark index of the Bombay Stock Exchange (BSE), while Nifty 50 is the flagship index of the National Stock Exchange (NSE) in India. Both track the performance of major listed companies and are key indicators of Indian stock market trends. Source: Borsa Italiana.

[6]Thomson Reuters News Analytics (TRNA) is a machine-readable news feed that provides real-time metadata and sentiment scores for financial news articles, enabling algorithmic trading systems to incorporate structured information from unstructured text sources.

Singh et al. (2021), for example, employed a BERT-based model to evaluate public sentiment during the COVID-19 pandemic. Their findings highlighted how these models can effectively manage the informal, emotionally intense content typical of social media, offering valuable data for crisis communication and policy evaluation. Similarly, Costola et al. (2023) showed that sentiment derived from over 200,000 COVID-related news articles—using a financial-domain-adapted BERT model—had a significant influence on the performance of the S&P 500, underscoring the sensitivity of financial markets to collective sentiment in times of uncertainty.

Rognone et al. (2020) underscored the increasing importance of sentiment analysis in financial and economic contexts, particularly in relation to decision-making processes. They focused on the integration of high-frequency intra-day data with sentiment derived from news sources to examine the influence of media coverage on both traditional currencies and cryptocurrencies. Their analysis revealed distinct differences in how Bitcoin and traditional currencies react to news sentiment, indicating that cryptocurrencies may follow different behavioral patterns. Similarly, Birjali et al. (2021) discussed the convergence of sentiment analysis with emerging technologies such as Big Data, cloud computing, and blockchain. According to their survey, such integrations are significantly improving predictive accuracy and facilitating real-time data processing across a range of industries.

Beyond the financial domain, there is a growing focus on adapting sentiment analysis models to specific linguistic and sociocultural contexts. For example, Pota et al. (2020) demonstrated that adapting BERT to Italian tweets significantly improved performance, illustrating the necessity of customizing sentiment models for local languages, dialects, and cultural nuances. This trend toward language-specific and domain-adapted models reflects a broader shift in sentiment analysis research, as highlighted by Birjali et al. (2021), who noted the importance of language resources and pre-processing pipelines adapted to the characteristics of informal, noisy, or short-form text common in social media and mobile communication platforms.

Moreover, the scope of opinion mining has expanded considerably, with growing applications in areas such as marketing, healthcare, political science, and public administration. In marketing, it is commonly employed to track brand perception, assess customer satisfaction, and improve targeting strategies. In the healthcare domain, it supports the interpretation of patient feedback and helps monitor public attitudes toward medical treatments and health campaigns. Within political science, sentiment analysis is used to measure public opinion on policy issues, analyze electoral trends, and investigate media bias (Benhabib and Spiegel, 2019; Birjali et al., 2021; Wankhade et al., 2022). Birjali et al. (2021) note that governments and institutions are progressively adopting sentiment analysis for real-time decision-making and policy development, particularly through the analysis of social media and online discussions. Similarly, Wankhade et al. (2022) highlight its value in informing long-term strategic planning across diverse sectors, ranging from consumer markets to public health systems.

In the context of social issues, models like MSA-COSR (Multi-aspect Sentiment Analysis for Chinese Online Social Reviews) have been proposed to conduct unsupervised aspect-specific sentiment analysis in Chinese online forums, demonstrating the potential of such approaches to capture the nuances of complex public discourse (Xianghua et al., 2013). Furthermore, Birjali et al. (2021) highlight the growing significance of aspect-level sentiment analysis, particularly in fields that demand detailed insights, such as product development and service quality assessment. They also discuss innovative techniques that integrate multiple layers of sentiment analysis to produce a more nuanced understanding of consumer sentiment.

Sentiment analysis is also being increasingly applied in the context of corporate crises and reputational risk management. Comyns and Franklin-Johnson (2018), in their study of the Rana Plaza disaster[7], found that when the media presents a

[7]The Rana Plaza disaster refers to the collapse of a commercial building in Dhaka, Bangladesh, in 2013, which housed garment factories supplying global fashion brands. Over 1,100 workers were

crisis as involving several companies, it can increase public attention and damage the reputation of those firms. Their findings suggest that crisis communication strategies must account for the emotional undertone of public discourse, which sentiment analysis can help quantify and interpret. In alignment with this, Birjali et al. (2021) advocate for the use of sentiment tracking as an integral part of corporate risk management systems, enabling companies to detect early signals of public discontent or backlash.

Finally, in business intelligence, the integration of sentiment analysis into data-driven decision systems is becoming standard practice. Companies use sentiment data to inform product design, tailor advertising strategies, and assess the impact of brand campaigns, while public institutions deploy sentiment dashboards to monitor social reactions to governance decisions. As Zhang and Liu (2017) and Wankhade et al. (2022) observe, this capacity to transform subjective, qualitative data into structured insights makes sentiment analysis a vital tool in the digital economy. Birjali et al. (2021) further expand on this point by outlining how sentiment analysis, when combined with other Artificial Intelligence-driven (AI) technologies, can be deployed at scale to deliver actionable insights across industries. Figure 2.2 illustrates this expanding landscape, showcasing applications across healthcare, product reviews, strategic planning, and beyond.

## 2.4 Social Media and Financial Markets

Investor sentiment has traditionally been measured through macroeconomic indicators and psychological models. However, with the rise of social media, new sources of data have emerged, offering more immediate and detailed insights into market behavior. Baker and Wurgler (2007) differentiate between two main approaches to

killed, making it one of the deadliest industrial accidents in history and a pivotal moment in global discussions on corporate social responsibility and supply chain ethics (Comyns and Franklin-Johnson, 2018).

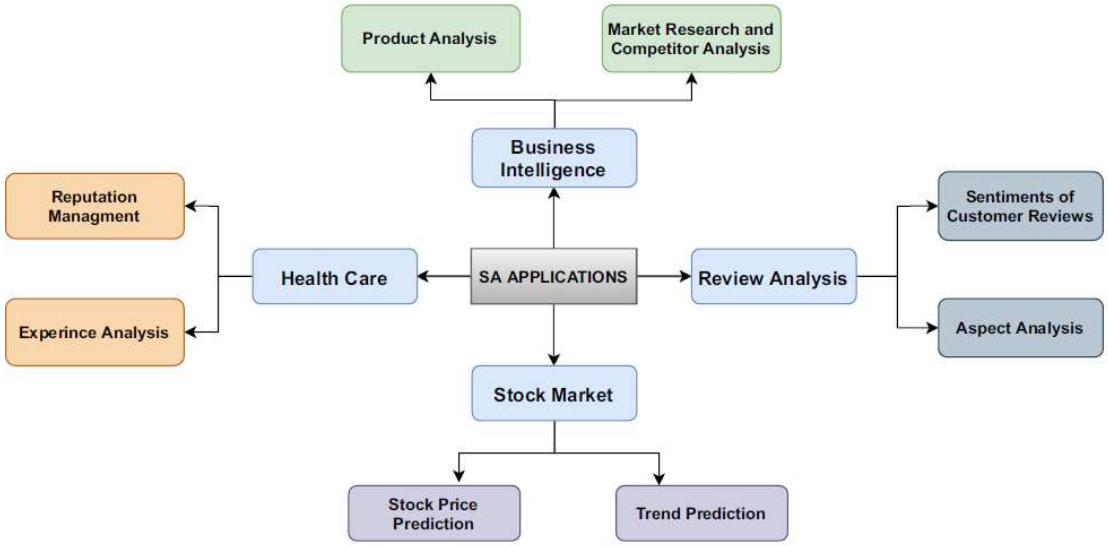



**Figure 2.2:** Sentiment analysis applications. Source: Wankhade et al. (2022).

sentiment analysis in finance: bottom-up and top-down models. The first examines individual investor biases, using psychological tendencies like overconfidence, representativeness, and conservatism to explain underreaction or overreaction to past returns or fundamentals Baker and Wurgler (2007). The second focuses on aggregate sentiment and its influence on asset prices, arguing that waves of collective optimism or pessimism can significantly affect markets, especially in contexts of limited arbitrage and persistent inefficiencies (Nofsinger, 2005; Baker and Wurgler, 2007).

Platforms like Twitter have become central to this new wave of sentiment research (Zhang et al., 2011; Metta et al., 2022). Social media provides large-scale, real-time data on public opinions and emotional reactions, allowing researchers to track shifts in investor psychology as events unfold. Unlike traditional sentiment indicators, which may lag behind actual market changes, social media offers a more immediate reflection of collective mood and behavior (Asur and Huberman, 2010). As Birjali et al. (2021) emphasize, social media serves as a dominant channel for user-generated content that captures a wide range of emotions and opinions relevant to financial sentiment. Twitter, in particular, has been leveraged in numerous studies to forecast

electoral outcomes, detect political leaning, and analyze investor sentiment and its impact on cryptocurrency and foreign exchange markets (Birjali et al., 2021).

Recent advancements in sentiment analysis methods have significantly improved the ability to process social media data in financial applications. Mishev et al. (2020) present a comprehensive evaluation of sentiment analysis models applied to financial texts, emphasizing the transition from traditional lexicon-based approaches to more advanced deep learning techniques, including transformer-based models like BERT. Their study highlights that while lexicon-based methods are interpretable and relatively easy to implement, they struggle with contextual nuances and domain-specific terminology, particularly in financial microblogs and short-text formats common on platforms like Twitter. In contrast, transformer models such as FinBERT, trained specifically on financial corpora, demonstrate superior performance in capturing sentiment polarity and emotional tone in social media content (Mishev et al., 2020). Mishev et al. (2020) also underscore the importance of domain adaptation and fine-tuning for achieving optimal results, noting that pre-trained general-purpose models often struggle to accurately understand the specialized language used in the financial domain. Their findings reinforce the value of combining high-frequency social media data with advanced NLP models to support sentiment-driven financial decision-making.

Initial attempts to measure investor sentiment using online sources highlight the fundamental importance of unstructured digital text in the development of financial models. Das and Chen (2007) pioneered the use of NLP to extract sentiment from internet message boards, notably Yahoo! Finance conversations surrounding Amazon. Their methodology classified user-generated messages into bullish, bearish, or neutral categories and demonstrated that aggregated sentiment had predictive value for stock price volatility and message volume.

Further reinforcing the importance of textual tone, Tetlock (2007) empirically demonstrated that the frequency of negative words in firm-specific news articles is a

significant predictor of both earnings performance and stock returns. Their study showed that linguistic negativity provides incremental information beyond traditional financial indicators, especially when focused on firm fundamentals. Notably, the authors observed that stock prices underreact to negative language in the short term, enabling temporary mispricings. These effects were strongest for high-frequency, fundamentals-related news and were more pronounced when using standardized metrics of linguistic negativity. Their findings support the notion that qualitative media content, when rigorously quantified, offers a powerful complement to numeric financial data in explaining asset price movements.

Understanding how people use platforms like Twitter is key to interpreting their role in financial markets. Java et al. (2007) conducted one of the earliest large-scale studies on Twitter behavior, identifying core user motivations such as sharing personal updates, spreading information, and reporting news. These findings support the idea that Twitter is not only a communication tool but also a rich source of public sentiment that can signal market movements. Birjali et al. (2021) further note that the temporal immediacy and emotional expressiveness of tweets make them highly suitable for real-time sentiment analysis in volatile financial environments. Rognone et al. (2020) found that tweet sentiment was correlated with fluctuations in Bitcoin prices and traditional currency values, indicating the growing relevance of high-frequency textual sentiment analysis in financial forecasting.

Kearney and Liu (2014) provide a comprehensive review of textual sentiment analysis in financial applications and highlight several limitations of early models, which often relied on simplistic BoW approaches or manually curated sentiment lexicons. Kraaijeveld and De Smedt (2020) underscored that sentiment extracted from Twitter and other platforms must be handled with care due to noise, irony, and context sensitivity, factors that complicate its interpretation and require advanced linguistic models.

Moreover, emotional expressions on social media often arise from instinctive

human responses rather than socially learned behaviors. As Ekman (1992) suggests, emotions like fear, anger, or surprise are biologically driven and universally expressed. During periods of financial uncertainty or market shocks, these emotions are likely to be visible in social media posts, making platforms like Twitter useful for capturing raw emotional reactions that may influence investor decisions. In this context, both Birjali et al. (2021) and Kearney and Liu (2014) point to the importance of incorporating emotion detection into financial sentiment models. While Birjali et al. (2021) highlight advances in machine learning techniques for classifying emotional tone, Kearney and Liu (2014) emphasize the need for sentiment frameworks that can differentiate between transient affect and sustained sentiment trends.

In summary, social media has transformed how sentiment is measured and analyzed in financial contexts. It enables real-time monitoring of public mood and offers valuable insights into behavioral finance, particularly during periods of volatility. As Birjali et al. (2021) observe, social media sentiment analytics are now integral to financial forecasting systems, portfolio management tools, and high-frequency trading strategies. Similarly, Kearney and Liu (2014) argue that although methodological challenges remain, the integration of social media data into financial modeling represents a critical step toward a more nuanced and responsive understanding of investor behavior in the digital era.

## 2.5 Elon Musk's Communication and Public Influence

Elon Musk is known for his direct and informal communication style on social media, especially on Twitter. His frequent and often provocative tweets have gained massive attention not only from the public but also from investors and analysts. Because Tesla is a company that is relatively young, highly volatile, and difficult to value, its stock is particularly sensitive to investor sentiment. According to Balaji et al.

(2021), companies with these characteristics are more likely to be influenced by shifts in market mood, and Musk's tweets often act as triggers for such shifts. Research has empirically confirmed this dynamic: Dam (2023) found that Musk's tweets lead to statistically significant movements in Tesla's stock price, particularly when the tweets are emotionally charged or newsworthy.

As a well-known and influential figure, Musk's public statements carry significant weight. His tweets often reach millions of people instantly and can cause strong reactions in financial markets. This direct form of communication allows him to influence public opinion and investor behavior without relying on traditional media. In this sense, his Twitter account functions almost like a live channel of sentiment transmission, making it an important source for analyzing how online communication can affect financial decisions (Metta et al., 2022). This effect is not limited to finance—Schivinski and Dabrowski (2016) found that social media communication, particularly user-generated content, plays a key role in shaping consumer perceptions of brand equity and loyalty. In Musk's case, the blurring of personal and corporate communication on platforms like Twitter allows his persona to directly influence both consumer attitudes and investor sentiment, reinforcing the idea that executive social media presence can have measurable effects on brand perception and market behavior. Additionally, Chou et al. (2009) emphasize that social media functions as a powerful channel for rapidly disseminating information and shaping public dialogue, particularly in emotionally charged or high-stakes contexts. This supports the view that Musk's tweets operate not only as market signals but also as influential narratives that can shape collective opinion and behavioral responses at scale[8].

From a psychological perspective, Ekman's theory of basic emotions helps explain

[8]While this analysis centers on Elon Musk's influence on market sentiment via social media, it is worth noting that broader socio-political events, such as the 2024 U.S. presidential election, can also impact public sentiment and financial markets. Musk's commentary on such events, whether direct or indirect, may interact with or amplify existing sentiment trends. However, these dynamics fall outside the scope of this study.

Musk's influence.Ekman (1992) argues that emotions like fear, joy, or surprise are universal and biologically driven. These emotions often guide how people react to information, especially in uncertain situations. When Musk shares emotionally charged content—whether it's optimistic, sarcastic, or unexpected—it may trigger quick and instinctive responses in his audience, including investors. Supporting this, sentiment analysis studies have shown that emotional content in his tweets correlates strongly with fluctuations in stock trading volumes and price volatility (Bhadkamar and Bhattacharya, 2022). Musk also uses language in ways that are informal and emotionally expressive. He often includes humor, emojis, or sarcasm, which adds complexity to analyzing the sentiment of his tweets but also makes them rich in emotional content (Schouten and Frasincar, 2015; Hussein, 2018). His posts generate strong engagement from the public, both positive and negative, further amplifying their effect on collective sentiment. In line with this, large language models such as RoBERTa and BERT have been used successfully to classify sentiment, including Tesla's social media presence (Sharma et al., 2024). Furthermore, predictive models combining sentiment analysis and machine learning, such as ARIMA-LSTM hybrids, have demonstrated the ability to forecast Tesla stock trends by integrating public sentiment signals from Musk's online activity (Jiang, 2024).

In summary, Elon Musk's communication shows how a single influential figure can shape market sentiment through social media. His tweets offer a unique and valuable case for studying the relationship between public influence, emotional expression, and financial market behavior (Dam, 2023).

# Chapter 3

# Literature Review

## 3.1 Sentiment Analysis in Finance

Sentiment analysis has become a central tool in financial research, particularly in understanding how investor mood and public perception influence asset prices. According to Baker and Wurgler (2007) sentiment does not merely explain contemporaneous mispricing but also holds predictive power for future returns. Their findings indicate that during periods of elevated market sentiment, speculative stocks tend to underperform in the medium term as sentiment normalizes or underlying fundamentals fail to justify inflated valuations. This concept is particularly relevant when analyzing stocks like Tesla, which often attract speculative trading and exhibit strong market reactions to public communication, especially via social media platforms like Twitter. The evidence supports the broader hypothesis that investor sentiment can act as a forward-looking indicator.

Building on this conceptual foundation, the literature has progressively incorporated textual data as a source of sentiment. One early contribution is by Das and Chen (2007), who analyzed message board discussions on platforms like Yahoo! Finance. They found that even informal, unstructured investor commentary, often dismissed as "noise", can offer predictive signals for stock performance, especially in

the case of small-cap or low-liquidity stocks.

The predictive value of sentiment is also evident in corporate communication. For instance, Li (2010) demonstrated that forward-looking statements in annual 10-Q and 10-K filings[1] carry meaningful information about both future earnings and stock returns. In an earlier work, Li (2006) also showed that the use of risk-related language in financial disclosures significantly impacts investor reaction and future volatility, thus reinforcing the notion that qualitative text data is informative for market forecasting.

Similarly, financial media plays a crucial role in shaping investor expectations. Tetlock (2007) found that the tone of daily financial news, particularly the prevalence of negative words, significantly affects short-term stock returns. This finding was further extended by Tetlock et al. (2008), who focused on firm-specific news and showed that negative sentiment in such articles is predictive of both earnings surprises and stock returns.

Another important application of sentiment analysis is found in the study of spoken corporate communication. Price et al. (2012) examined earnings conference calls and discovered that the tone used, especially during the unscripted Q&A portions, provides useful signals that help explain abnormal returns following these events.

A psychological perspective also contributes to the theoretical understanding of sentiment in finance. According to Ekman (1992), emotional expressions are biologically determined and universally recognized, supporting the view that aggregated public emotion, when measured systematically, can be a stable indicator of market sentiment. This rationale strengthens the methodological use of sentiment analysis in modeling investor behavior.

[1]10-Q is a quarterly report, while 10-K is an annual report that U.S. public companies must file with the Securities and Exchange Commission (SEC). It provides a detailed overview of the company's financial performance, risks, and operations, including audited financial statements and the Management's Discussion and Analysis (MD&A) (Li, 2010).

More recently, Mishev et al. (2020) provided a comprehensive review of sentiment analysis techniques in finance, highlighting a methodological evolution from basic lexicon-based approaches to modern deep learning models. Their study emphasizes the superior performance of contextual models, such as FinBERT, over traditional methods. This finding aligns with the approach taken in this thesis, which incorporates transformer-based architectures to analyze sentiment from both social media and financial documents.

In summary, the literature confirms that sentiment, whether derived from investor forums, corporate filings, media reports, or spoken communication, has a measurable impact on financial markets. The methodological progression toward machine learning and deep learning approaches reflects both the complexity of sentiment itself and the growing availability of unstructured financial text data (Mishev et al., 2020).

## 3.2 Twitter Sentiment Analysis

Twitter has emerged as one of the most widely used platforms for sentiment analysis, thanks to its real-time nature and the availability of large-scale public data. It has been applied across various domains, such as predicting real-world outcomes like movie box-office revenues (Asur and Huberman, 2010), or analyzing public response to global events. For example, Singh et al. (2021) examined sentiment related to the COVID-19 pandemic and its impact on social life, suggesting that emotional trends on Twitter can reveal public perception of government policies and support decision-making during health crises.

In the financial domain, Twitter sentiment has shown predictive potential. Bollen et al. (2011b) and Zhang et al. (2011) observed that an increased percentage of emotionally charged tweets correlates with declines in major stock indices such as the Dow Jones, NASDAQ, and S&P 500, whereas lower emotional intensity is often linked to market gains. Further, Kraaijeveld and De Smedt (2020) demonstrated

that Twitter sentiment can predict cryptocurrency movements—such as Bitcoin, Litecoin, and TRON—with varying degrees of accuracy depending on the timeframe and asset.

Another relevant study by Valencia et al. (2019) emphasized the value of integrating public sentiment with market indicators, showing that doing so generally improves financial forecasting models. Pagolu et al. (2016) reinforced this insight by applying machine learning models to Twitter data and achieving over 70% accuracy in predicting Microsoft stock price movements using N-gram and Word2Vec-based sentiment features.

Importantly, the quality of sentiment analysis on Twitter is influenced by several linguistic and technical factors. Tweets often contain abbreviations, sarcasm, and spelling errors due to character limits, making accurate interpretation challenging (Acosta et al., 2017). The presence of bot accounts further complicates analysis; for example, Kraaijeveld and De Smedt (2020) noted that bots accounted for 1–14% of crypto-related tweets, potentially biasing sentiment metrics.

Foundational work by Bollen et al. (2011a) analyzed nearly 10 million tweets to detect public mood responses to major socio-political events, such as the 2008 U.S. election and financial crisis. They found that Twitter users' emotions shifted significantly in response to such events, although these changes were typically short-lived.

Further advancements include Sailunaz and Alhajj (2019)'s model, which incorporated not only original tweets but also their replies, allowing for a more nuanced understanding of sentiment evolution in conversation threads. Their multidimensional framework introduced metrics such as agreement, sentiment, and emotion scores to capture user interactions more effectively.

Other studies have explored the influence of individual users. Metta et al. (2022) and Bhadkamar and Bhattacharya (2022) showed that tweets by influential figures like Elon Musk can significantly affect stock prices, particularly for companies he

mentions, including Tesla. Additionally, Basiri et al. (2021) proposed a deep learning architecture for sentiment classification, validated on several datasets including Twitter. Although the study focused on model development, it underscored Twitter's relevance in evaluating state-of-the-art sentiment analysis techniques.

Earlier contributions laid the groundwork for these developments. Java et al. (2007) analyzed user intentions rather than sentiment, offering a crucial distinction between behavioral intent and emotional expression. Similarly, Das and Chen (2007) demonstrated that online sentiment—originally extracted from message boards—could be quantified using lexicon-based techniques. Kolchyna et al. (2015) further developed this approach by combining lexicon-based and machine learning methods, showing that incorporating emoticons, slang, and abbreviations into sentiment lexicons improves classification performance in social media contexts.

## 3.3 Sentiment Extraction Techniques

Sentiment extraction techniques can be broadly categorized into three main approaches: lexicon-based methods, machine learning-based techniques, and hybrid models (Wankhade et al., 2022). This classification is used as a guiding structure for the present review and is visually summarized in Figure 3.1.

Lexicon-based techniques rely on predefined dictionaries of opinion-bearing terms and typically employ rule-based logic to assess sentiment polarity. These methods are valued for their interpretability and ease of use, especially when labeled data is scarce. However, their inability to capture context, handle negation, or adapt across domains often limits their effectiveness (Zhang and Liu, 2017).

In contrast, machine learning approaches build models that learn from labeled data using features derived from the text. These include syntactic elements, short sequences of words (known as n-grams), or numerical representations of words (called word embeddings) that capture their meaning based on context. While traditional

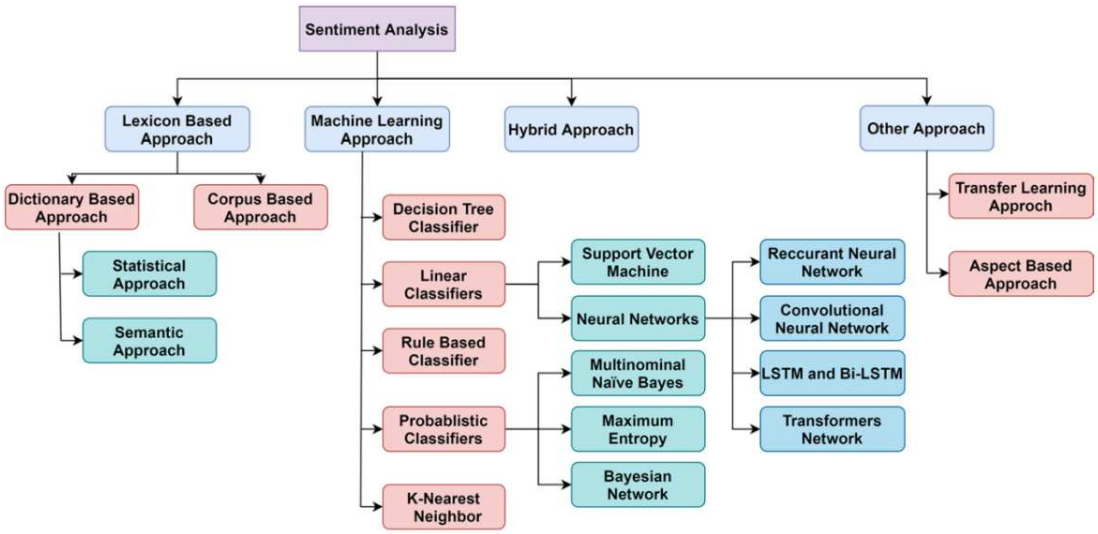



**Figure 3.1:** Different approaches to sentiment analysis. Source: Wankhade et al. (2022).

machine learning models, such as naïve bayes or SVM, depend on manually engineered features, recent developments in deep learning allow these representations to be learned directly from data. Techniques such as BoW, TF-IDF, and Word2Vec have historically played a central role in enabling such models. In the literature, these techniques have proven especially useful in early sentiment analysis applications across news, product reviews, and social media (Qader et al., 2019; Aizawa, 2003; Goldberg and Levy, 2014).

Hybrid approaches aim to combine the semantic richness of lexicon-based methods with the learning capabilities of machine learning models. Lexicons can be used to generate features, guide labeling, or filter noisy inputs, thus improving both model interpretability and robustness. Notable work in this space includes that of Prabowo and Thelwall (2009), who developed a multi-stage framework integrating rule-based filtering and probabilistic classification to improve sentiment detection in political discourse.

Over time, sentiment extraction has evolved from symbolic and statistical approaches to neural and context-aware architectures. For instance, Nasukawa and Yi (2003) demonstrated one of the earliest integrated systems combining syntactic

parsing with sentiment lexicons, setting a precedent for future hybrid and grammar-sensitive models. While their work predates the era of deep learning, the emphasis on semantic structure remains influential.

More recently, emotion classification has emerged as a nuanced extension of polarity-based sentiment analysis. Grounded in Ekman (1992)'s theory of basic emotions, this direction seeks to identify discrete emotional states, such as joy, anger, or fear, based on linguistic patterns. This granularity has proven valuable in applications where simple polarity labels are insufficient, particularly in the analysis of social media content or political communication.

In the following sections, we will discuss and analyze the main approaches to sentiment analysis, namely lexicon-based, machine learning-based, hybrid, and deep learning methods[2]. Particular attention will be given to their use in sentiment analysis within financial and social media contexts.

### 3.3.1 Lexicon-Based Methods

Lexicon-based sentiment analysis methods operate by leveraging sentiment dictionaries, predefined lists of words annotated with polarity scores, to infer the sentiment of a given text. These methods function by matching the words in the input to entries in the lexicon and aggregating their associated sentiment values. Although this approach is generally straightforward and highly interpretable, it often fails to account for contextual nuances such as negation, domain-specific meanings, and syntactic structures (Zhang and Liu, 2017).

The origins of lexicon-based sentiment analysis can be traced back to early systems like the General Inquirer (Stone et al., 1966), which systematically mapped textual content to psychological and thematic categories using curated word lists.

[2]Although deep learning is technically a subfield of machine learning (Wankhade et al., 2022), it can be treated it as a distinct category due to its rapid development and growing relevance in both academic research and practical applications.

This rule-based framework demonstrated that meaningful affective information could be extracted even in the absence of statistical modeling, thus laying a foundational methodology for sentiment research (Stone et al., 1966).

Lexicon-based approaches gained particular traction in financial research. One of the most influential studies in this context is that of Tetlock (2007), who applied the Harvard IV-4 dictionary via the General Inquirer software to quantify the pessimistic tone of financial news articles. He found that elevated negativity scores were predictive of lower subsequent market returns. Building on this, Tetlock et al. (2008) further established the value of dictionary-based sentiment indicators in forecasting firm-level outcomes. These early applications sparked a wave of interest in adapting lexicon-driven techniques for financial text analysis.

However, important limitations of general-purpose sentiment dictionaries have emerged, especially when applied to specialized domains like finance. Loughran and McDonald (2011) argued that the Harvard IV-4 dictionary was poorly suited for financial discourse, where terms like liability or capital, often labeled as negative, are in fact neutral or positive. In response, they developed a domain-specific financial sentiment lexicon tailored to corporate filings and earnings announcements, achieving significantly improved classification accuracy.

Several works have since reinforced the relevance of lexicon-based techniques in financial contexts. For example, Das and Chen (2007) employed sentiment word lists to classify investor forum posts as bullish, bearish or neutral[3], showing that even simple rule-based approaches could uncover predictive signals in unstructured textual data. Similarly, Li (2006) demonstrated how the frequency of uncertainty-related words in corporate reports could serve as a proxy for perceived risk and a predictor of future stock performance. Price et al. (2012) extended this line of research to the

[3]Messages are classified by their algorithms into one of three types: bullish (optimistic), bearish (pessimistic), and neutral (comprising either spam or messages that are neither bullish nor bearish) (Das and Chen, 2007).

analysis of earnings calls, applying frequency-based methods to identify tone shifts in executive speech and linking these to post-call market reactions.

Despite their widespread use, lexicon-based methods are not without shortcomings. Their reliance on static word lists limits their capacity to capture subtle sentiment cues in informal, ambiguous, or context-sensitive language, issues particularly salient in social media data. They are generally ill-equipped to handle sarcasm, polysemy, and negation structures, which can lead to sentiment misclassification (Zhang and Liu, 2017).

Moreover, the development and maintenance of high-quality lexicons can be labor-intensive and require domain expertise. Efforts to improve upon traditional lexicon-based frameworks have included more nuanced semantic modeling. For instance, the MSA-COSR system incorporates the HowNet lexicon, which is structured around semantic units known as sememes, enabling the calculation of similarity between input terms and benchmark sentiment concepts. This design introduces an element of semantic reasoning into sentiment classification, enhancing both accuracy and interpretability (Xianghua et al., 2013).

A particularly notable contribution in this domain is the work by Nasukawa and Yi (2003), who proposed a rule-based system that not only identified sentiment-laden expressions but also linked them to specific subject entities via syntactic parsing. Their model was capable of determining localized polarity within a document, offering greater precision than document-level classification. However, the system's dependency on manually constructed lexicons and handcrafted rules limited its scalability and adaptability to complex or noisy linguistic input.

In summary, lexicon-based sentiment analysis has played a formative role in the development of the field, especially in structured domains such as finance. Its strengths lie in interpretability, transparency, and low data requirements. However, its limitations in handling context, ambiguity, and dynamic language use have led to a gradual shift toward data-driven models (Zhang and Liu, 2017). As such, this

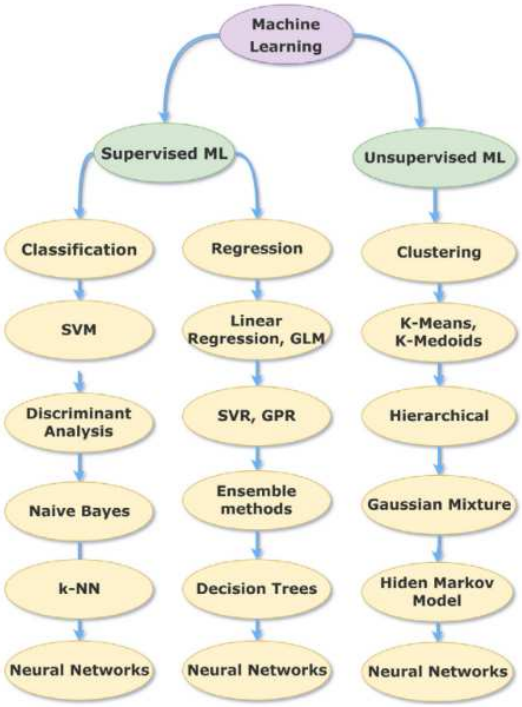



**Figure 3.2:** Taxonomy of machine learning algorithms relevant to social media sentiment analysis. Source: Balaji et al. (2021).

thesis does not adopt static lexicon-based methods for sentiment extraction tasks.

### 3.3.2 Machine Learning-Based Methods

Machine learning models offer a flexible and scalable framework for sentiment classification by learning patterns directly from data. These methods have been widely adopted in the literature, ranging from simple probabilistic classifiers to ensemble-based algorithms and neural networks, often relying on feature representations such as n-grams[4], TF-IDF vectors, or word embeddings (James et al., 2013).

Figure 3.2 illustrates a taxonomy of machine learning algorithms, adapted from Balaji et al. (2021). The diagram distinguishes between supervised and unsupervised learning approaches. Supervised machine learning is divided into classification and regression tasks, which include methods such as SVM, naïve bayes, KNN, decision trees, and ensemble methods. These are often used when labeled data is available.

[4]N-grams are contiguous sequences of n items (typically words) in text (Bergsma et al., 2009).

On the other hand, unsupervised machine learning focuses on clustering techniques like K-Means, hierarchical clustering, and Gaussian mixture models, which are used to identify hidden patterns in unlabeled data. The figure also highlights that neural networks can be applied across both supervised and unsupervised settings, bridging traditional machine learning with more complex deep learning architectures. This classification provides a structured overview of the algorithmic options available for analyzing sentiment in large-scale, unstructured data from platforms like Twitter (James et al., 2013; Balaji et al., 2021).

Early applications of machine learning in sentiment analysis typically employed supervised algorithms like naïve bayes, SVM, logistic regression, and random forest. These models demonstrated strong baseline performance across multiple domains due to their balance between interpretability and effectiveness (James et al., 2013; Balaji et al., 2021). Sailunaz and Alhajj (2019), for instance, employed a Naïve Bayes classifier to predict both polarity and emotion in tweets. By augmenting their training set with tweet replies and refining feature selection, they achieved competitive results, showing that even traditional models can perform well with proper data preparation.

More recent approaches have explored the integration of dense representations to improve classifier input. Pagolu et al. (2016) compared N-gram-based features with Word2Vec embeddings in SVM and Random Forest models for Twitter sentiment classification, concluding that distributed embeddings improved performance. This aligns with current best practices, where embeddings like Word2Vec or BERT are frequently adopted to replace sparse input formats. Efforts have also been made to improve the efficiency of machine learning models in large-scale settings. Joulin et al. (2016) introduced fastText, a shallow neural network architecture combining bag-of-ngrams with hierarchical softmax. Despite its simplicity, fastText achieved accuracy comparable to deeper models, making it an effective solution in contexts with computational constraints. Innovative alternatives to sequential learning have

also emerged. Yao et al. (2019) proposed the Text GCN model, which constructs a graph structure of words and documents and applies Graph Convolutional Networks for classification. This approach bypasses the need for external embeddings and sequence modeling by learning structural relationships directly from the text graph.

Overall, machine learning-based sentiment analysis has evolved from frequency-driven models to more complex architectures that incorporate semantic and structural information. In this thesis, we will use machine learning techniques, which remain relevant as baselines and continue to play a vital role in applications where interpretability, computational efficiency, or domain-specific adaptation are key, particularly in financial sentiment research (James et al., 2013).

### 3.3.3 Hybrid Approaches

Hybrid approaches to sentiment analysis aim to integrate the interpretability and domain knowledge of lexicon-based methods with the adaptability and predictive accuracy of machine learning models. By combining these complementary strengths, hybrid frameworks can offer improved robustness, particularly in contexts characterized by data sparsity, linguistic ambiguity, or domain specificity. For instance, sentiment lexicons are frequently used to generate additional features or pre-label training data, thereby supporting supervised classifiers when labeled datasets are limited or incomplete (Zhang and Liu, 2017; Wankhade et al., 2022).

Early hybrid models emerged as promising solutions to overcome the limitations of using either lexicon-based or machine learning techniques in isolation. A typical design involved computing initial sentiment scores from dictionaries such as SentiWordNet or AFINN, which were then integrated into the input feature space of classifiers like Naïve Bayes, logistic regression, or SVM. Kolchyna et al. (2015), among others, demonstrated that these combinations could outperform standalone approaches on various benchmark datasets, especially when lexicons contributed complementary semantic information.

An illustrative example is the multi-stage framework proposed by Prabowo and Thelwall (2009), where an initial rule-based module classifies clearly polarized inputs, while more ambiguous instances are handled by a machine learning classifier. Their evaluation on political discussion forums highlighted how this division of labor helped mitigate the rigidity of rule-based systems and the data dependence of statistical models, resulting in improved classification performance.

Xianghua et al. (2013) further extended the hybrid paradigm through the MSA-COSR system, which combines unsupervised topic modeling using LDA with lexicon-based sentiment scoring via the HowNet dictionary. By associating discovered topics with sentiment scores, this framework enabled nuanced analysis of unstructured, informal content like tweets, without relying on labeled data. The inclusion of semantic reasoning elements through "sememes"[5] added interpretability to an otherwise statistical pipeline.

While such hybrid systems were especially prominent in the mid-2010s, their role has evolved with the rise of deep learning and contextual language models. Recent advances, particularly the introduction of transformer-based architectures such as BERT, RoBERTa, and their derivatives, have significantly diminished the reliance on handcrafted lexicons. These models inherently capture syntactic and semantic relationships in text, and can effectively deal with phenomena like sarcasm, negation, and slang, common in user-generated content, without the need for manual sentiment resources (Mishev et al., 2020).

Nevertheless, hybridization remains relevant, particularly within deep learning itself. One notable example is the Attention-based Bidirectional CNN-RNN Deep Model (ABCDM) introduced by Basiri et al. (2021). This architecture integrates various deep learning methods, including recurrent networks like Bi-LSTM and

[5]There are two basic terminologies in HowNet, which are “concept” and “sememe”. Concept is a description of the semantic of words. Every word can express one concept or a few concepts. Sememe denotes the minimum semantic unit to describe the concept (Xianghua et al., 2013).

Bi-GRU, alongside an attention mechanism to capture both long-range dependencies and localized patterns in text. By combining multiple network types, ABCDM allows the model to selectively focus on sentiment-bearing elements in the input, achieving state-of-the-art results across several sentiment benchmarks, including Twitter and product reviews.

In a broader survey, Balaji et al. (2021) discuss the growing preference for deep ensemble and hybrid neural architectures in social media analysis. These systems outperform traditional and lexicon-based models in most benchmarks, offering enhanced adaptability across domains. Given such developments, this thesis does not adopt lexicon-based hybrid models as part of its methodological framework.

### 3.3.4 Deep Learning Methods

Deep learning models are a subset of machine learning techniques, and in recent years they have gained increasing importance in the academic field due to their ability to achieve state-of-the-art results, especially in opinion mining tasks (Mishev et al., 2020). For this reason, we decided to dedicate a specific section of the literature review to this topic.

Deep learning architectures such as LSTM networks (Hochreiter and Schmidhuber, 1997) and transformer-based models provide powerful tools for modeling sequential and contextual information in text. In recent years, these architectures have become increasingly central in sentiment analysis research, especially in domains characterized by informal, noisy, or unstructured language, such as social media (Mishev et al., 2020; Balaji et al., 2021).

One of the earliest and most widely adopted deep learning approaches for sentiment classification is the LSTM model (Hochreiter and Schmidhuber, 1997), which has been extensively applied to tasks involving time-dependent textual data. In particular, attention-enhanced LSTMs have been used to selectively focus on sentiment-relevant words or phrases, improving performance in fine-grained sentiment tasks such as

aspect-based sentiment analysis (Wang et al., 2016). These models have been successfully applied to domains like financial communication and product reviews, where temporal and contextual cues are essential for accurate sentiment attribution.

Convolutional neural networks, although not sequential in nature, have also been adapted to sentiment analysis due to their ability to detect local n-gram patterns and sentiment-bearing phrases. Convolutional neural networks-based models have proven especially effective for classifying short texts such as tweets, where relevant sentiment expressions are often compact and positionally salient. Their relatively low computational cost and strong performance on benchmark datasets have made them a common baseline in comparative studies (Wang et al., 2016; Jiang, 2021).

The most transformative advancement in recent sentiment analysis research has been the adoption of transformer-based models. It is worth noting that these architectures, especially BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), have redefined state-of-the-art performance on a range of sentiment benchmarks. Their ability to dynamically adjust word representations based on context allows them to handle complex linguistic structures, including sarcasm, negation, and idiomatic expressions (Mishev et al., 2020).

Transformer models have also been adapted for fine-grained sentiment tasks. For example, the RACSA model (Liao et al., 2021) extends RoBERTa with attention mechanisms and convolutional layers to perform aspect-category sentiment analysis, enabling the model to associate distinct sentiment polarities with specific aspects within a single input. While this thesis does not focus on aspect-based classification, RACSA illustrates how transformer-based models can be tailored for more interpretable and targeted sentiment analysis.

In the financial domain, Transformer models have demonstrated superior performance over both lexicon-based and classical machine learning methods. Mishev et al. (2020) show that BERT and FinBERT significantly outperform earlier approaches in classifying sentiment in financial texts. These models capture domain-specific

nuances that traditional methods often miss, making them especially useful for analyzing sentiment in corporate news, earnings calls, or tweets by influential figures.

Similarly, Singh et al. (2021) apply BERT to classify COVID-19-related tweets by emotional category, highlighting its effectiveness in real-time, emotionally charged discourse. Jiang (2021) reviews the broader use of deep learning models, LSTM and Transformer, in financial forecasting and sentiment-aware market prediction, showing that these architectures are well suited to integrate textual and numeric data sources. Alternative approaches such as Text GCN (Yao et al., 2019) offer graph-based representations for sentiment classification, moving beyond sequential modeling by structuring words and documents in a global graph. This method demonstrates that deep learning for sentiment can extend beyond RNNs and Transformers, particularly when document structure or inter-text relationships are important.

Finally, the importance of high-quality input representations remains a central theme in all deep learning approaches. As detailed by Mikolov et al. (2013a,b), distributed word embeddings such as those generated by Word2Vec form the basis for many deep architectures.

In summary, deep learning, especially Transformer-based models, currently defines the cutting edge in sentiment analysis research. Their ability to handle linguistic complexity, contextual variation, and informal expressions makes them particularly suitable for analyzing sentiment in tweets, especially when the source is a high-profile individual like Elon Musk (Mishev et al., 2020).

Given the limitations of lexicon-based and hybrid approaches in handling informal, high-context language such as tweets, this thesis focuses exclusively on machine learning and deep learning methods. Transformer-based models like RoBERTa are particularly well suited for capturing the semantic and contextual richness of Elon Musk's Twitter activity, offering a more adaptable and accurate framework for sentiment analysis in financial contexts.

## 3.4 Research Gap and Contribution of the Thesis

While sentiment analysis in finance has seen significant advances, several key gaps remain unaddressed in the current literature. This section outlines how the present thesis aims to bridge those gaps through novel data sources, methodological rigor, and targeted analysis.

Baker and Wurgler (2007) influential framework for measuring investor sentiment relies on macro-level financial indicators such as IPO volume, trading turnover, and dividend premiums. While effective in capturing broad market sentiment, their approach does not incorporate unstructured and high-frequency data sources like social media. This thesis addresses that limitation by leveraging Elon Musk's Twitter activity as a novel, bottom-up proxy for investor sentiment. Unlike traditional sentiment indices, tweets offer real-time, personalized insights into how a single influential voice may impact financial markets. By processing this unstructured content using advanced natural language models, the study extends the Baker and Wurgler (2007) framework into the domain of micro-level, behaviorally driven sentiment analysis.

Although previous research has demonstrated the effectiveness of deep learning techniques in financial prediction tasks, the specific influence of sentiment expressed by high-profile individuals remains underexplored. Jiang (2021) highlights the need to further investigate how sentiment from identifiable and impactful sources affects market behavior. Responding to this gap, this thesis focuses specifically on Elon Musk's tweets and their relationship with Tesla's stock performance.

Another gap lies in the fragmentation of methodological choices across sentiment analysis studies. Many works adopt individual machine learning models without offering a structured comparison across algorithm families or learning paradigms. By situating its methods within a broader taxonomy of supervised, unsupervised, and regression models, as proposed by Balaji et al. (2021), this thesis contributes a more

systematic understanding of the strengths and trade-offs associated with each class of algorithm in the context of financial sentiment analysis.

Furthermore, while early rule-based sentiment extraction techniques, such as the system proposed by Nasukawa and Yi (2003), laid important groundwork for linking sentiment expressions to specific subjects, these methods suffer from limited adaptability to the informal, dynamic language of platforms like Twitter. Their reliance on handcrafted lexicons and shallow syntactic parsing constrained scalability and made them ill-suited for detecting linguistic nuances such as sarcasm, slang, or novel phraseology. This study overcomes such limitations by implementing modern machine learning and deep learning architectures that generalize more effectively across informal and evolving text sources, while retaining high levels of predictive accuracy (Mishev et al., 2020).

In sum, this thesis contributes to the literature by integrating domain-specific sentiment (Elon Musk's tweets), advanced text modeling techniques (e.g., RoBERTa), and financial forecasting tools within a unified analytical framework. It extends existing research, offering new insight into how individual-level communication can influence asset prices in real time.

## 3.5 Limitations of current literature

Despite notable advances in sentiment analysis, several methodological and contextual challenges persist. This section reviews the key limitations that hinder current models from achieving robust and domain-adaptable performance (Zhang and Liu, 2017; Wankhade et al., 2022).

Domain dependence represents another critical barrier: models trained on one type of text, such as product reviews or general news, tend to perform poorly when applied to specialized domains like finance. Loughran and McDonald (2011) provide a salient example of this issue by showing how standard sentiment lexicons misclassify

financially neutral terms like liability or capital as negative. Their work highlights the need for domain-specific tools that can accommodate specialized vocabulary and discourse structures.

The limitations of lexicon-based methods, although foundational to the field, are well documented. Tools like the General Inquirer (Stone et al., 1966), while interpretable and rule-based, rely on static word lists and fail to consider context, negation, or syntactic variation. These approaches struggle to detect sentiment shifts in more complex linguistic forms, such as rhetorical questions or multi-topic statements, and are particularly ineffective when faced with sarcastic or nuanced language. Hussein (2018) further underscores the impact of linguistic complexity, noting that irony, multilingual variation, and domain sensitivity continue to hinder classification accuracy. Her comprehensive survey suggests that the next generation of sentiment systems must move beyond surface-level text analysis and adopt context-aware models capable of managing subtle emotional and pragmatic cues.

Traditional machine learning approaches often focus on local sentence structures and treat sentences as isolated units. This limits their ability to capture global semantic dependencies across documents. To address this, Yao et al. (2019) propose a graph-based model that represents an important step toward more holistic text understanding, though it still depends on graph construction and may struggle with scalability in real-time applications.

Another challenge lies in data availability and quality, particularly for low-resource languages or domains lacking labeled datasets. Without sufficient annotated data, even advanced models risk overfitting or misinterpreting sentiment (James et al., 2013). Additionally, opinion spam and manipulated sentiment, especially prevalent in commercial and financial domains, further complicate classification tasks by introducing noise and intentional bias (Zhang and Liu, 2017).

In sum, although the field has progressed considerably, from lexicon-based and shallow classifiers to deep learning and graph-based methods, significant limitations

remain. These include the inability to fully interpret figurative language, adapt across domains, process multilingual input, and function effectively in real-time, high-volume settings. Addressing these gaps requires continued innovation in model architecture, training data diversity, and evaluation methods tailored to dynamic and high-stakes contexts such as financial markets and social media discourse (Hussein, 2018; Mishev et al., 2020).

# Chapter 4

# Data Collection and Pre-processing

## 4.1 Data Sources

### 4.1.1 Elon Musk's Tweets

As outlined in Section 3.2, Twitter is one of the primary sources of sentiment data. This is not only due to its widespread global use, but also because of its well-documented communication patterns and community dynamics. For this reason, it was selected as a main data source for this thesis (Bollen et al., 2011a).

Java et al. (2007) analyzed over a million tweets to understand user behavior on the platform. Their findings highlighted Twitter's support for a variety of communication purposes, from personal commentary to public information sharing. This early research underscores Twitter's value as a rich and multifaceted dataset, particularly relevant for studies on sentiment analysis and market impact.

The main dataset used for the sentiment analysis in this thesis is a freely available dataset from the Kaggle platform[1], which includes tweets from Elon Musk spanning the period from June 2010 to April 2025. While the dataset has not been indepen-

[1]Kaggle is an online platform for data science and machine learning competitions, offering datasets, notebooks, and tools that enable users to collaborate and develop models in a shared environment.

dently verified, it has been compiled and published by the authors of a journalistic investigation (The Insider, 2025), and appears to be consistent, well-documented, and suitable for analytical purposes.

Since the aim of this thesis is to investigate the impact of Elon Musk's tweets on Tesla stock prices, while controlling for confounding factors such as the COVID-19 pandemic and the protectionist policies from the U.S.-China trade war during Trump's presidency, the analysis is restricted to the period from January 1, 2023, to February 28, 2025. This time window spans two years and two months.

Within this timeframe, the dataset includes 31,513 tweets authored by Elon Musk. These are categorized as general tweets, meaning they cover a wide range of topics beyond Tesla and its financial performance. The content includes discussions about U.S. politics, the presidency, geopolitical issues in Europe and globally, and often also less structured or off-topic commentary, typical of Musk's communication style on the platform.

Before conducting the sentiment analysis, an initial preprocessing and data cleaning phase was performed on the tweet dataset, as detailed in Chapter 4.2. This process included the removal of duplicate entries and the filtering of non-textual elements such as URLs, emojis, and special characters. Additional steps involved standardizing the text format, such as converting all characters to lowercase, and removing stopwords to ensure the data were suitable for NLP. The resulting cleaned and preprocessed tweet corpus served as the textual foundation for sentiment extraction and subsequent analysis (Wankhade et al., 2022).

### 4.1.2 Tesla Stock Price Data

Historical stock price data for Tesla Inc. (TSLA) were retrieved using the Yahoo Finance API[2]. This includes historical stock prices, real-time quotes, trading volumes,

[2]The Yahoo Finance API is a service that provides access to financial market data, including stock quotes, historical prices, and company information, commonly used for financial analysis and

and company-level financial information. For the purposes of this thesis, the dataset was constructed to cover the time period from January 1, 2023, to February 28, 2025. This time frame aligns with the selected Twitter data window and spans approximately two years and two months of daily trading activity.

The dataset includes several key financial variables: opening and closing prices, intraday highs and lows, trading volume, and adjusted closing prices (which account for dividends and stock splits). These indicators are essential for conducting time series analysis and for evaluating the potential relationship between social media sentiment and stock price fluctuations (Bollen et al., 2011b).

Before proceeding with the analysis, an initial data exploration and cleaning phase was performed. This involved checking for missing values, anomalies, and inconsistencies in the time series to ensure data quality and reliability. The cleaned and validated dataset was then used as the financial foundation for the empirical analysis presented in the following chapters.

## 4.2 Data Cleaning and Pre-processing

### 4.2.1 Text Cleaning: Handling Slang, Emojis, and Sarcasm

The textual data collected from Twitter is inherently unstructured and often noisy, which can hinder the performance of sentiment analysis models. In order to prepare the dataset for analysis, a structured data cleaning and pre-processing phase was applied, as illustrated in this section (Bollen et al., 2011b; Zhang and Liu, 2017).

The first step consisted in removing duplicate tweets based on their full text content. This reduced the dataset from 31,513 to 28,264 unique tweets during the time-period considerated, meaning that more than 10% of the original data consisted of repeated content. This step was essential to avoid data redundancy and ensure

research.

that the analysis would not be biased by replicated information.

Subsequently, a dedicated cleaning function was implemented in Python to process the tweet text. This function removed URLs, user mentions, hashtags, numbers, punctuation, emojis, "RT"[3] and "amp"[4], elements that typically do not contribute directly to sentiment analysis and could introduce noise. The text was also normalized by converting all characters to lowercase and removing any leading or trailing whitespace. These operations were carried out using regular expressions and string manipulation tools from the standard Python libraries, and applied efficiently across the entire dataset. For instance, the tweets becomes:

```
RT @Tesla:  Holiday Update rolling out now 🎅 →
holiday update rolling out now

RT @SpaceX: Watch Falcon 9 launch 54 Starlink satellites →
watch falcon launch starlink satellites
```

These examples illustrate how the cleaning function simplifies the input while retaining the core semantic content relevant for sentiment analysis.

At this stage, the focus was limited to cleaning and simplifying the raw text, without altering its semantic content. Tokenization and feature extraction (e.g., BoW, TF-IDF, Word2Vec, BERT embeddings) were not yet applied, as they are addressed in the next phase of the pipeline (Section 5.1). The cleaned text was stored in a new column of the dataset and served as the basis for all further textual analysis and model input.

While this approach helps reduce noise and improve model performance, it does not fully address the complexity of informal or creative language often found on

[3]"RT" is automatically added at the beginning of retweets in the dataset. Since it does not carry meaningful sentiment information, it was removed during the text cleaning process.

[4]The token "amp" appears as a result of HTML encoding, where the symbol "&" is represented as "&" in the raw tweet text. This artifact was removed during pre-processing using a decoding function.

social media, such as slang or sarcasm. These aspects are not explicitly treated in the cleaning phase, but are expected to be partially managed by the use of robust pre-trained transformer models, such as RoBERTa (see section 5.2.), during the modelling stage (Schouten and Frasincar, 2015; Hussein, 2018).

### 4.2.2 Financial Data Quality Checks

Before proceeding with the analysis, we carried out a series of preliminary data quality checks on the historical stock price data for Tesla Inc. (TSLA), retrieved via the Yahoo Finance API using the `quantmod` package in R. These checks were important to ensure that the dataset was clean, consistent, and ready for use in the subsequent time series analysis.

First, we verified the presence of missing values across all variables, including opening price, closing price, daily highs and lows, volume, and adjusted close. The results showed that there were no missing values in any of the columns. We then checked for duplicate entries based on the trading dates, and found that no duplicates were present, each date in the dataset was unique.

To visually assess the presence of outliers, a boxplot of Tesla's closing prices was generated. As shown in Appendix D, the chart highlights several high-value outliers, which are common in financial time series due to normal market volatility. For this reason, they were retained in the dataset.

We also checked whether all business days within the selected time range were included. By comparing the dataset's trading dates to a complete sequence of expected business days (excluding weekends and official holidays), 22 missing trading days were identified. These gaps are considered acceptable, as they likely correspond to stock market holidays or exceptional suspension days.

Lastly, we made sure that all the columns were correctly recognized as numeric time series data, ensuring compatibility with the time series analysis techniques applied later in the thesis. With these quality checks completed, the dataset was

considered clean, consistent, and suitable for the empirical analysis.

## 4.3 Exploratory Data Analysis

In this section, we present an exploratory analysis of both datasets, highlighting their main characteristics and providing initial insights into their structure and content.

### 4.3.1 Tweet Corpus Analysis

To develop a preliminary understanding of the dataset, an exploratory analysis was conducted on Elon Musk's tweets posted over the selected time period, from January 1, 2023, to February 28, 2025. As previously mentioned, the dataset used in this analysis contains only unique tweets, following the removal of duplicates, resulting in a total of 28,264 entries.

A monthly tweet frequency chart was created to identify posting trends over time. The results reveal a clear upward trajectory in Musk's activity on the platform, with notable spikes during key public moments. The lowest number of tweets posted in a single month was 581, while the highest peak occurred in November 2024, with a total of 2,054 tweets. On average, Elon Musk published approximately 1,087 tweets or retweets per month throughout the selected period.

Figure 4.1 illustrates the distribution of tweet activity across the timeline. The trend suggests a progressively increasing use of Twitter, indicating not only a sustained presence but also a growing reliance on the platform as a channel for communication and influence.

A second exploratory analysis examined the distribution of tweet lengths, measured by the number of words in the cleaned text. As illustrated in Figure 4.2, the vast majority of tweets fall within a short range, with most containing fewer than 15 words. The mean tweet length is approximately 9.2 words, and the distribution is strongly right-skewed, indicating that short tweets are significantly more common

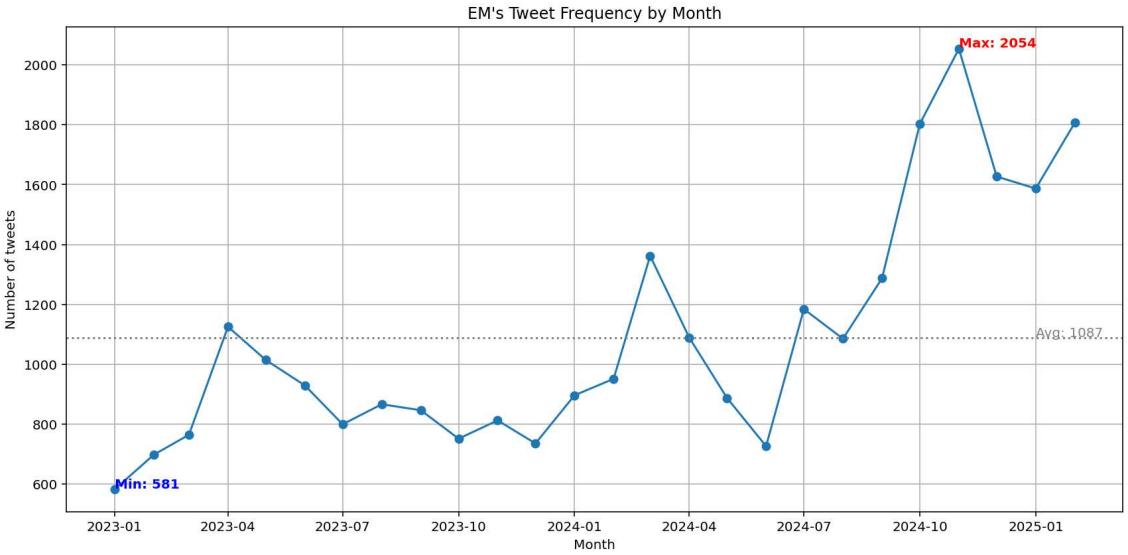



**Figure 4.1:** Elon Musk's Tweet Frequency by Month.

than longer ones. A corresponding boxplot illustrating this distribution is included in Appendix D, providing additional insight into the variability and presence of outliers in tweet length.

This brevity is consistent with the nature of the platform, which originally imposed a 140-character limit prior to 2017, later extended to 280 characters. The data suggest that Elon Musk tends to use concise language in his posts, often expressing ideas in short, impactful phrases. This characteristic may imply a preference for immediacy and clarity, and it also suggests that the words used in each tweet carry a relatively high semantic weight, a point particularly relevant for subsequent sentiment analysis.

In addition to tweet frequency and content analysis, this exploratory phase also examined the engagement metrics associated with Elon Musk's tweets. These include the total monthly counts of likes, retweets, replies, and quote tweets, as illustrated in Figure 4.3.

The graph reveals a clear upward trend in engagement, particularly during the second half of 2024. This surge could plausibly be linked to the political climate surrounding the U.S. presidential elections, although this topic lies outside the scope of the present study. Notably, the number of likes peaked in November 2024 before

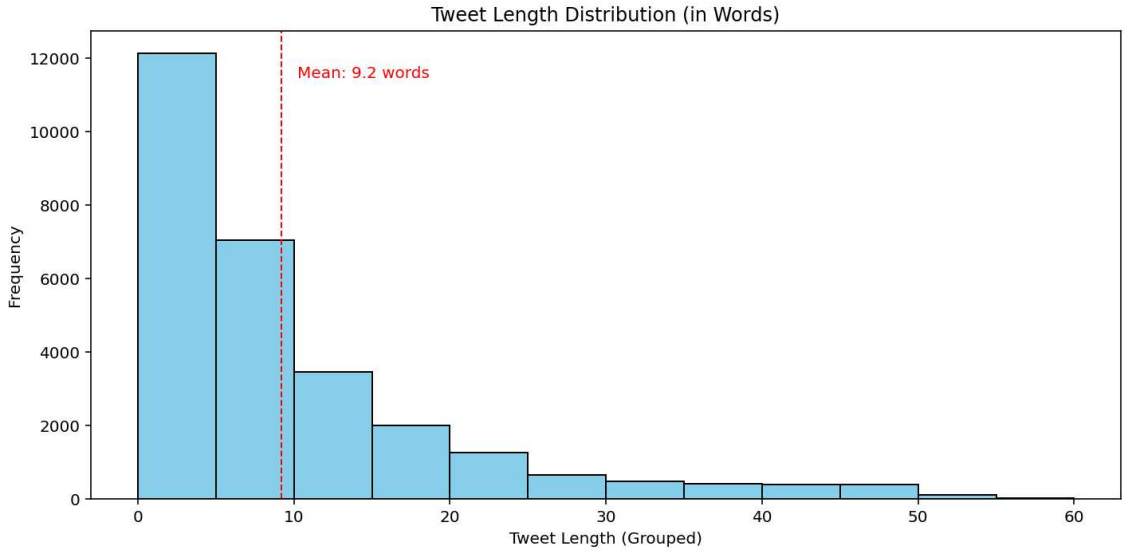



**Figure 4.2:** Tweet Length Distribution.

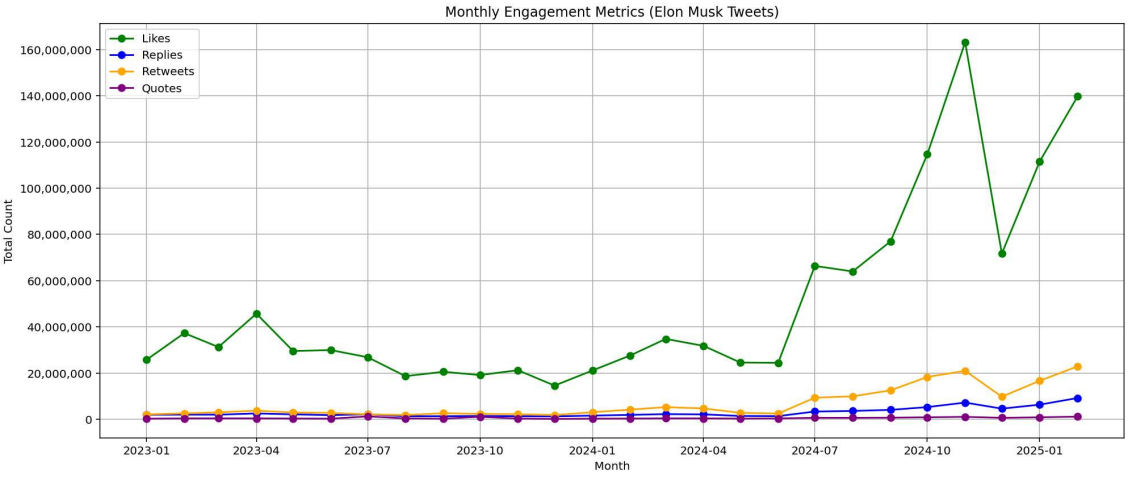



**Figure 4.3:** Elon Musk's Engagement metrics per Month.

dropping significantly in December. However, engagement appeared to recover in early 2025, with likes rising again, and all other metrics, replies, retweets, and quotes, showing a modest but steady increase beginning mid-2024.

These patterns suggest a growing responsiveness from the public to Musk's online presence, especially in periods of heightened social or political attention.

To further explore user engagement, we analyzed the number of views (impressions) received by Elon Musk's tweets. Due to the significantly higher magnitude of view counts compared to other engagement metrics (such as likes, replies, or retweets),

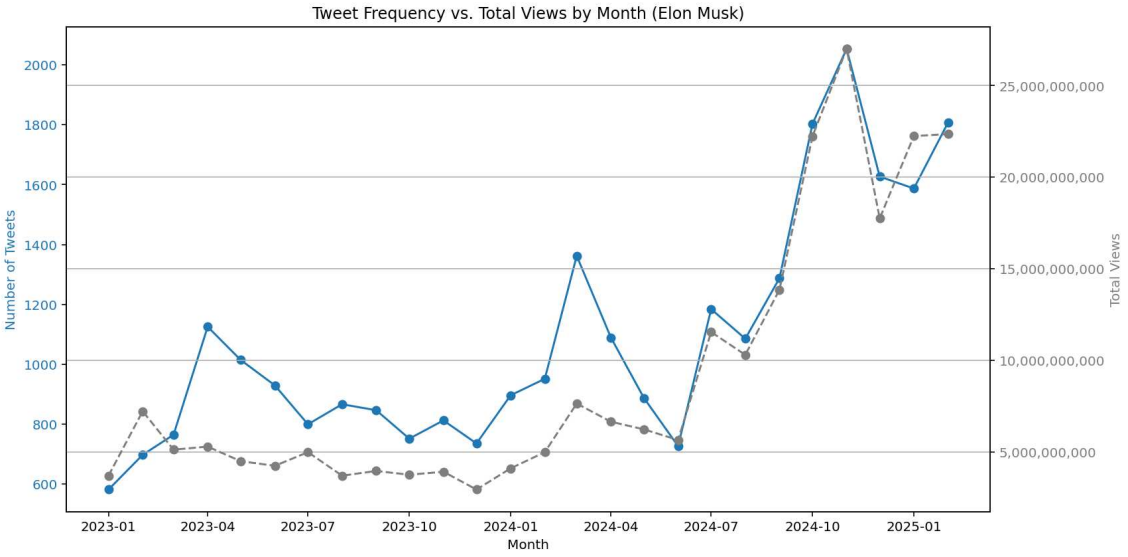



**Figure 4.4:** Tweet Frequency vs. Total Views by Month.

these were examined separately. Specifically, we created a dual-axis plot to compare the total monthly tweet counts with the corresponding monthly view counts, as shown in figure Figure 4.4. This visualization helps evaluate Musk's communicative reach and efficiency over time, and it allows for the identification of anomalies or potential correlations between tweet activity and audience exposure. It appears to be a general alignment between tweet frequency and view count, particularly during the latter half of 2024, a period that also corresponded to an increase in overall engagement. Interestingly, while both metrics peak in November 2024, the subsequent decline is sharper for views than for tweet volume, suggesting possible shifts in user attention or platform dynamics. Earlier periods in the timeline reveal greater discrepancies between the two variables, indicating that higher posting frequency did not always result in proportionally higher visibility.

For a more detailed breakdown, Appendix D includes an additional graph showing the monthly view count on its own, as well as a comparative visualization including all engagement metrics.

Then, I proceed with the creation of different boxplots of tweet engagement metrics in logarithmic scale. Figure 4.5 presents the distribution of five core engagement




**Figure 4.5:** Distribution of Tweet Engagement Metrics (Logarithmic Scale).

metrics for Elon Musk's tweets: likes, replies, retweets, quotes, and views. Given the high variance and the presence of extreme outliers, such as tweets exceeding 2 million likes or 600 million views, the data are visualized using a logarithmic transformation $log(1+x)$. This transformation allows a clearer comparison of the typical distribution for each metric while preserving visibility of the long-tail nature of the data. The boxplots reveal that likes, replies, and retweets tend to follow similar distributions, with most values concentrated in the lower range and a limited set of tweets generating disproportionately high engagement. Views show the largest spread, suggesting a broader variability in reach, likely influenced by both platform algorithms and the virality of individual tweets. For reference, the raw (non-transformed) version of these boxplots is available in Appendix D.

Moreover, a basic word frequency analysis was conducted to highlight the most commonly used terms in Elon Musk's tweets. This initial step offers a high-level overview of his communication style and provides useful insight for the upcoming sentiment modeling phase. To enhance the lexical analysis, standard English stop-words, such as "*the*", "*is*", and "*to*", were removed, enabling a clearer focus on the most semantically meaningful words. The resulting bar chart, shown in Figure 4.6, displays the top content words based on frequency. This reveals Musk's recurring

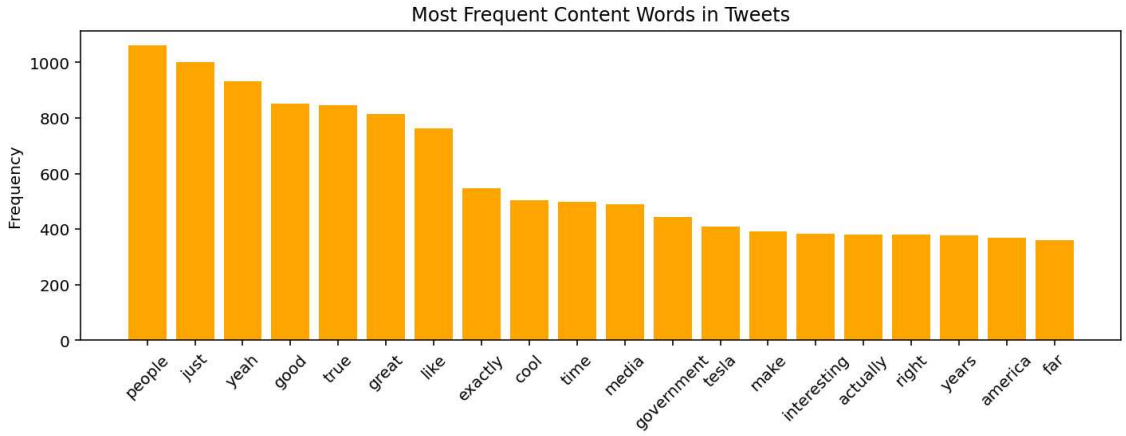



**Figure 4.6:** Most Frequent Words in EM's Tweets.

themes and preferred expressions over the observed period.

Although full tokenization and embedding techniques are introduced in Chapter 5, a preliminary tokenization was applied here exclusively for exploratory purposes. This allowed for the generation of word frequency distributions across the cleaned tweet corpus. Interestingly, the distribution shows that words like "*government*", "*Tesla*", and "*America*" appear with similar frequency, reflecting recurring macro-level topics in Musk's discourse. Meanwhile, words such as "*yeah*", "*great*", or "*cool*" are stylistically characteristic of his informal and conversational tone. While these words carry limited semantic weight, they were retained at this stage to illustrate aspects of Musk's unique language style on social media. However, for downstream tasks such as sentiment classification and model training, these stylistic markers were later excluded as stopwords, given their low discriminative power in identifying sentiment polarity.

To conclude this exploratory phase, we generated a word cloud based on the most frequent n-grams. Appendix D includes a word cloud that visually represents the most frequently used terms, as also shown in Figure 4.6. Subsequently, we created an additional word cloud by excluding shorter words with fewer than four letters. Words such as yeah, just, good, true, like, and cool, which often appear in casual online discourse but contribute little to deeper semantic analysis, were filtered out

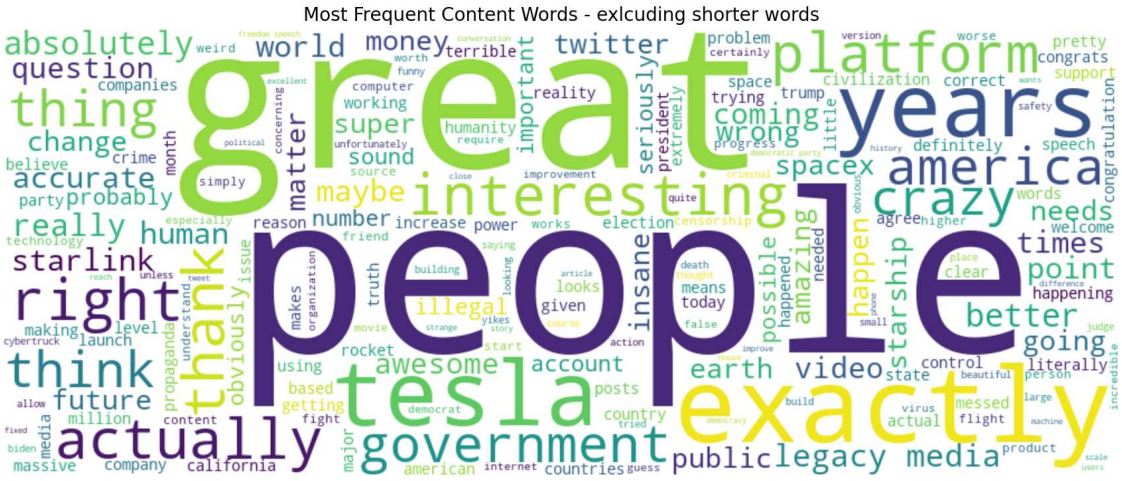

Most Frequent Content Words - exlcuding shorter words
great
people
platform
years
america
crazy
interesting
tesla
exactly
actually
government
media
legacy
public
absolutely
question
thing
change
accurate
probably
really
human
starlink
right
think
thank
obviously
future
million
massive
company
california
world
money
terrible
twitter
problem
civilization
correct
pretty
congrats
support
congratulation
trying
trump
coming
wrong
space
spacex
speech
definitely
needs
welcome
times
point
happening
better
going
literally
control
video
earth
state
messed
actual
flight
virus
product
amazing
happen
starship
insane
seriously
president
important
matter
super
working
computer
humanity
sound
maybe
reason
number
increase
power
election
censorship
agree
higher
words
looks
given
illegal
means
today
possible
happened
needed
clear
place
awesome
account
using
rocket
based
getting
fight
posts
country
democrat
american
internet
countries
major
reality
worth
funny
unfortunately
source
improvement
quite
criminal
democratic party
progress
little
safety
history
beautiful
person
large
machine
build
tried
democracy
weird
freedom speech
excellent
concerning
political
simply
close
believe
party
month
crime
technology
especially
friend
saying
looking
article
building
truth
makes
organization
understand
issue
propaganda
unless
tweet
reach
making
cybertruck
launch
level
allow
biden
media
fixed
content
based
using
rocket
awesome
account
earth
video
state
control
literally
person
beautiful
large
machine
build
virus
flight
messed
actual
product
scale
users
incredible
judge
difference
happening
welcome
congratulation
wants
safety
speech
definitely
history
democratic party
progress
little
criminal
obvious
quite
improvement
close
source
unfortunately
computer
working
super
humanity
require
sound
maybe
reason
number
increase
power
works
election
censorship
agree
higher
words
needs
welcome
times
point
happening
better
going
literally
control
video
earth
state
messed
actual
flight
virus
product
amazing
happen
starship
insane
seriously
president
important
matter




**Figure 4.8:** Bi-gram and Tri-gram of EM's most frequent words.

including the minimum, maximum, mean, median, standard deviation, and the first and third quartiles for each variable. As mentioned in Section 4.2, data cleaning and preprocessing were performed prior to this step. The dataset was filtered to include only the relevant time period, spanning from January 1, 2023, to February 28, 2025.

By analyzing the summary statistics in Table 4.1, it is possible to gain a clearer understanding of the structure and behavior of the key variables in the dataset. The Open Price reflects the value at which the stock begins trading when the market opens, providing insight into initial market sentiment and investor expectations. The High Price corresponds to the highest trading value reached during the day, while the Low Price captures the lowest value recorded in the same time frame. The Close Price represents the final price at which the stock is traded before market closure and is widely used in financial analysis. In contrast, the Adjusted Close Price accounts for corporate actions such as dividends or stock splits, offering a more accurate representation of the stock's value over time (Hamilton, 2020).

In this specific dataset, the close price and the adjusted close price are identical across all observations, indicating that no corporate actions requiring adjustment occurred during the considered time period. As a result, the two variables can be used interchangeably in this analysis.

These statistics allow for some preliminary insights. The mean value across the different price variables is approximately 234, but the substantial gap between the minimum and maximum values indicates a high level of volatility throughout the observed period. The average daily trading volume stands at around 113.3 million shares, with a standard deviation exceeding 40 million, further suggesting significant variability in trading activity and investor behavior.

Moreover, the fact that the mean exceeds the median for all variables suggests a slight positive skewness in their distributions, an expected characteristic in financial time series, often driven by outliers or abrupt price increases (Hamilton, 2020).

**Table 4.1:** Descriptive Statistics of Tesla Stock Data (2023–2025).

| Variable | Min | 1st Quartile | Median | Mean | 3rd Quartile | Max | Std Dev |
|---|---|---|---|---|---|---|---|
| TSLA.Open | 103.0 | 184.9 | 220.9 | 234.8 | 256.0 | 475.9 | 69.99 |
| TSLA.High | 111.8 | 187.7 | 224.7 | 239.8 | 260.9 | 488.5 | 71.55 |
| TSLA.Low | 101.8 | 181.8 | 216.0 | 229.7 | 251.4 | 457.5 | 67.72 |
| TSLA.Close | 108.1 | 184.8 | 220.2 | 234.8 | 256.8 | 479.9 | 69.50 |
| TSLA.Adjusted | 108.1 | 184.8 | 220.2 | 234.8 | 256.8 | 479.9 | 69.50 |
| TSLA.Volume | 37.2M | 83.4M | 107.8M | 113.3M | 133.9M | 306.6M | 40.6M |

Figure 4.9 illustrates the time series of Tesla's daily closing prices over the observed period. The blue line represents the actual closing values. As shown in the graph, the stock experienced several distinct phases of volatility. In particular, there was a marked upward trend during the second half of 2024, followed by a sharp correction in early 2025. This pattern reflects typical market dynamics, where periods of sustained growth are often accompanied by corrections triggered by shifts in investor sentiment or macroeconomic events.

To complement the closing price visualization, an additional Open-High-Low-Close (OHLC) chart is provided in Appendix D. This chart offers a more detailed view of intraday price fluctuations, as well as trading volume, enabling a more comprehensive understanding of market behavior throughout the period under study.

Although the time series of Tesla's closing price captures periods of both growth and decline, it does not fully represent the underlying dynamics of the dataset, as it exhibits a general upward trend over time. This trend, while expected, limits our ability to analyze certain statistical properties of the series, such as volatility clustering or stationarity. For this reason, and given that time series data often exhibit features such as seasonality and autocorrelation, we applied a logarithmic transformation to the closing prices. This transformation helps to stabilize the variance and better highlight the structural characteristics of the data (Tsay, 2005; Hamilton, 2020).




**Figure 4.9:** TSLA Closing Prices (2023-2025).

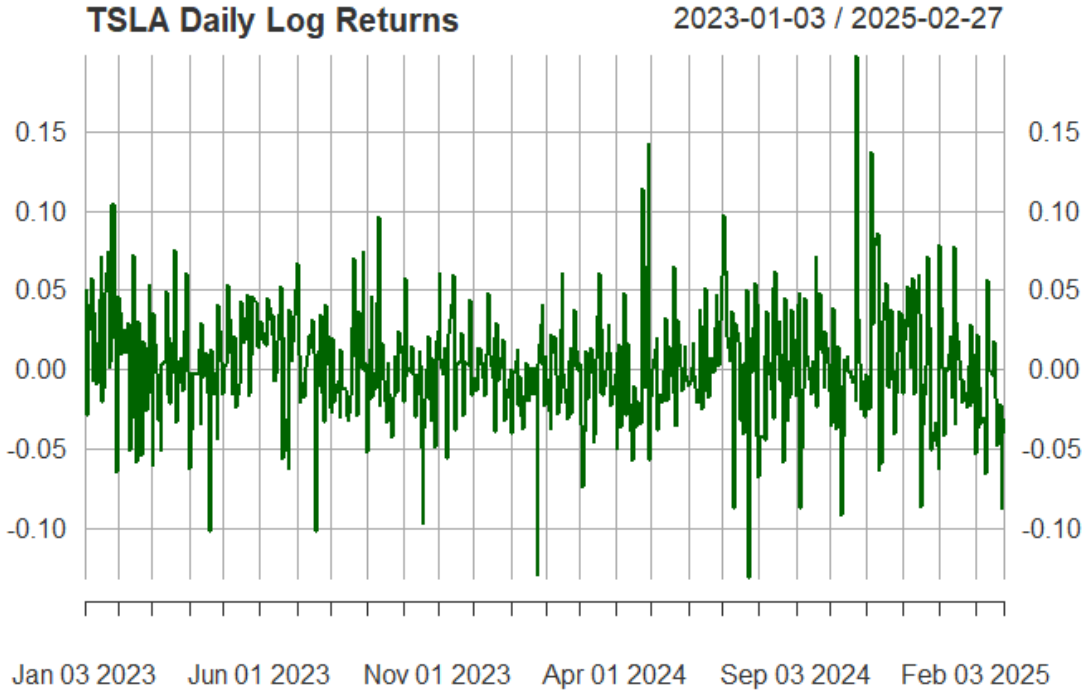



**Figure 4.10:** TSLA Daily Log-Returns (2023-2025).

Figure 4.10 displays the time series of Tesla's daily log returns from January 3, 2023, to February 27, 2025[5]. The series appears relatively stationary, yet it shows clear periods of increased volatility, particularly during the latter part of 2024 and the beginning of 2025. As discussed in Section 4.3.1, this period coincides with a noticeable rise in the visibility of Elon Musk's tweets, as well as a significant increase in his posting activity. These fluctuations suggest the presence of volatility clustering, a common phenomenon in financial time series where high-volatility periods tend to be followed by similar ones.

To further explore the characteristics of the return distribution, we constructed a histogram of the daily log returns. Figure 4.11 presents this histogram. The distribution is approximately symmetric and centered around zero, but it exhibits heavier tails than the normal distribution. This indicates that extreme returns, both positive and negative, occur more frequently than would be expected under a Gaussian assumption, a finding that aligns with typical features of financial return series.

To further assess the distribution of Tesla's daily log returns, skewness and kurtosis were calculated. The skewness value of approximately 0.37 indicates a slight positive asymmetry, suggesting that positive returns are slightly more frequent or extreme than negative ones. The kurtosis is around 5.60, which is significantly higher than the kurtosis of a normal distribution (equal to 3). This implies that the distribution has heavier tails and a sharper peak. In practical terms, this suggests a greater probability of extreme daily returns than would be expected under a normal distribution.

As an additional check, a quantile-quantile (QQ) plot was used to compare the empirical distribution of returns to a normal distribution. The slight tail deviations, observed in Figure 4.12, visually confirm that the empirical data exhibits heavier

[5]The dates do not exactly correspond to January 1, 2023, and February 28, 2025, as they refer to trading days only, excluding weekends and public holidays.

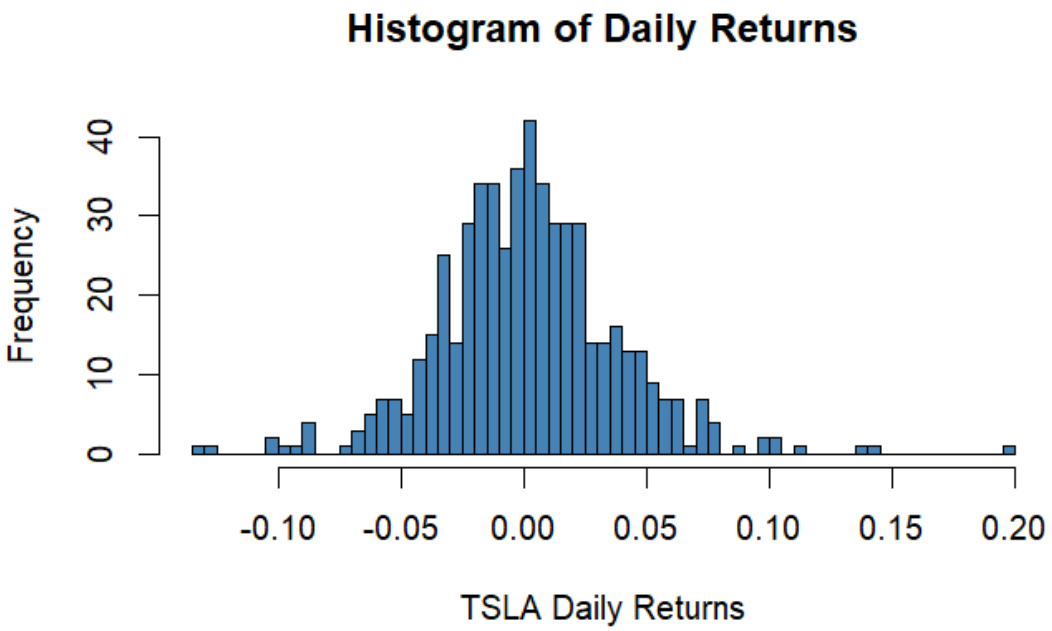



**Figure 4.11:** Distribution of TSLA Daily Returns (2023-2025).

tails, in line with the elevated kurtosis value (5.60). A full density comparison is provided in Appendix D for completeness.

The analysis then turned to the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to investigate potential temporal dependencies in the return series. Figure 4.13 shows the ACF of Tesla's daily log returns (as expected, there is a large spike at lag 0, which reflects the perfect correlation of the series with itself and is therefore not considered meaningful for interpretation). All subsequent autocorrelation coefficients lie within the significance bounds, suggesting no statistically significant autocorrelation at higher lags. This indicates that the returns are not strongly correlated over time, which is consistent with the efficient market hypothesis.

Figure 4.14 presents the Partial Autocorrelation Function (PACF) of Tesla's daily log returns. The plot shows no statistically significant spikes, indicating the absence of a strong partial autocorrelation structure in the return series. This finding supports the hypothesis of weak-form market efficiency, suggesting that past returns

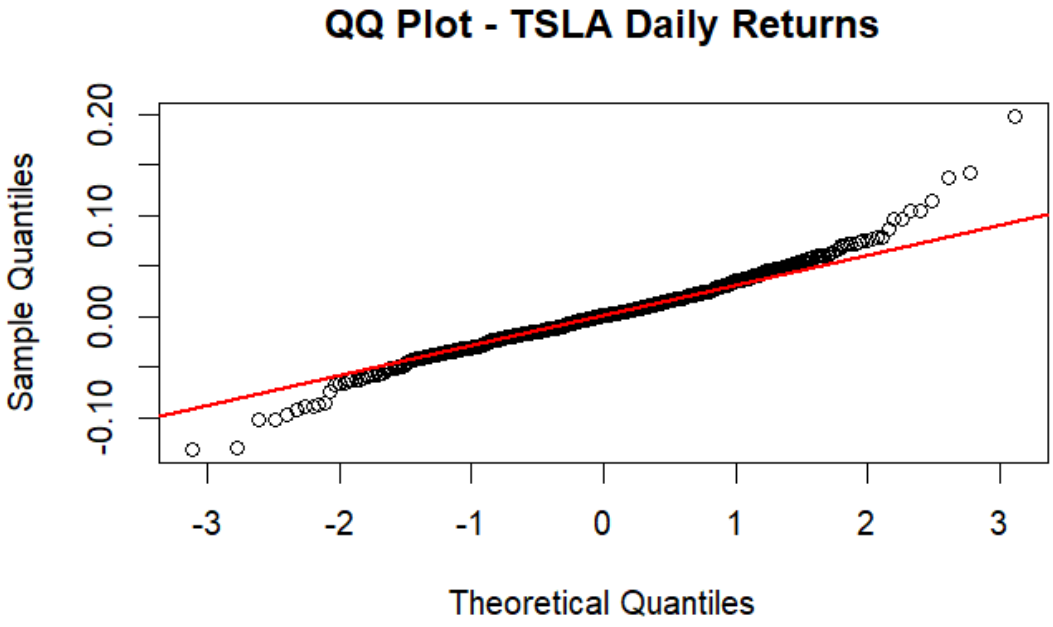



**Figure 4.12:** QQ Plot of TSLA Daily Returns (2023-2025).

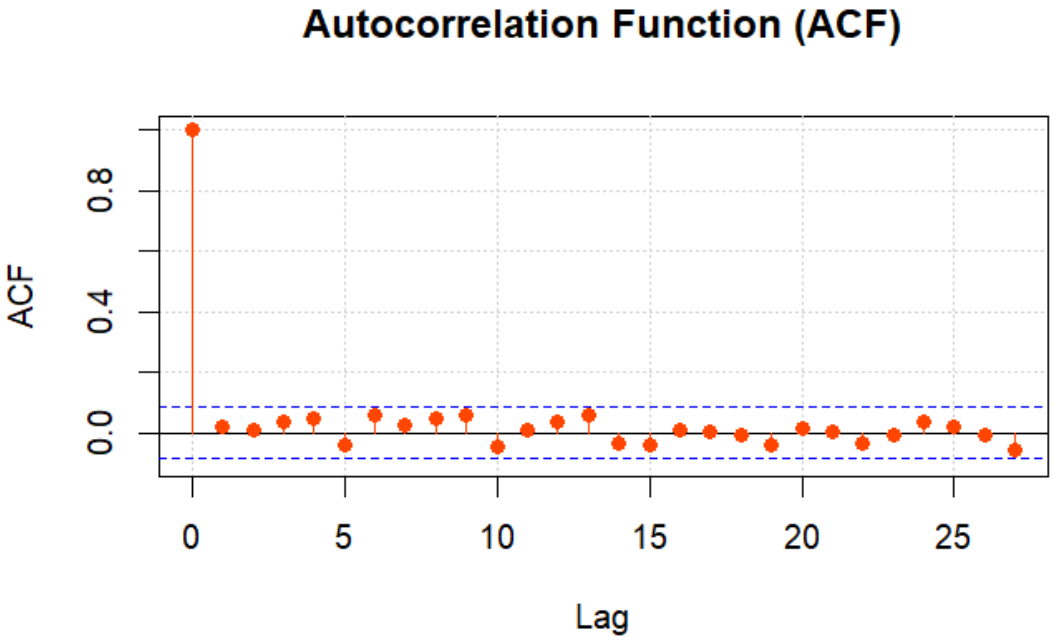



**Figure 4.13:** TSLA's ACF (2023-2025).

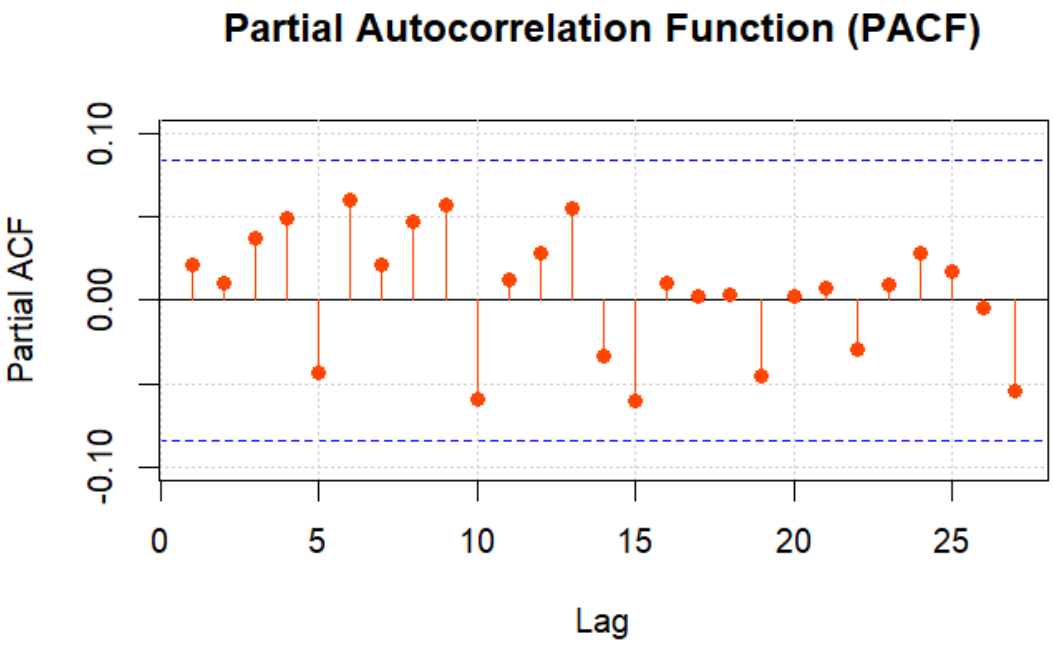



**Figure 4.14:** TSLA's PACF (2023-2025).

do not carry predictive power for future price movements and that returns follow a random walk process. Moreover, the lack of significant autocorrelation in both the ACF and PACF plots implies that the return series may exhibit a Markovian structure, in which the future value depends solely on the present state, rather than the full historical path[6]. This behavior aligns with the theoretical foundations of random walk models and further reinforces the notion that Tesla's daily log returns are largely unpredictable based on their own past values.

The analysis then focused on the volatility dynamics of Tesla's daily log returns, as initially highlighted in the descriptive summary presented in Table 4.1. Figure 4.15 illustrates the 20-day rolling volatility of the return series over the observed period[7].

[6]In a Markov process, the probability distribution of the next state depends only on the current state (Tsay, 2005; Hamilton, 2020). It is commonly expressed as:

$$P(X_{t+1} \mid X_t, X_{t-1}, \ldots, X_0) = P(X_{t+1} \mid X_t)$$

.

[7]The 20-day rolling volatility is computed as the standard deviation of daily returns over a moving 20-day window. It serves as a dynamic measure of short-term market risk, capturing

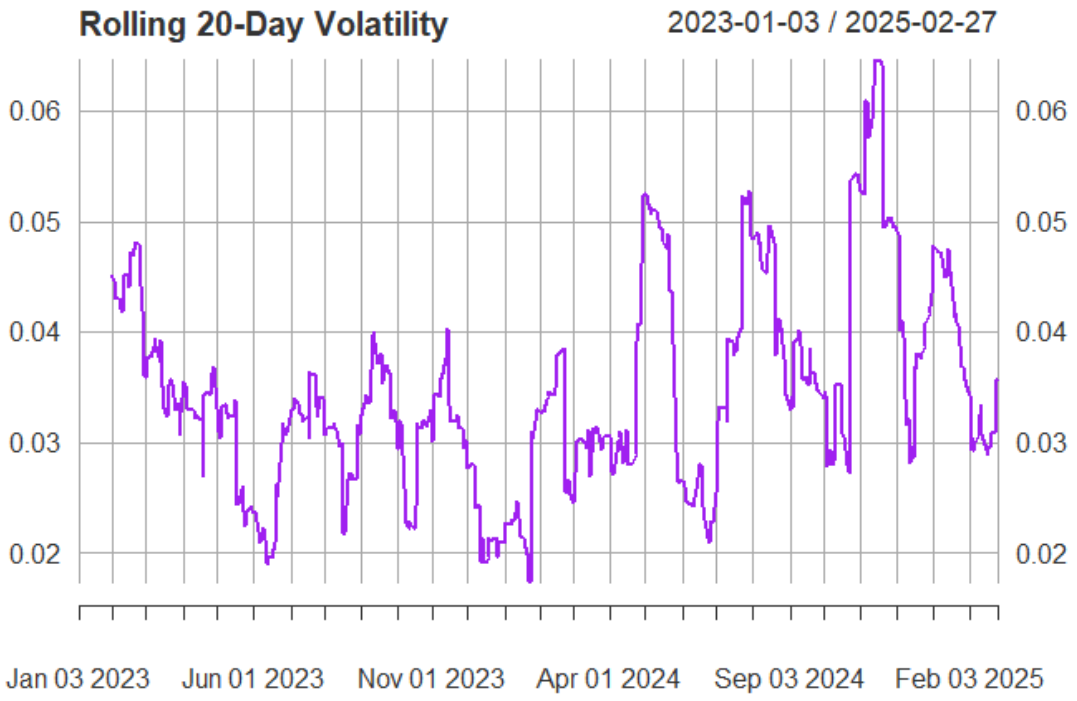



**Figure 4.15:** TSLA Rolling 20-Day Volatility (2023-2025).

This measure acts as a proxy for short-term market uncertainty and provides insights into the changing risk profile of the stock across time (Andreou and Ghysels, 2002b). As the figure shows, volatility is far from constant, it fluctuates substantially throughout the sample. In particular, pronounced spikes in volatility are observed during the second half of 2024 and the beginning of 2025, suggesting increased market turbulence during those periods. This time-varying volatility is a well-documented characteristic of financial markets and is consistent with the phenomenon of volatility clustering, where periods of high volatility are likely to be followed by further high volatility, and periods of calm tend to persist.

Figure 4.16 presents the distribution of Tesla's daily log returns grouped by calendar month, providing a visual means of assessing potential seasonal patterns in the data. Each boxplot represents the interquartile range (IQR) for a given month, with the horizontal line indicating the median. The whiskers capture the range of

how volatility evolves over time by continuously updating the estimate using the most recent 20 observations (Andreou and Ghysels, 2002b).

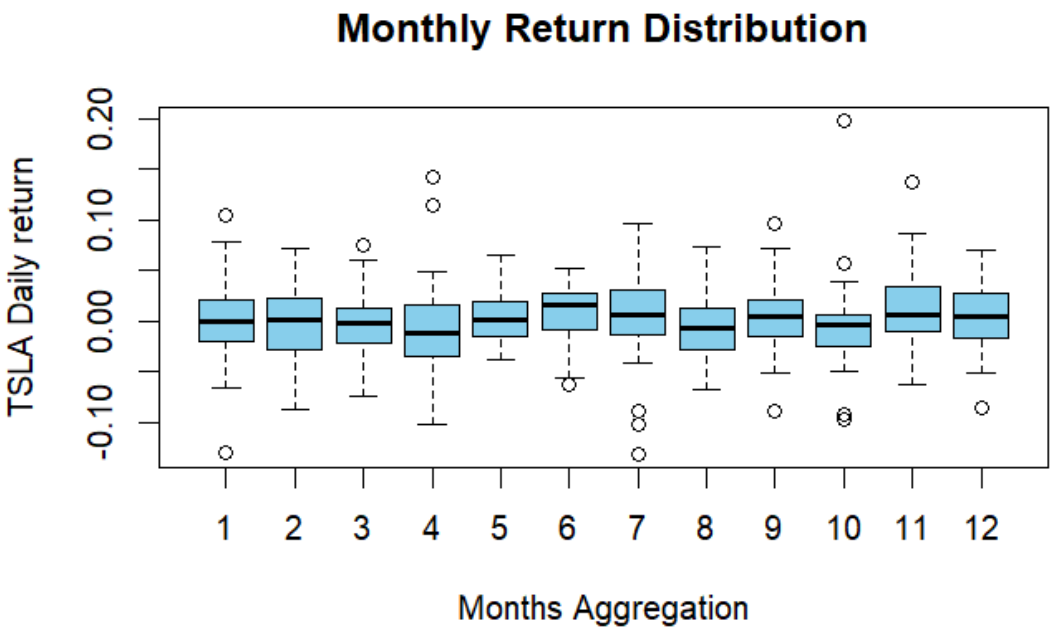



**Figure 4.16:** Monthly Return Distribution (2023-2025).

most observations, while individual points denote outliers.

From a visual inspection, the distribution of returns appears fairly consistent across months, with no clear or systematic seasonal effects. Nonetheless, certain months, most notably January and October, show slightly wider IQRs and a greater number of outliers, suggesting occasional periods of heightened volatility. Overall, however, the variation is not strong enough to support the presence of a calendar-based return anomaly within the observed period. These findings suggest that Tesla's stock returns do not exhibit a persistent seasonal structure, at least at the monthly level.

Figure 4.17 illustrates the relationship between Tesla's daily log returns and trading volume (measured in millions of shares) over the period from January 2023 to February 2025. Each point on the scatter plot corresponds to an individual trading day. The majority of returns are clustered around zero, while trading volumes generally range around 100 million shares, with occasional spikes exceeding 300 million. The correlation coefficient between daily returns and volume is approximately

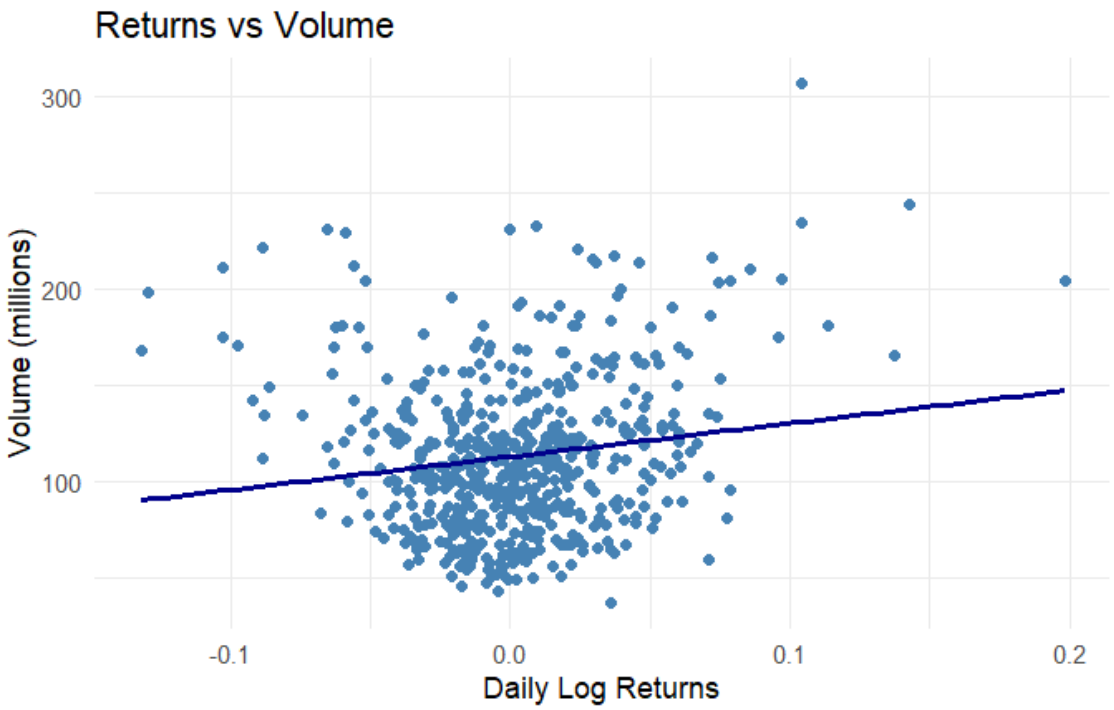



**Figure 4.17:** TSLA's returns vs. volume (2023-2025).

0.15, indicating a weak positive association. This result suggests that trading days with larger absolute returns are often accompanied by slightly higher trading volumes, although the relationship is not particularly strong. The scatter plot, combined with the correlation measure, implies that returns and volume are only loosely linked in Tesla's trading activity during this period. Outliers observed along both axes reflect sporadic episodes of elevated volatility or abnormal investor interest.

Overall, the exploratory analysis reveals that Tesla's stock returns are characterized by high volatility and a slight positive skewness, along with heavier tails than would be expected under a normal distribution. This indicates that extreme price movements, both upward and downward, occur more frequently than predicted by a Gaussian model. Volatility appears to cluster over time, with high-volatility periods followed by continued turbulence. In contrast, the return series displays minimal autocorrelation, which supports the weak-form efficiency hypothesis, suggesting that past prices do not contain significant predictive power for future price movements. Furthermore, monthly seasonal effects appear limited, and the relationship between

trading volume and returns remains marginal.

These findings offer a strong empirical foundation for the econometric analysis that follows, particularly in terms of understanding and modeling volatility, risk, and potential market reactions to external signals such as sentiment.

### 4.3.3 Temporal Alignment of Tweets and Market Data

Before applying sentiment models and econometric estimation, the tweets of Elon Musk and Tesla's stock market performances needed to be synchronized based on a common time dimension. Synchronization ensures that any sentiment score can be appropriately matched to pertinent financial outcomes to facilitate effective correlation and causality analyses in the rest of the chapters.

In order to combine these two very different sources of data, a daily-level temporal alignment process was implemented. Because it is possible for tweets to occur at any time, that is, at non-trading hours or during weekends, a complete calendar for a day was built covering the entire analysis period (January 1st, 2023 to February 28th, 2025) including weekends and bank holidays.

On the content front, tweets were cleaned and aggregated utilizing Python. Time stamps were converted to New York time (EST/EDT) to align with the standard of the financial market. Tweets were grouped by date, with daily tweet counts computed and the cleaned text retained in list format. Aggregated tweet data were then stored as a CSV file.

Simultaneously, Tesla's stock data was collected in R via the quantmod package. Closing prices were utilized to calculate daily log returns and a rolling volatility measure of 20 days. Raw opening and closing prices were saved as well for enabling any event study analysis. The processed finance data was saved to CSV as well.

Both of the datasets were merged in R on the complete daily calendar as a base, thus even the non-trading or non-tweet days would be included. This is to ensure temporal correspondence among the variables and allow for the analysis to account

for the lagged influence of off-hours or weekend tweets on market performance.

The resulting dataset includes one row per calendar day, containing variables such as Tesla's daily return, rolling volatility, tweet count, and tweet text content. This unified dataset serves as the input for the sentiment extraction pipeline (see Chapter 5) and the econometric analyses (see Chapter 6). A sample of the first six rows of the final merged dataset is presented in Table 4.2.

**Table 4.2:** Excerpt from the temporally aligned dataset (tweets and market data).

| **Date** | **Open** | **Close** | **Return** | **Volatility** | **# Tweets** | **Tweet Texts (excerpt)** |
|---|---|---|---|---|---|---|
| 2023-01-01 | NA | NA | NA | NA | 19 | 'lol but maybe', 'yikes', 'later this week', 'the cdc', ... |
| 2023-01-02 | NA | NA | NA | NA | 8 | 'absolutely', 'ehrlich despises humanity', ... |
| 2023-01-03 | 118.47 | 108.10 | NA | NA | 23 | 'seems like our image recognition needs some work', ... |
| 2023-01-04 | 109.11 | 113.64 | 0.0499 | NA | 5 | 'it's a new world goodbye homework', 'public likes', ... |
| 2023-01-05 | 110.51 | 110.34 | -0.0295 | NA | 20 | 'this demands an explanation', 'the drone wars are...' |
| 2023-01-06 | 103.00 | 113.06 | 0.0243 | NA | 5 | 'an actual jurassic park would be awesome', ... |

# Chapter 5

# Sentiment Analysis in Python: Techniques and Results

Sentiment analysis has become an essential technique for extracting actionable insights from unstructured text in fields such as finance (Tetlock, 2007), political science (Gentzkow et al., 2019), and marketing (Pang et al., 2008). While manual annotation remains the gold standard for small-scale studies (Krippendorff, 2018), it suffers from inherent limitations: annotator subjectivity (Hovy and Lavid, 2010), poor scalability (Nowak and Rüger, 2010), and high costs for large datasets. These challenges are particularly evident in financial contexts, where the real-time analysis of executive communications and social media requires both high precision and the capacity to process substantial volumes of data (Loughran and McDonald, 2011).

To address these constraints, this chapter presents a hybrid methodology that integrates advanced NLP techniques with an evaluation of simpler, computationally efficient alternatives. At the core of this approach is RoBERTa (Liu et al., 2019), a transformer-based model that has shown strong performance in capturing nuanced sentiment, including domain-specific jargon, negation, and sarcasm (Barbieri et al., 2020; Özkurt, 2024). By leveraging RoBERTa both as a high-accuracy benchmark and as a source of pseudo-labeled data for lighter models, the study aims to balance

predictive performance with interpretability and processing speed. This dual focus not only addresses the limitations of manual annotation and resource-intensive deep learning but also provides a framework adaptable to real-time, large-scale sentiment monitoring in the financial domain.

## 5.1 Methodological Overview

Sentiment analysis is valuable for understanding opinions and trends across disciplines, yet the manual labeling of large datasets is rarely practical. Ensuring consistency in annotation demands considerable human effort, and disagreements often arise in the presence of sarcasm or ambiguity (Hovy and Lavid, 2010; Nowak and Rüger, 2010). Within the scope of this thesis, a fully manual annotation process would have required excessive time and resources, making it unfeasible.

The limitations of manual methods have driven the adoption of automated sentiment classification. Recent advances in transformer-based models, such as BERT (Devlin et al., 2019) and its optimised variant RoBERTa, have significantly improved classification accuracy while maintaining strong generalisation across datasets. Despite the computational cost and potential biases inherent in these models, their ability to process large-scale, real-time data streams makes them indispensable for many applications.

RoBERTa was selected as the benchmark for this study due to its consistently strong results on established sentiment benchmarks (Liu et al., 2019) and its proven ability to capture complex contextual cues relevant to financial language. Its outputs serve not only as a standalone classification solution but also as the foundation for pseudo-labeling, enabling the creation of training data for simpler models. This approach makes it possible to assess whether less complex classifiers, when trained on high-quality pseudo-labeled data, can achieve performance levels comparable to RoBERTa while offering advantages in computational efficiency and interpretability.

The analytical process begins with a review of feature extraction methods, starting from traditional approaches such as BoW and TF-IDF, and progressing to embedding-based techniques including Word2Vec and BERT-derived representations. RoBERTa is then employed to generate pseudo-labels for a large corpus, which are subsequently used to train logistic regression, random forest, and LSTM models. These models are evaluated not only for predictive accuracy but also for processing speed and ease of interpretation. Potential issues arising from pseudo-labeling, such as the overrepresentation of neutral sentiment, are addressed through targeted mitigation strategies.

By combining the strengths of advanced deep learning methods with the practicality of simpler algorithms, this chapter aims to offer both a rigorous methodological contribution and actionable insights for deploying sentiment analysis in financial contexts where efficiency, accuracy, and interpretability must be carefully balanced. The sections that follow describe each stage of the methodology and present the corresponding results.

## 5.2 Feature Extraction Techniques

In this section, we introduce and implement several feature extraction techniques that are essential for transforming raw textual data into structured numerical representations. These representations are required for building machine learning models to perform sentiment analysis. The techniques presented here differ in their nature, some are traditional and more interpretable, while others are more complex and semantically rich. Traditional methods, such as BoW and TF-IDF, are discussed in Section 5.2.1., whereas semantic embedding approaches will be addressed in Section 5.2.2.

Before delving into the implementation, it is important to understand the goal of feature extraction in NLP. Feature extraction involves converting raw text into

informative numerical vectors that can be used by algorithms for classification, clustering, and other downstream tasks. This process generally includes several preprocessing and transformation steps such as tokenization, lemmatization, stopword removal, and vectorization (Birjali et al., 2021; Wankhade et al., 2022).

Tokenization is the process of breaking text into individual units known as tokens, which typically correspond to words. These tokens form the base units on which further NLP operations are applied. Lemmatization reduces words to their canonical or base form, known as the lemma. For instance, terms like "running", "ran", and "runs" are reduced to "run", thereby improving consistency and reducing vocabulary size. Stopword removal eliminates common words such as "the", "is", and "and", which usually do not carry significant semantic content and may introduce noise into the data. Vectorization then transforms the cleaned and preprocessed text into numerical vectors that can be interpreted by machine learning algorithms. Techniques range from simple sparse representations like BoW and TF-IDF to more advanced dense embeddings like Word2Vec and contextual embeddings derived from transformer-based models such as BERT (Birjali et al., 2021; Wankhade et al., 2022).

The choice of feature extraction method can significantly impact model performance and interpretability. Hence, a comparative evaluation of these techniques is conducted in Section 5.5.

### 5.2.1 Bag-of-Words (BoW) and TF-IDF

BoW is one of the most foundational and widely used text representation models in NLP. The core idea is to treat a document as an unordered collection of words and to represent it as a frequency vector based on a predefined vocabulary (Qader et al., 2019). This method disregards word order and syntax, focusing purely on word occurrence. The implementation typically involves tokenizing the text, removing stopwords, and counting the occurrences of each term. The output is a high-dimensional, sparse matrix where each row corresponds to a document (in this

case, a day of tweets), and each column represents a term from the vocabulary. BoW was considered in this study due to its simplicity, interpretability, and efficiency. Despite its limitations, particularly the loss of context, it serves as a strong baseline for initial model development and allows for rapid experimentation (Qader et al., 2019).

TF-IDF (Term Frequency–Inverse Document Frequency) builds upon the BoW model by introducing a weighting mechanism that reflects the relative importance of a term within a document and across the entire corpus. It is computed using the following formula:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \cdot \log\left(\frac{N}{n_t}\right)$$

In this formulation, $TF(t, d)$ represents the frequency of term $t$ in document $d$, $N$ is the total number of documents in the corpus, and $n_t$ is the number of documents in which the term $t$ appears. This approach reduces the weight of common but uninformative words, thereby emphasizing terms that are more characteristic of specific documents. TF-IDF is particularly effective in tasks like sentiment analysis, where capturing nuanced term importance is crucial (Qaiser and Ali, 2018).

Qaiser and Ali (2018) highlight the mathematical foundations of TF-IDF and demonstrate its utility in various real-world web domains (.com, .edu, .org, .biz). While they acknowledge its effectiveness in identifying key terms, they also note some limitations, including its lack of semantic depth and sensitivity to word form variations.

In this study, both BoW and TF-IDF were applied to daily aggregated tweets. The text was tokenized, stopwords were removed, and the data was transformed using `CountVectorizer` and `TfidfVectorizer` from the `scikit-learn` library. Both vectorizers were configured to extract the top 500 most frequent or informative

**Table 5.1:** Top 10 most informative terms by BoW and TF-IDF.

| Rank | BoW Term | TF-IDF Term |
|---|---|---|
| 1 | people (1085) | people (0.0612) |
| 2 | just (1015) | yeah (0.0577) |
| 3 | yeah (934) | just (0.0576) |
| 4 | good (854) | true (0.0547) |
| 5 | true (848) | good (0.0536) |
| 6 | great (817) | great (0.0519) |
| 7 | like (763) | like (0.0491) |
| 8 | exactly (546) | exactly (0.0408) |
| 9 | cool (505) | cool (0.0391) |
| 10 | time (500) | **tesla** (0.0377) |

features[1]. The resulting feature matrices were stored and will be used in subsequent sentiment classification experiments.

To better illustrate the difference between the two representations, Table 5.1 compares the top 10 terms ranked by frequency in the BoW model and by average weight in the TF-IDF model. While both methods highlight commonly used words such as "people", "just", and "good", TF-IDF gives higher relative importance to domain-specific or infrequent-but-informative terms like "tesla", demonstrating its strength in identifying discriminative vocabulary. Overall, the majority of the top-ranked words are similar across both models, indicating a general consistency in capturing prominent terms despite differences in weighting strategies.

These two techniques lay the foundation for our comparative analysis of traditional versus semantic embeddings, enabling a comprehensive evaluation of their effectiveness in real-world sentiment analysis tasks.

### 5.2.2 Word Embeddings: Word2Vec, BERT

Word embeddings are a way to represent words as continuous numerical vectors that capture their underlying meaning. Unlike traditional methods such as BoW

[1]The maximum features parameter was set to 500 in order to limit the dimensionality of the feature space. This threshold ensures that the most frequent (or informative) 500 terms across the corpus are retained, while discarding less significant words that may contribute noise or sparsity.

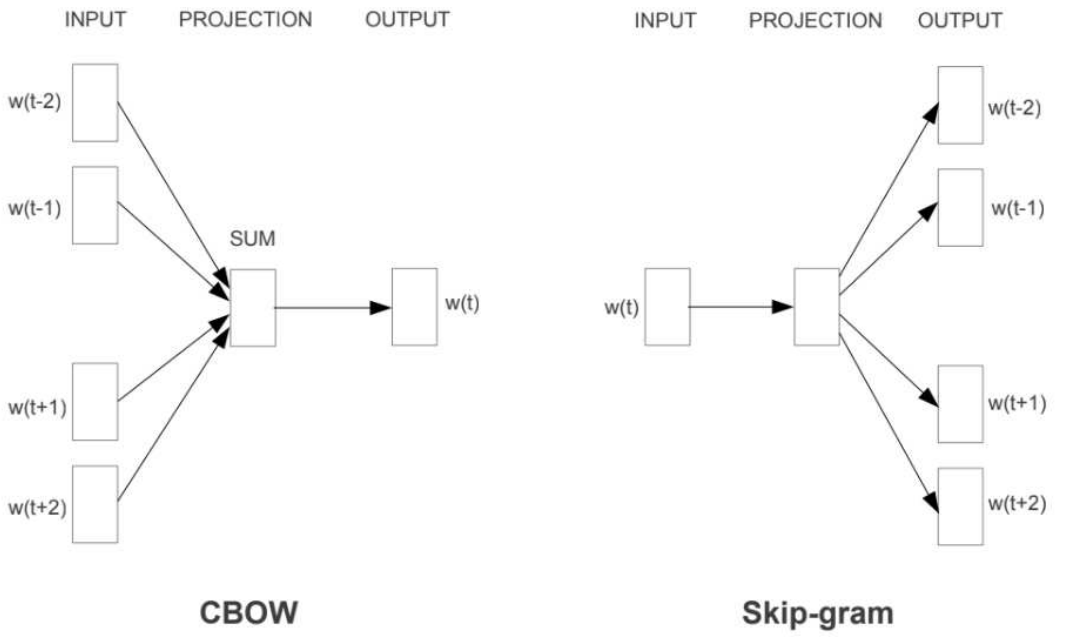



**Figure 5.1:** Word2Vec architectures. The functioning of CBOW and Skip-gram (Mikolov et al., 2013a).

or TF-IDF, which treat words as isolated units, embeddings consider relationships between words based on their co-occurrence patterns in language (Jiang, 2021). This allows the model to retain richer contextual and semantic information.

One of the most well-known embedding models is Word2Vec, which learns word vectors from large text corpora using shallow neural networks. It includes two main architectures: Continuous Bag-of-Words (CBOW) and Skip-Gram (Mikolov et al., 2013b,a).

CBOW predicts a target word based on its surrounding context, while Skip-Gram does the opposite: it predicts the surrounding words given a single target word. These two architectures are illustrated in Figure 5.1. In practice, the Skip-Gram model is often combined with a training technique called negative sampling, which improves efficiency by updating only a small subset of weights at each step (Goldberg and Levy, 2014).

Previous research by Pagolu et al. (2016) found Word2Vec to be particularly effective in sentiment classification within financial domains, supporting its inclusion

in the comparative experiments of this work.

In this study, a pre-trained Word2Vec model was used to generate embeddings from the textual data. Each day's tweets were preprocessed through tokenization and stopword removal. Then, for each day, the average vector of all valid tokens was computed, resulting in a single embedding per day. These daily embeddings were stored in a matrix, with one row per day and one column per vector dimension.

Another powerful embedding model used in this thesis is BERT. BERT improves upon earlier embedding methods by incorporating both left and right context using a deep transformer-based architecture. It is pre-trained on large text corpora through two self-supervised tasks: masked language modeling (predicting masked-out words) and next sentence prediction. BERT can then be fine-tuned for specific tasks, such as sentiment analysis, with strong performance. Thanks to its contextualized word representations, BERT is especially effective when working with short, informal, and noisy text such as tweets (Devlin et al., 2019; Xu et al., 2019; Sharma et al., 2024).

For BERT embeddings, each day's tweets were processed by tokenizing the text and applying mean pooling over the model's hidden representations, producing a fixed-dimensional vector per day. These embeddings were saved and used in subsequent sentiment classification.

Both resulting data frames include a date column for temporal alignment and were validated to contain the expected shape: one row per day and the respective feature columns for each embedding method. These embeddings serve as input for the sentiment classification models presented in the following sections.

Figure 5.2 presents the cosine similarity[2] heatmaps of the first 10 daily tweet embeddings generated using Word2Vec and BERT. As shown, Word2Vec embeddings maintain high similarity across all days, suggesting stable representations

[2]Cosine similarity is a measure of similarity between two vectors based on the cosine of the angle between them. It ranges from –1 to 1 and is commonly used to assess the semantic closeness of text embeddings in NLP (Xia et al., 2015).

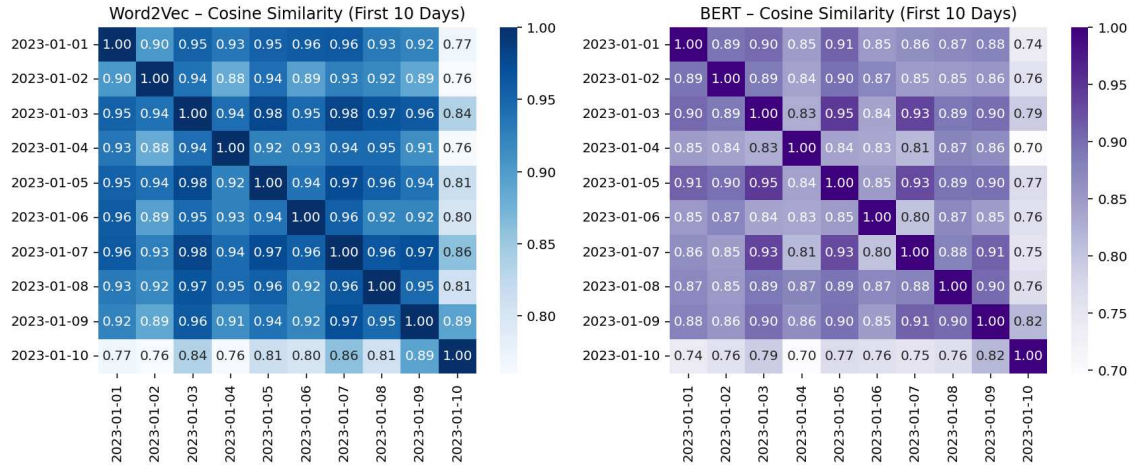



**Figure 5.2:** Cosine Similarity: Word2Vec vs. BERT.

but potentially limited sensitivity to subtle semantic shifts. In contrast, BERT shows greater variability, reflecting its capacity to capture deeper contextual nuances. This distinction supports the idea that BERT embeddings may be better suited for capturing day-to-day variations in sentiment or discourse in noisy social media data (Acheampong et al., 2021; Singh et al., 2021).

Overall, this approach provides a compact yet expressive representation of the textual data, capturing both local and contextual semantic patterns, and makes the dataset suitable for downstream supervised learning tasks. These embedding models are employed in this thesis because they enable a more nuanced understanding of language compared to frequency-based methods, ultimately supporting more accurate sentiment classification (Pagolu et al., 2016; Jiang, 2021).

## 5.3 Pseudo-labeling with RoBERTa

RoBERTa, an optimized variant of BERT, achieves state-of-the-art performance in sentiment analysis by refining the pretraining process through dynamic masking, extended training duration, and the use of larger datasets, while omitting the next-sentence prediction task (Liu et al., 2019). Its bidirectional transformer architecture is particularly effective at capturing contextual nuances in informal text, making it

well-suited for social media sentiment analysis (Sharma et al., 2024; Jiang, 2021).

In this study, RoBERTa is employed as a benchmark for two primary reasons. The manual annotation of daily Twitter data at scale would be prohibitively time-consuming and inherently subjective (Hovy and Lavid, 2010; Nowak and Rüger, 2010). Given RoBERTa's high accuracy in prior sentiment classification tasks (Barbieri et al., 2020; Özkurt, 2024), its use for generating reliable pseudo-labels is well justified. Furthermore, by treating RoBERTa's outputs as a form of "ground truth," it becomes possible to evaluate whether simpler models, such as logistic regression or LSTM architectures, can approximate its performance with significantly lower computational costs, a crucial factor for real-time financial applications.

Pseudo-labeling here refers to a semi-supervised learning approach in which RoBERTa's predictions on unlabeled data serve as training labels for other models (Arazo et al., 2020). The process begins with the aggregation of daily tweets, preprocessed as described in Chapter 4, into single documents. A pretrained RoBERTa model, fine-tuned for social media data, then classifies each document's sentiment as negative (0), neutral (1), or positive (2). The final daily sentiment label is determined through majority voting, where the most frequent sentiment across all tweets for a given day is assigned as the pseudo-label. This transformation converts the original task into a supervised learning problem, allowing traditional models to be trained for direct comparison. It is important to note that pseudo-labels are not treated as equivalent to human annotations; rather, they provide a consistent reference derived from a high-performance model.

The main strengths of this approach lie in its scalability and contextual sensitivity. RoBERTa can process large volumes of data far more rapidly than manual labeling, and its attention mechanisms are capable of capturing subtleties such as sarcasm and domain-specific jargon more effectively than lexicon-based methods (Sharma et al., 2024). However, pseudo-labeling also presents certain limitations. Chief among these is bias propagation, as the pseudo-labels inevitably reflect any systematic biases

in RoBERTa's predictions, such as a tendency to overpredict neutral sentiment in ambiguous cases. Moreover, the performance of simpler downstream models remains dependent on RoBERTa's accuracy, necessitating careful validation checks (Arazo et al., 2020; Sharma et al., 2024).

The pseudo-labeled dataset serves a single primary role within the experimental design: it provides training data for lightweight models, enabling an assessment of whether such feature-based methods can approach transformer-level performance. In this context, RoBERTa's outputs are treated as a consistent reference rather than a verified ground truth, as no additional accuracy checks against human annotations were performed. This approach offers a practical balance between accuracy and efficiency, an essential consideration in financial applications, where latency and interpretability often take precedence over marginal improvements in metrics such as F1-score (Loughran and McDonald, 2011).

## 5.4 Compared Methods

To benchmark the performance of more advanced neural models, this study employs two traditional supervised learning algorithms (logistic regression and random forest) and a deep supervised model (LSTM) as baselines. These models are well-established in the field of text classification due to their interpretability, computational efficiency, and solid performance in high-dimensional feature spaces (James et al., 2013).

Supervised classification algorithms aim to map input features to discrete sentiment classes, typically positive, neutral, or negative, based on a labeled training set. In this case, the labels were generated through a pseudo-labeling process using RoBERTa. This aggregated dataset served as the training and evaluation base for those models. As introduced in Section 3.3.2 (Balaji et al., 2021), both algorithms represent key components of supervised machine learning and offer a meaningful comparison against deep learning-based approaches. Their results provide a foundational

benchmark to assess whether the added complexity of more advanced supervised models leads to measurable improvements in sentiment classification performance.

### 5.4.1 Multinomial Logistic Regression

Multinomial logistic regression is a linear classification algorithm used for multiclass problems ($K \geq 2$). Unlike linear regression, it estimates the probability that a given observation belongs to each of several categories (James et al., 2013). Since the sentiment labels in our dataset comprise three classes (positive, neutral, negative), multinomial logistic regression is employed, which naturally extends the binary framework to multiclass settings.

For a response variable $Y \in \{1, \ldots, K\}$ and predictor variables $X = (X_1, \ldots, X_p)$, the model specifies the probability $P(Y = k \mid X)$ as:

$$P(Y = k \mid X) = \frac{e^{\beta_{0k} + \beta_{1k} X_1 + \cdots + \beta_{pk} X_p}}{\sum_{j=1}^{K} e^{\beta_{0j} + \beta_{1j} X_1 + \cdots + \beta_{pj} X_p}} \tag{5.1}$$

where $k$ indexes the sentiment classes (e.g., $k \in \{\text{negative}, \text{neutral}, \text{positive}\}$), and $\beta_{0k}, \beta_{1k}, \ldots, \beta_{pk}$ are coefficients estimated from the training data.

To implement the model, Python was used to split the labeled dataset into training and test sets with a 50/50 random split[3]. The daily sentiment labels generated by RoBERTa served as the target variable. Logistic regression was applied with four feature extraction techniques: BoW, TF-IDF, Word2Vec, and BERT embeddings. For each combination, key evaluation metrics, accuracy, macro F1-score, precision, and recall, were computed[4]. Due to the dataset's imbalance (neutral tweets dominating), macro F1-score is a particularly important measure, as it treats all classes equally

[3]A random 50/50 train-test split helps reduce overfitting, where a model performs well on training data but fails to generalize. Overfitting occurs when the model captures noise instead of underlying patterns (James et al., 2013).

[4]Accuracy measures overall correctness, precision evaluates the reliability of positive predictions, recall assesses the model's ability to find all relevant cases, and the F1-score balances precision and recall (Bishop and Nasrabadi, 2006).

**Table 5.2:** Multinomial logistic regression performance across feature extraction techniques.

| Feature | Accuracy | F1-score | Precision | Recall |
|---|---|---|---|---|
| **BoW** | 0.81 | **0.48** | **0.56** | **0.45** |
| TF-IDF | **0.83** | 0.30 | 0.28 | 0.33 |
| Word2Vec | 0.78 | 0.45 | 0.49 | 0.44 |
| BERT | 0.79 | 0.40 | 0.49 | 0.39 |

*Test set: 394 tweets*

regardless of their frequency. The results are summarized in Table 5.2.

Although TF-IDF achieves the highest accuracy, this result is skewed by its tendency to classify most tweets as neutral, reflecting the dataset's class imbalance. In contrast, BoW demonstrates the most balanced performance across all sentiment classes, with the highest macro F1-score, precision, and recall. This suggests that BoW's sparse binary representation, which captures word presence/absence without weighting, is better suited for logistic regression in this context. The model likely benefits from BoW's simplicity, as more complex embeddings (e.g., Word2Vec, BERT) may introduce noise or lose discriminative power for short-text sentiment classification.

The confusion matrix for BoW is shown in Figure 5.3; additional matrices for TF-IDF, Word2Vec, and BERT are provided in Appendix D.

The dataset exhibits class imbalance, with most tweets labeled as neutral. While this reflects the natural distribution of public communications, it biases models toward the dominant class. Metrics like macro F1-score and confusion matrices mitigate this by assessing performance across all sentiment categories. From an econometric perspective, non-neutral sentiment events are of primary interest, as clear positive/negative polarity may have stronger financial market implications (Atkins et al., 2018; Metta et al., 2022).

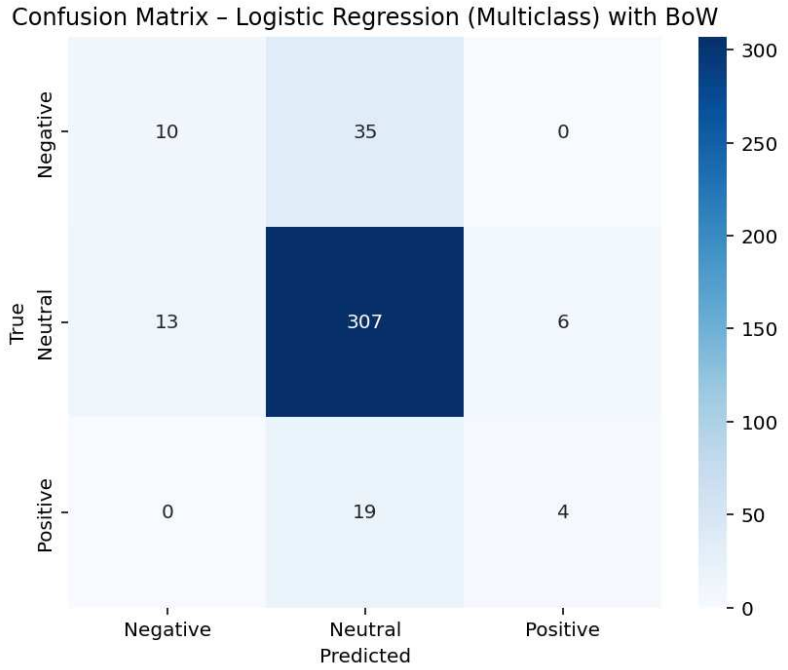



**Figure 5.3:** Confusion matrix for multinomial logistic regression trained with BoW.

### 5.4.2 Random Forest

Random forest is an machine learning method that builds upon the principles of bagging[5] to improve prediction accuracy and model robustness. It constructs a large collection of decision trees, each trained on a different bootstrapped sample[6] of the training data. However, unlike standard bagging, random forest introduces an additional layer of randomness: at each split in a tree, only a randomly selected subset of predictors is considered. This decorrelates the individual trees, as it prevents dominant predictors from being selected at every split, leading to a more diverse ensemble. The final prediction is obtained by aggregating the outputs of all trees, through majority voting for classification tasks or averaging for regression. This

[5]Bagging (Bootstrap Aggregating) is an ensemble learning technique that reduces variance and improves prediction accuracy by training multiple models on bootstrapped samples of the training data and averaging their predictions (for regression) or taking a majority vote (for classification). It is especially effective with high-variance models like decision trees (James et al., 2013).

[6]The bootstrap is a resampling technique used to estimate the uncertainty (such as standard errors or confidence intervals) of a statistical estimator by repeatedly drawing samples with replacement from the original dataset and recalculating the estimator for each sample. It is especially useful when theoretical formulas for variability are unavailable or complex (James et al., 2013).

**Table 5.3:** Random forest's performance with different feature extraction techniques.

| Feature | Accuracy | F1-score | Precision | Recall |
|---|---|---|---|---|
| BoW | **0.83** | 0.30 | 0.28 | 0.33 |
| TF-IDF | 0.83 | 0.30 | 0.28 | 0.33 |
| **Word2Vec** | 0.82 | **0.35** | **0.43** | **0.36** |
| BERT | 0.82 | 0.33 | 0.41 | 0.35 |

*Test set: 394 tweets*

strategy effectively reduces variance without significantly increasing bias, making random forest a powerful tool, particularly in high-dimensional or correlated feature spaces (James et al., 2013).

In this study, random forest is implemented as one of the baseline models for sentiment classification of Elon Musk's tweets. The algorithm's strength lies in its ensemble nature, which reduces overfitting and enhances generalization by averaging the outputs of multiple randomized decision trees. As explained by Biau and Scornet (2016), the model's design enables effective handling of high-dimensional input spaces and provides interpretability through variable importance scores. These characteristics are advantageous in tweet-level sentiment analysis, where textual inputs are noisy and sparse, and understanding feature impact is often valuable (Biau and Scornet, 2016).

The dataset was randomly split 50/50 into training and test sets. We tested the model using four different feature extraction techniques, BoW, TF-IDF, Word2Vec, and BERT, and evaluated performance using standard classification metrics (accuracy, F1-score, precision, and recall). As shown in Table 5.3, while BoW and TF-IDF achieved the highest accuracy (0.83), they performed poorly in terms of macro-averaged F1-score (0.30), indicating a bias toward the majority class (neutral sentiment). In contrast, Word2Vec, with only a slight drop in accuracy (0.82), achieved the highest F1-score (0.35), precision (0.43), and recall (0.36). This suggests Word2Vec provides a more balanced performance across all sentiment classes, making it the best feature representation for this task. BERT, despite its advanced contextual embeddings, did not outperform Word2Vec here, likely due to the relatively small

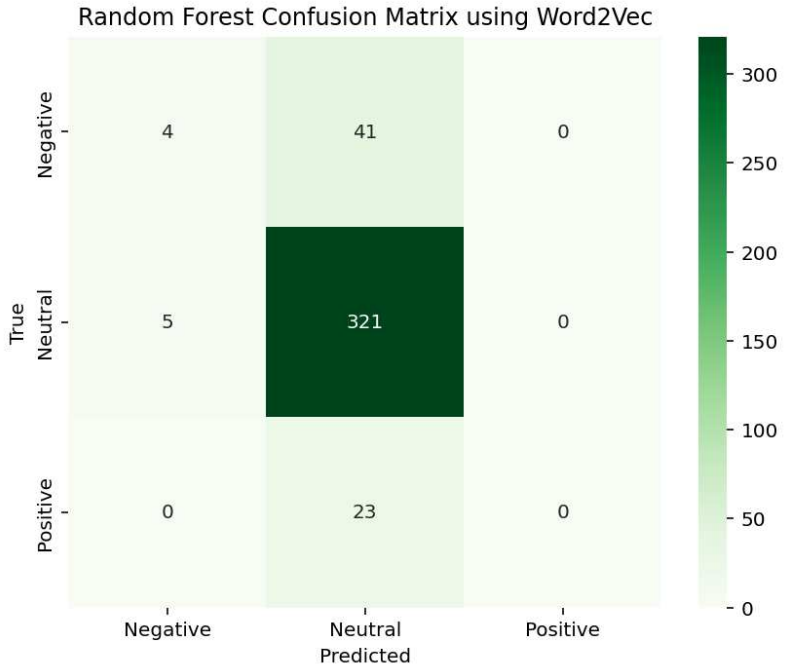



**Figure 5.4:** Confusion matrix for a random forest model trained using Word2Vec.

dataset size or the nature of tweet-based sentiment analysis.

Given these results, we selected Word2Vec as the optimal feature representation for random forest in our tweet sentiment classification task. Its superior F1-score and balanced precision-recall trade-off make it more reliable for capturing nuanced sentiment patterns in Elon Musk's tweets compared to other methods.

The confusion matrix for the Word2Vec model is shown in Figure 5.4, while confusion matrices for the other feature extraction techniques can be found in Appendix D. As observed, the model still exhibits a tendency to classify most tweets as neutral, reflecting the class imbalance in the data. However, Word2Vec's marginally better performance in identifying positive and negative sentiments (compared to BoW/TF-IDF) justifies its use in subsequent analyses.

### 5.4.3 LSTM (Long Short-Term Memory)

Deep learning has significantly advanced sentiment analysis by enabling models to automatically learn and extract hierarchical patterns from large volumes of

textual data. Among the most effective architectures for modeling sequential and contextual information are Long Short-Term Memory (LSTM) networks. LSTM networks are a specialized type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential data. Unlike traditional RNNs, LSTMs address the vanishing gradient problem[7], enabling them to retain relevant information over extended time periods (Hochreiter and Schmidhuber, 1997). This property makes LSTMs particularly well-suited for modeling time-series data or linguistic sequences, such as tweet streams. In sentiment analysis, LSTMs can effectively capture the progression and tone of emotional content across sequences of words, which is especially important in domains like finance.

Enhanced LSTM architectures that incorporate attention mechanisms have shown improved performance by allowing the model to focus on the most sentiment-relevant parts of a sentence. This is particularly beneficial in aspect-based sentiment analysis, where certain words contribute more significantly to the sentiment of a specific entity (Wang et al., 2016). In contrast, transformer models (see Section 3.3.4), based entirely on self-attention mechanisms rather than recurrence, have transformed NLP by offering greater scalability and parallelization. These models capture dependencies across entire sequences simultaneously, rather than sequentially (Birjali et al., 2021).

In this study, an LSTM-based neural network was implemented in Python for daily-level sentiment classification. The dataset was split into training and test sets with a 50/50 ratio. The model was trained and used to predict sentiment labels on the test set. Each input consisted of a 100-dimensional Word2Vec vector representing the aggregated content of tweets for a single day. The final model architecture comprised one LSTM layer, followed by fully connected dense layers, and ended with a softmax output layer for three-class sentiment classification (negative, neutral, positive).

[7]The vanishing gradient problem occurs when neural networks struggle to learn from inputs far back in a sequence (Hochreiter and Schmidhuber, 1997).

**Table 5.4:** LSTM's performance with Word2Vec.

| Feature | Accuracy | F1-score | Precision | Recall |
|---|---|---|---|---|
| **Word2Vec** | 0.83 | 0.30 | 0.28 | 0.33 |

*Test set: 394 tweets*

Differently from before, Word2Vec was selected as the embedding method due to its ability to produce dense, low-dimensional vector representations that capture semantic similarity between words (Mikolov et al., 2013b). These embeddings are particularly well-suited for LSTM models, as they preserve meaningful word-level relationships in a sequentially compatible format (Schuster and Paliwal, 1997). In contrast, approaches such as BoW or TF-IDF generate sparse, high-dimensional vectors that disregard word order and are therefore not ideal for sequence-based models like LSTMs. Although contextual embeddings such as BERT offer more advanced representations by dynamically capturing context through self-attention (Devlin et al., 2019), they are typically not used in conjunction with LSTMs, since BERT's transformer architecture already models sequential dependencies inherently (Vaswani et al., 2017). As a result, Word2Vec represents an effective balance between semantic richness and computational efficiency (Le and Mikolov, 2014), making it a suitable embedding method for the LSTM-based sentiment classification task.

Performance was evaluated using the same metrics used above. As shown in Table 5.4, the model achieved a high overall accuracy of 0.83; however, precision was lower for the positive and negative sentiment classes, suggesting a bias toward predicting the neutral class. The confusion matrix in Figure 5.5 provides a visual summary of classification performance across sentiment categories.

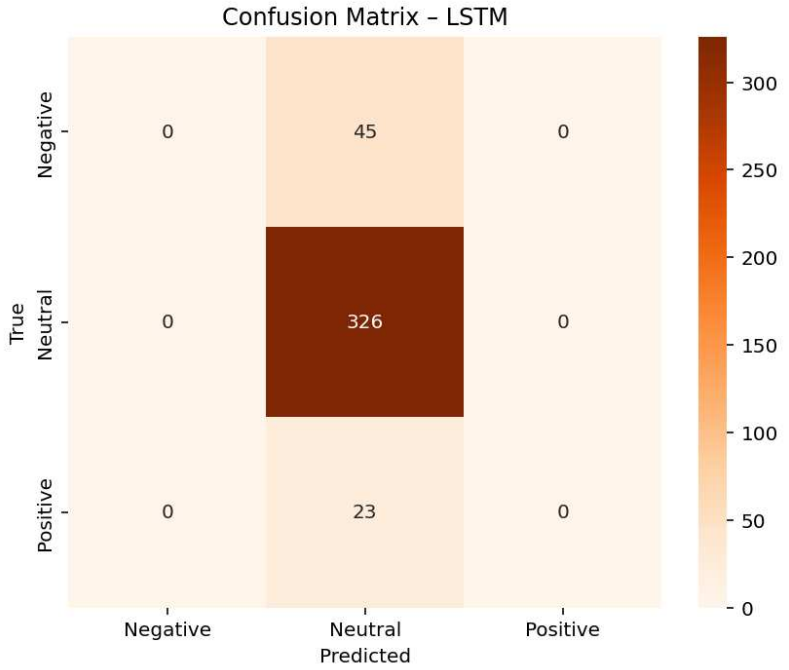



**Figure 5.5:** Confusion matrix for a LSTM model trained using a Word2Vec.

## 5.5 Mitigating Neutral Bias in Pseudo-Labeled Sentiment Data

As highlighted in the previous sections, the confusion matrices consistently show a strong bias toward the neutral sentiment class. This recurring outcome can be attributed to a combination of factors related to both the nature of the dataset and the preprocessing pipeline.

Firstly, the data consists of tweets and retweets from Elon Musk. As discussed during the exploratory data analysis in Chapter 4, many of these entries are extremely short, often consisting of single-word responses such as “maybe”, “nice”, or “yeah”. These short, context-less messages carry little semantic content and are typically interpreted by the model as neutral due to their lack of sentiment-bearing information. As a result, days with a high frequency of such tweets or retweets tend to skew the overall sentiment classification toward neutrality.

Moreover, the method of aggregating tweets on a daily basis may have unin-

tentionally amplified the neutrality bias. If the majority of tweets on a given day are neutral, due either to their short length or non-sentiment-bearing nature, the overall daily sentiment label, especially when derived using an aggregate RoBERTa prediction, is likely to be neutral as well. This aggregation, while simplifying the modeling process, may dilute more nuanced sentiment signals present in individual tweets.

Given these observations, and based on the optimal model and feature extraction method identified in the previous section, a new approach can be adopted. In order to reduce the impact of neutral bias, the models will be re-trained and evaluated after removing neutral samples from the dataset. These tweets often carry minimal semantic weight and do not contribute meaningful information for sentiment analysis. The neutral-filtered dataset was successfully saved for further analysis. After filtering, the dataset contains 7,109 positive tweets and 7,099 negative tweets, for a total of 14,208 sentiment-bearing entries. This balanced distribution provides a more robust foundation for retraining sentiment classification models without the distorting influence of neutral content.

The next section presents the results of this adjusted experiment and compares model performance after the exclusion of neutral-labeled data.

### 5.5.1 Performance Comparison and Model Evaluation

Following the removal of neutral-labeled tweets from the dataset, a series of classification models were re-trained using only sentiment-bearing data. This adjustment aimed to mitigate the previously observed bias toward neutrality. The evaluation presented in Table 5.5 compares the performance of three supervised learning models, logistic regression, random forest, and LSTM, across different feature extraction strategies, BoW and Word2Vec embeddings. All models were evaluated on the same training/test set splitted as 50/50, as before.

As shown in the table, logistic regression with BoW features achieved the highest

**Table 5.5:** Performance comparison of sentiment classification models using different feature extraction techniques. Results refer to non-aggregated, non-neutral tweets.

| Model | Feature | Accuracy | F1-score | Precision | Recall |
|---|---|---|---|---|---|
| Logistic Regression | BoW | 0.88 | 0.88 | 0.89 | 0.88 |
| Random Forest | Word2Vec | 0.87 | 0.87 | 0.87 | 0.87 |
| LSTM | Word2Vec | 0.85 | 0.85 | 0.86 | 0.85 |

overall performance, yielding an accuracy and F1-score of 0.88. This result confirms the robustness and simplicity of BoW-based models when applied to sentiment classification tasks involving short and informal text such as tweets. Despite its limitations in capturing semantic relationships between words, the BoW approach proves highly effective in this binary classification setting, likely due to the relatively distinct vocabulary patterns associated with positive and negative sentiment.

The random forest model trained on Word2Vec embeddings achieves competitive performance, with an accuracy of 0.87 and an F1-score of 0.87. This strong performance stems from Word2Vec's dense, semantically rich embeddings, which effectively capture word similarities and contextual relationships, advantages over sparse BoW representations (Mikolov et al., 2013b). However, its slight underperformance relative to logistic regression (which achieved 0.88 accuracy) may reflect limitations of the validation set approach used here. As noted by (James et al., 2013), single train-test splits can yield high-variance performance estimates, especially with smaller datasets. A k-fold cross-validation (e.g., 5-fold or 10-fold) could provide more robust evaluation by reducing variability in performance metrics. Additionally, random forests' greater sensitivity to noise or sparse features, common in short-text sentiment analysis, might explain the marginal gap, as simpler models like logistic regression can outperform complex ensembles when data is limited (Breiman, 2001).

The LSTM model, also leveraging Word2Vec features, achieves an accuracy and F1-score of 0.85. While LSTM networks are well-suited for sequential data and have demonstrated success in NLP tasks, their relative underperformance in this experiment may stem from several factors. First, the data used is not strictly

sequential or syntactically rich, limiting the advantages of recurrent architectures. Second, the model was trained on reshaped Word2Vec vectors rather than raw token sequences, which may have constrained its ability to learn complex temporal dependencies. Finally, the relatively small size of the dataset may have hindered the LSTM's generalization capacity, especially given its larger number of parameters compared to the other models.

To further illustrate the classification performance, Figure 5.6 display the confusion matrices for each model. These visualizations provide insight into how well each classifier distinguishes between positive and negative sentiment. Notably, the confusion matrix for logistic regression shows a balanced distribution of true positives and true negatives with minimal misclassifications. The random forest and LSTM models also show strong performance but slightly more frequent misclassification of negative tweets as positive, suggesting a marginal tendency toward optimism in those predictions.

Overall, these results confirm that removing neutral tweets enhances model discriminability, and highlight the trade-offs between simple linear models, tree-based ensembles, and deep learning approaches. While more sophisticated models like LSTMs offer theoretical advantages, they may not always yield superior performance in practice, particularly when working with short-text data and limited sample sizes. The next section presents an analysis of sentiment trends over time, using RoBERTa, to examine how public sentiment toward Elon Musk evolved throughout the study period.

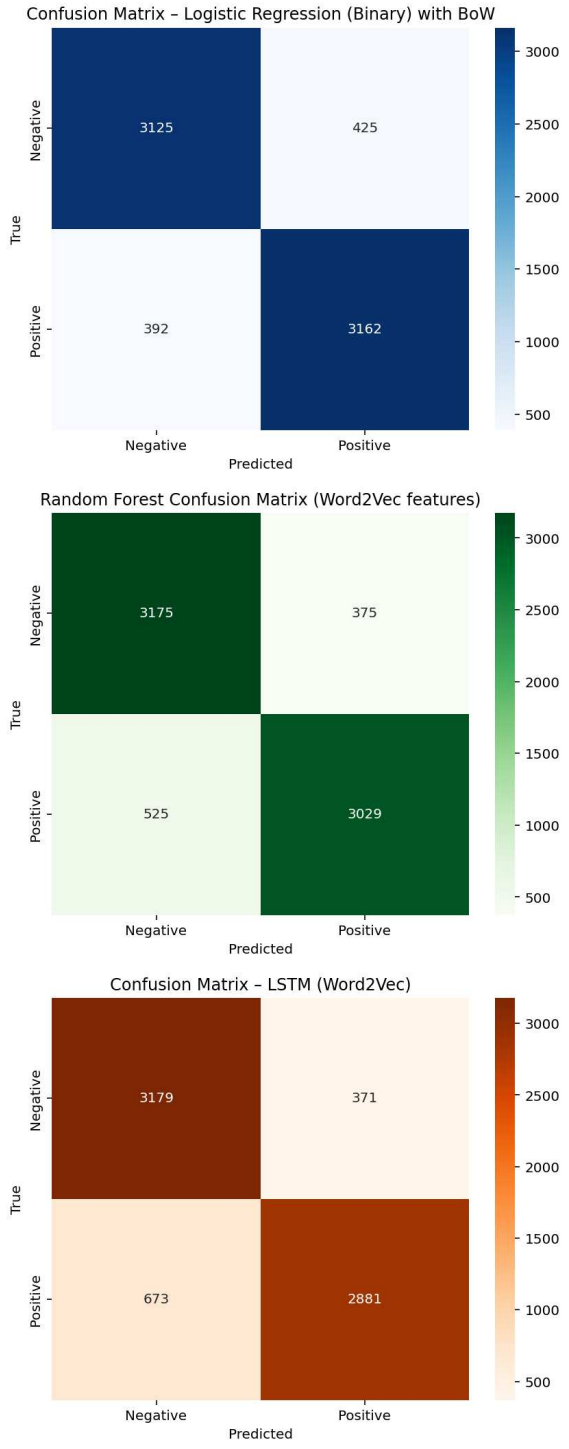



**Figure 5.6:** Confusion Matrices of Logistic Regression, Random Forest and LSTM. Results refer to non-aggregated, non-neutral tweets.

## 5.6 Sentiment Trends Over Time

To investigate how the sentiment of Elon Musk's non-neutral tweets evolved over time, a temporal sentiment analysis was performed by aggregating and visualizing sentiment-labeled tweet data over different time intervals. This analysis aimed to uncover potential patterns or shifts in public discourse, emotional tone, or communication style over the two-year period from January 2023 to March 2025. Three different types of plots were generated to capture complementary aspects of the sentiment dynamics.

Figure 5.7 presents the monthly distribution of non-neutral tweets, disaggregated by sentiment. This stacked bar plot provides an overview of the total number of tweets per month, classified as either positive or negative. One clear trend is the steady increase in tweet volume over time, particularly from mid-2024 onwards. During the early months of 2023, the total number of sentiment-laden tweets remained relatively low and stable. However, beginning in the last quarter of 2023 and continuing through 2024, the frequency of both positive and negative tweets began to rise more sharply. Notably, negative sentiment tweets consistently outnumber positive ones in most high-volume months, with peak activity observed in October and November 2024, as well as February 2025. This trend may reflect escalating public attention or polarizing events related to Musk during this period, though further contextual analysis would be necessary to link these spikes to specific events.

Figure 5.8 illustrates the 7-day rolling average of daily tweet counts for both positive and negative sentiments. This smoothed line plot reduces the noise present in raw daily data, making short-term trends more discernible. Both sentiment lines exhibit periodic increases and decreases, but starting around mid-2024, there is a marked increase in the frequency and intensity of negative sentiment peaks, surpassing the positive trend in both amplitude and consistency. This pattern may indicate phases of intensified public criticism or polarizing statements from Musk,

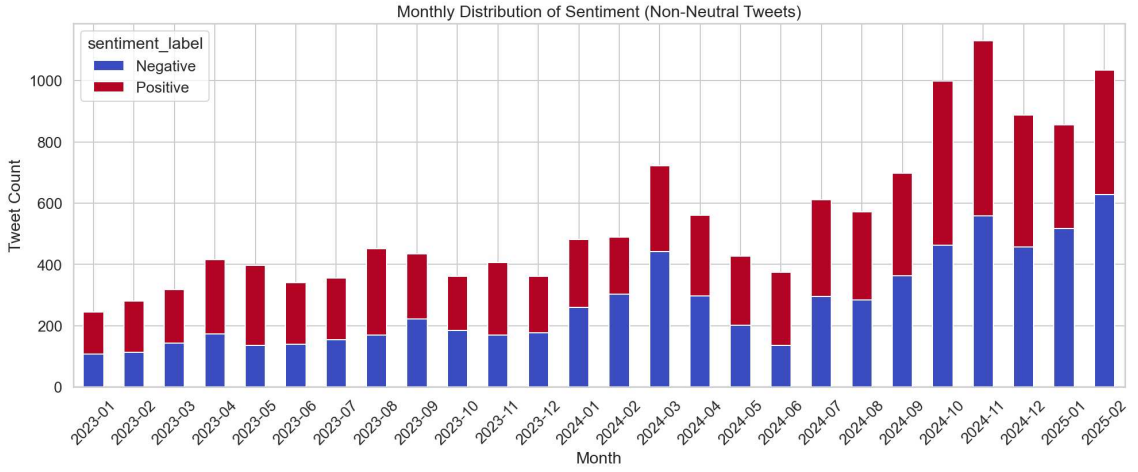



**Figure 5.7:** Monthly distribution of sentiment (non-neutral tweets).

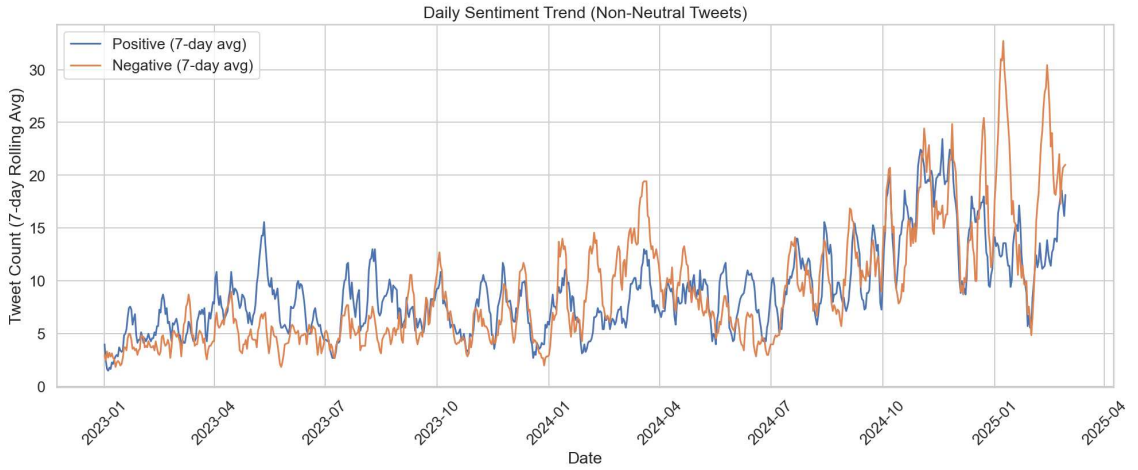



**Figure 5.8:** Daily sentiment trend (non-neutral tweets).

leading to more reactive, emotionally charged discussions online.

To better understand the relative sentiment composition over time, Figure 5.9 shows a stacked area chart of daily sentiment proportions, excluding neutral tweets. This plot emphasizes the proportion of positive versus negative sentiment rather than absolute counts, providing insight into how the emotional tone shifted independent of tweet volume. Throughout the observation period, the balance between positive and negative tweets varies significantly from day to day. However, the overall tendency leans toward a higher proportion of negative sentiment, particularly in mid to late 2024. There are also several short periods in which positive sentiment briefly dominates, though these episodes are generally less sustained. The fluctuations

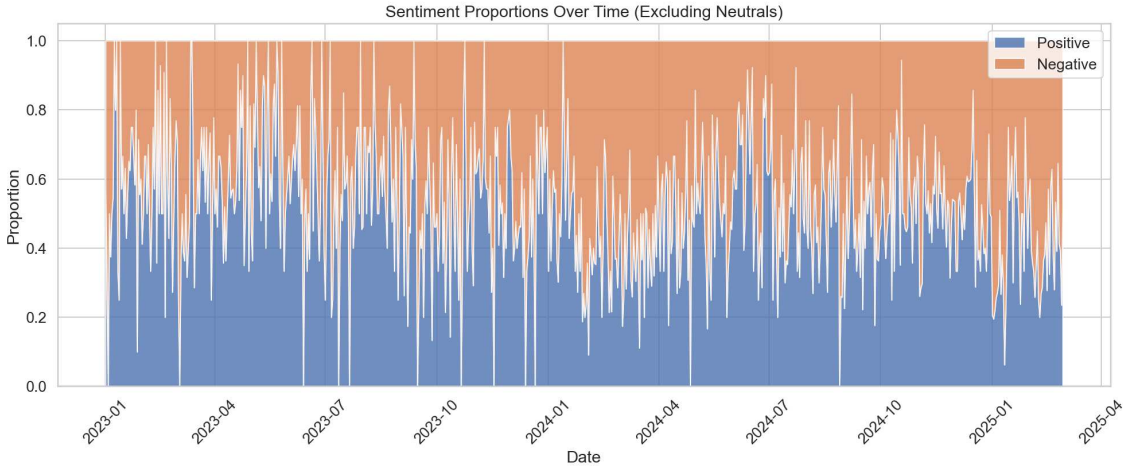



**Figure 5.9:** Sentiment proportions over time (non-neutral tweets).

in proportion suggest that sentiment is not only reactive but potentially cyclical or event-driven, aligning with external news cycles, product announcements, or controversies involving Musk.

In summary, the combination of these visualizations reveals a multifaceted view of sentiment trends in Musk's tweets. There is clear evidence of growing volume, heightened emotional polarization, and an increasing dominance of negative sentiment over the analyzed period. These findings suggest that sentiment around Musk has become more intense and perhaps more critical, particularly during the latter half of the dataset. Understanding these patterns can provide useful context for downstream analyses, such as assessing sentiment impact on financial markets or exploring causal relationships with media events.

## 5.7 Summary of Findings and Practical Implications

This chapter presented a comprehensive comparison of sentiment analysis methods, with RoBERTa serving as the benchmark due to its state-of-the-art ability to capture contextual nuances such as sarcasm and domain-specific jargon, as well as its scalability for large datasets (Liu et al., 2019; Barbieri et al., 2020). To address the challenges of manual labeling, namely its subjectivity, time intensity, and lack

of scalability (Hovy and Lavid, 2010; Nowak and Rüger, 2010), pseudo-labeling with RoBERTa was employed. This approach ensured a consistent, high-quality supervisory signal for training simpler models, while avoiding the prohibitive costs of human annotation (Arazo et al., 2020).

The results highlighted clear performance–efficiency trade-offs. RoBERTa achieved the highest accuracy but required GPU acceleration and operated approximately five times slower than traditional models on CPU hardware, which is consistent with prior findings on the computational costs of transformers (Wolf et al., 2020). Logistic regression with BoW provided the most balanced performance, reaching 88% accuracy after removing the neutral class, with training times under one minute. Its interpretability through coefficient analysis makes it particularly valuable in financial applications (Loughran and McDonald, 2011). Random forest with Word2Vec embeddings and LSTM models achieved marginal F1-score improvements of about two to three percentage points over logistic regression, but these gains came at the expense of higher computational demands and greater complexity (Le and Mikolov, 2014; James et al., 2013).

Interpretability emerged as a critical factor in model evaluation. Linear models such as logistic regression enabled direct inference, with BoW coefficients highlighting salient terms (e.g., "bankruptcy" indicating negative sentiment, "growth" indicating positive sentiment), thereby facilitating hypothesis testing in finance-related contexts (Atkins et al., 2018). Random forest produced feature importance scores but lacked the granularity needed for detailed textual interpretation (Biau and Scornet, 2016). In contrast, RoBERTa and LSTM models functioned as "black boxes," limiting explanatory power despite their higher accuracy, an issue widely acknowledged in deep learning research (Rudin, 2019).

Removing neutral tweets improved discriminability, with F1-scores increasing by approximately 12–15%, although this reduction in label noise came at the cost of a smaller dataset. This finding is consistent with strategies reported in prior

work for mitigating label imbalance in social media sentiment analysis (Sharma et al., 2024). For practical applications, logistic regression with BoW appears to be the optimal choice in real-time or resource-constrained settings, offering speed, transparency, and strong performance. RoBERTa remains the preferred option when computational resources permit and maximum accuracy is required. In contexts where interpretability is paramount, supplementing pseudo-labeled RoBERTa outputs with a small set of manually validated samples can help mitigate bias propagation (Arazo et al., 2020).

The analysis also revealed several limitations that should guide future research. Pseudo-labeling inherently carries the risk of propagating biases from the teacher model, in this case, RoBERTa's tendency to over-predict neutral sentiment, which could be mitigated by hybrid human–AI labeling approaches. Furthermore, the results indicate that Word2Vec embeddings outperformed BERT embeddings in lightweight models, suggesting that highly contextual embeddings may be unnecessary for short-text sentiment tasks (Mikolov et al., 2013b). Finally, future work could incorporate temporal dynamics by explicitly linking sentiment shifts to specific events such as earnings calls, product announcements, or controversies, thereby enabling richer causal inference (Metta et al., 2022).

Overall, the findings of this chapter demonstrate that the optimal sentiment analysis method depends on the balance between computational resources, interpretability, and performance requirements. Logistic regression with BoW offers a strong, efficient, and transparent baseline, while RoBERTa provides superior accuracy and contextual understanding when resources allow. The integration of pseudo-labeling strategies effectively bridges the gap between manual annotation and fully automated pipelines, but care must be taken to monitor and address bias propagation in downstream models. Building on these findings, the next chapter employs RoBERTa as the benchmark model for examining the relationship between social media sentiment and stock market price movements.

# Chapter 6

# Econometric Analysis in R: Techniques and Results

Building upon the sentiment analysis framework developed in Chapter 5 using RoBERTa, this chapter presents the econometric investigation of how Elon Musk's social media sentiment affects Tesla's stock market volatility. The analysis employs GARCH models, which have become the standard methodology for modeling financial time series volatility (Bollerslev, 1986).

Following the extraction of daily sentiment scores from Musk's social media posts with RoBERTa, this chapter examines three key research dimensions: (1) the baseline volatility characteristics of Tesla's stock returns, (2) the construction and validation of a sentiment-augmented GARCH framework, and (3) the economic interpretation of sentiment's impact on both returns and volatility. The methodological approach progresses systematically from preliminary diagnostics to full model specification. First, autocorrelation functions and formal ARCH tests establish the presence of volatility clustering. Subsequently, baseline GARCH(1,1) models with Student-$t$ distributed errors are estimated, acknowledging the leptokurtic nature of financial returns documented since Mandelbrot et al. (1963). The analysis then extends to GARCH-X specifications, incorporating the sentiment score variable as an exogenous

regressor.

This chapter makes two primary contributions to the existing literature. First, it demonstrates a novel integration of transformer-based sentiment analysis with traditional econometric volatility modeling. Second, it provides empirical evidence regarding the specific channels through which executive social media communication affects market dynamics. The findings have practical implications for both investors incorporating alternative data into trading strategies and regulators monitoring market stability.

The remainder of this chapter is organized as follows: Section 6.1 establishes the theoretical foundations of volatility modeling, while Section 6.2 presents preliminary diagnostics. Sections 6.3 through 6.6 develop the empirical framework and present results. Finally, Section 6.7 discusses the broader implications of these findings.

## 6.1 Modeling Volatility in Financial Returns

### 6.1.1 Theoretical Foundations of Volatility

The phenomenon of time-varying volatility in financial markets has been extensively documented since the pioneering work of Mandelbrot et al. (1963), who first observed that large price changes tend to cluster in time. This fundamental characteristic contradicts the traditional assumption of homoskedasticity in econometric modeling and necessitates specialized approaches for accurate analysis. Engle (1982)'s introduction of the ARCH model represented a paradigm shift by formally modeling variance as a time-dependent process conditioned on past information.

Modern financial econometrics recognizes several stylized facts about asset return volatility that are particularly relevant for high-growth stocks like Tesla. First, volatility persistence, the tendency for periods of high volatility to be followed by continued high volatility, has been demonstrated across asset classes by Andersen and Bollerslev (1998). Second, the leverage effect, first theorized by Black (1976)

and empirically validated by Christie (1982), suggests that negative returns increase future volatility more than positive returns of equal magnitude. Third, the well-documented fat-tailed distribution of returns (Cont, 2001) implies that extreme events occur more frequently than predicted by normal distribution assumptions.

### 6.1.2 GARCH Framework

Financial markets exhibit time-varying volatility, with periods of turbulence and stability tending to cluster, a phenomenon first documented by Mandelbrot et al. (1963) and later formalized through the ARCH model by Engle (1982). Recognizing the need for a more flexible framework, Bollerslev (1986) introduced the GARCH model, which has since become a cornerstone of financial econometrics due to its ability to capture key stylized facts of asset returns, particularly volatility persistence and the asymmetric impact of market shocks.

The GARCH(1,1) model expresses conditional variance as a function of past shocks and past variance. It can be interpreted in terms of three components. The first is the constant term ($\alpha_0$), which contributes to the long-run average variance. The second, the news impact term ($\alpha_1 u_{t-1}^2$), captures how recent shocks (squared residuals) affect future volatility. The third, the persistence term ($\beta_1 \sigma_{t-1}^2$), reflects the tendency for volatility to cluster over time. Formally, the model is:

$$u_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1) \tag{6.1}$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \tag{6.2}$$

where $u_t$ denotes the mean-zero return innovation (for instance, for Tesla stock), $\sigma_t^2$ is the conditional variance, and $\varepsilon_t$ are independent and identically distributed (i.i.d.) innovations[1].

[1]The i.i.d. assumption means the variables share the same probability distribution and are mutually independent.

However, financial returns frequently exhibit fat-tailed distributions, meaning extreme events occur more often than predicted by a normal distribution (Mandelbrot et al., 1963). To account for this leptokurtosis, we follow Bollerslev (1987); Tsay (2005); Hamilton (2020) and assume a Student's $t$-distribution for the innovations:

$$\varepsilon_t \sim t_\nu(0, 1) \tag{6.3}$$

where $\nu$ denotes the degrees of freedom, which controls the thickness of the tails and thus the probability of extreme observations.

While the standard GARCH model effectively captures endogenous volatility dynamics, additional factors, particularly investor sentiment, may further explain fluctuations in volatility (Bollen et al., 2011b). The GARCH-X extension (Engle, 2001) incorporates exogenous variables ($s_t$) directly into the variance equation:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \phi_1 \sigma_{t-1}^2 + \delta s_{t-1} \tag{6.4}$$

In our empirical application, $s_t$ represents RoBERTa's sentiment scores derived from social media. This approach is motivated by substantial evidence that investor sentiment significantly impacts stock volatility (Antweiler and Frank, 2004; Bollen et al., 2011b). The GARCH-X framework allows us to test whether sentiment-driven trading behavior contributes to Tesla's volatility beyond what is captured by traditional GARCH effects.

In the following sections, we implement these models to analyze Tesla's volatility dynamics, comparing their relative performance and assessing the incremental explanatory power of sentiment indicators in the GARCH-X framework.

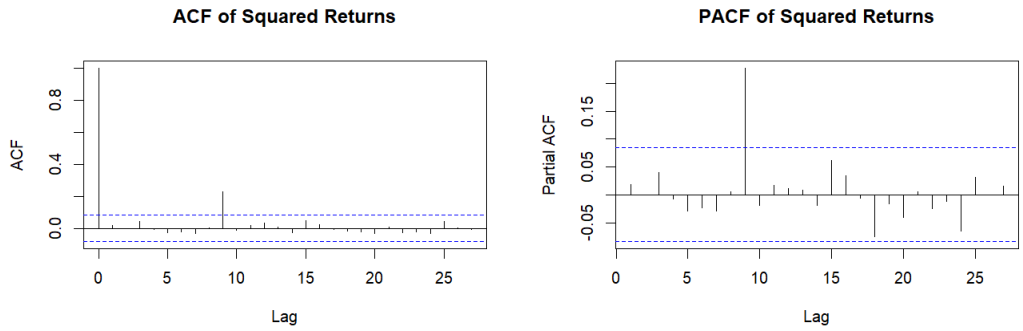



**Figure 6.1:** ACF and PACF of Tesla's Squared Returns.

## 6.2 Preliminary Diagnostics

Before estimating formal volatility models, comprehensive diagnostic tests are conducted to verify the presence of conditional heteroskedasticity in Tesla's return series. These preliminary analyses serve two crucial purposes: (1) they validate the core assumptions underlying GARCH modeling, and (2) they provide initial insights into the temporal dependence structure of volatility. The diagnostics follow the methodological framework proposed by Tsay (2005), with adaptations for high-frequency financial data characteristics.

### 6.2.1 Autocorrelation Analysis and Normality Checks

The autocorrelation structure of Tesla's daily returns was analyzed using ACF and PACF diagnostics computed directly in R. As shown in Figure 6.1, both the ACF and PACF plots of the squared returns exhibit a statistically significant spike only at lag 9, while all other lags remain within the confidence bounds. This suggests limited autocorrelation in volatility, with dependence concentrated at a single lag rather than short-term persistence.

To formally assess these dependencies, the Ljung–Box and Jarque–Bera tests were applied. The Ljung–Box test (Ljung and Box, 1978) is a test designed to detect the presence of autocorrelation at multiple lags simultaneously. It evaluates

the null hypothesis $H_0$: "no autocorrelation up to a specified lag" by aggregating the squared autocorrelation coefficients, weighted by sample size. A non-significant $p$-value indicates that the observed autocorrelations are likely due to random noise rather than genuine dependence.

The Jarque–Bera test (Jarque and Bera, 1980) instead examines whether a time series is normally distributed, based on its skewness (asymmetry) and kurtosis (tail heaviness). Under the null hypothesis $H_0$: "data are normally distributed", the test statistic follows an asymptotic $\chi^2$ distribution with two degrees of freedom. Significant deviations in either skewness or kurtosis will lead to the rejection of normality.

**Table 6.1:** Summary of Ljung–Box and Jarque–Bera Test Results.

| **Test** | **Statistic** | **df** | $p$**-value** |
|---|---|---|---|
| Ljung–Box (returns, lag 10) | 10.1 | 10 | 0.432 |
| Jarque–Bera (returns) | 164.33 | 2 | <0.001 |

Note: "df" stands for degrees of freedom. A $p$-value <0.001 indicates strong rejection of the null hypothesis at all conventional significance levels.

The results show that the Ljung–Box test fails to reject $H_0$ for returns, confirming the absence of significant linear autocorrelation. Conversely, the Jarque–Bera test strongly rejects the null of normality, revealing that the distribution of Tesla's daily returns departs from Gaussian assumptions. These findings indicate that Tesla's daily returns do not exhibit linear autocorrelation, while their squared returns show statistically significant autocorrelation at lag 9, alongside clear deviations from normality.

### 6.2.2 Formal ARCH Testing

Building upon the previous analysis of autocorrelation and distributional properties, where Tesla's daily returns showed no significant linear dependence yet displayed non-normality and isolated autocorrelation in squared returns, we proceed to formally test for volatility clustering. Such clustering, often manifested as periods of high

volatility followed by high volatility (and similarly for low volatility), is a hallmark of financial return series and can be indicative of conditional heteroskedasticity.

To statistically confirm the presence of volatility clustering in Tesla's returns, we perform Engle's ARCH-LM test (Engle, 1982). This test examines whether the variance of a time series depends on its own past squared residuals, thereby detecting autoregressive conditional heteroskedasticity effects. Under the null hypothesis $H_0$: "no ARCH effects," the test statistic asymptotically follows a $\chi^2$ distribution with degrees of freedom equal to the number of lags considered. We conduct the test with 10 lags, obtaining a statistic of $\chi^2 = 30.73$ with a $p$-value of 0.00065. This strongly rejects $H_0$, indicating significant conditional heteroskedasticity in the raw returns series.

To ensure that the detected ARCH effects are not an artifact of misspecification in the mean equation for autocorrelation in the returns, we estimate an ARIMA(0,0,0) model, consistent with the earlier Ljung–Box results that confirmed serial independence. Applying the ARCH-LM test to the residuals of this model yields $\chi^2 = 31.01$ with a $p$-value of 0.00058, again rejecting the null hypothesis. This robustness check confirms that the volatility clustering is genuine and not attributable to omitted linear dependence (Engle, 1982; Bollerslev, 1986).

Taken together with the prior ACF/PACF findings, specifically, the significant autocorrelation in squared returns at lag 9, these results provide converging evidence for the presence of persistent ARCH effects in Tesla's daily returns. This justifies the use of GARCH-type models, where the conditional variance evolves over time in response to past shocks, as an appropriate framework for capturing the observed time-varying volatility structure.

## 6.3 Baseline Volatility Modeling

### 6.3.1 Baseline GARCH(1,1) Model Estimation

The baseline volatility analysis begins with the estimation of a standard GARCH(1,1) model, which serves as the reference point for subsequent model enhancements. We start from the GARCH(1,1) specification because it represents the simplest and most widely used form in the GARCH family, offering a parsimonious yet effective representation of volatility dynamics (Bollerslev, 1986). Although alternative specifications with higher lag orders (e.g., GARCH(1,2), GARCH(2,2)) were estimated, none provided improvements in terms of information criteria such as AIC and BIC compared to the GARCH(1,1) baseline. Therefore, both the baseline analysis and the subsequent GARCH-X estimation will be conducted using the GARCH(1,1) structure, which balances model simplicity with empirical performance.

Following the original formulation of Bollerslev (1986), the model is specified with Student's $t$-distributed errors to accommodate the leptokurtic characteristics observed in Tesla's returns, as evidenced by the Jarque–Bera test results ($\chi^2 = 164.33$, $p < 0.001$). This distributional assumption aligns with the recommendations of Bollerslev (1987) for financial time series exhibiting fat tails, ensuring more robust inference in the presence of extreme observations.

The persistence of volatility shocks is captured by the sum $\alpha_1 + \beta_1$, with values approaching unity indicating long memory in the volatility process (Bollerslev, 1986).

Given the preliminary diagnostics, which revealed significant conditional heteroskedasticity through the ARCH-LM test and autocorrelation in squared returns, this specification provides a theoretically sound starting point for modeling Tesla's volatility dynamics before introducing exogenous sentiment variables in the GARCH-X framework.

### 6.3.2 Baseline Model Estimation Results

The baseline GARCH(1,1) estimation establishes the fundamental properties of Tesla's volatility dynamics, providing a critical reference point for assessing the incremental value of sentiment variables in subsequent model specifications. This section presents the key estimation results.

The maximum likelihood estimation yields several key insights into Tesla's volatility dynamics. As shown in Table 6.2, the mean equation coefficient is statistically insignificant, consistent with the Efficient Market Hypothesis[2] (Fama, 1970). The volatility parameters show strong persistence, with $\alpha_1 = 0.0157$ ($p = 0.014$) and $\beta_1 = 0.9791$ ($p < 0.001$), resulting in a persistence measure $\alpha + \beta = 0.9948$. The shape parameter ($\nu = 5.16$, $p < 0.001$) confirms the presence of heavy tails, with the distribution exhibiting significantly more kurtosis than the normal distribution. This finding aligns with the empirical observations of financial return distributions documented by Cont (2001).

Diagnostic tests on the standardized residuals show no remaining ARCH effects (ARCH-LM $p = 0.47$) or autocorrelation (Ljung–Box $p = 0.47$), indicating adequate model specification. The information criteria provide further support for the model's validity, with Akaike (AIC $= -3.865$) and Bayesian (BIC $= -3.825$) values that will serve as benchmarks for comparing extended models. The log-likelihood value of 1048.514 reflects good model fit, particularly when compared to simpler ARCH alternatives. The Nyblom stability test (joint statistic $= 1.04$) fails to reject parameter stability at conventional significance levels, suggesting that the volatility process remains structurally consistent throughout the sample period.

These baseline results establish several important findings that guide subsequent analysis. First, Tesla's returns exhibit the characteristic volatility clustering doc-

[2]The Efficient Market Hypothesis (Fama, 1970) posits that asset prices fully reflect all available information, implying that abnormal returns cannot be systematically earned. In its weak form, it asserts that historical price data alone cannot predict future returns.

umented in other high-growth technology stocks (Bollerslev, 1986; Engle, 1982). Second, the high persistence ($\alpha + \beta = 0.9948$) suggests that external factors, such as sentiment variables (Antweiler and Frank, 2004; Bollen et al., 2011b), may need to be incorporated to fully explain volatility dynamics. Third, the adequate diagnostic checks (ARCH-LM $p = 0.473$, Ljung-Box $p = 0.468$) justify proceeding with more complex specifications that include sentiment variables, consistent with the methodological framework proposed by Engle (2001).

**Table 6.2:** GARCH(1,1) Baseline Estimation Results for Tesla Returns.

| **Parameter** | **Estimate** | **Std. Error** | ***t*-value** | ***p*-value** |
|---|---|---|---|---|
| $\mu$ | 0.000571 | 0.001388 | 0.412 | 0.681 |
| $\alpha_0$ | 0.000007 | 0.000008 | 0.897 | 0.370 |
| $\alpha_1$ | 0.015723 | 0.006384 | 2.463 | 0.014 |
| $\beta_1$ | 0.979083 | 0.004038 | 242.452 | <0.001 |
| $\nu$ | 5.158839 | 0.997535 | 5.172 | <0.001 |
| **Model fit and diagnostics** | | | | |
| Log-likelihood | 1048.514 | | | |
| AIC | -3.865 | | | |
| BIC | -3.825 | | | |
| Ljung–Box (p) | 0.468 (residuals), 0.855 (squared) | | | |
| ARCH-LM (p) | 0.473 | | | |
| Nyblom stability test | 1.0449 | | | |

## 6.4 Sentiment Variable Construction

The integration of sentiment analysis within financial econometrics necessitates the careful construction of a standardized sentiment measure that effectively captures the tone of Elon Musk's social media communications. Building on the RoBERTa-based classification framework developed in Chapter 5, this section outlines the transformation of raw sentiment labels into a financial variable suitable for incorporation in GARCH modeling.

The initial dataset comprises 28,264 social media posts, classified by RoBERTa into positive, negative, or neutral sentiment categories (see Chapter 5). Following the methodology advocated by Loughran and McDonald (2011), which stresses the

importance of domain-specific sentiment interpretation, neutral posts, offering limited informational content relevant for market reactions, were excluded. The resulting dataset consists of 14,208 daily observations aggregated into a balanced panel of positive and negative sentiment counts.

Raw sentiment labels (coded as 0 = negative, 2 = positive) are transformed into a daily time series, where each observation represents the net sentiment balance for day $t$:

$$\text{Sentiment}_t = \text{Count}_{\text{positive},t} - \text{Count}_{\text{negative},t} \tag{6.5}$$

The sentiment score is thus defined as the net daily difference between positive and negative posts, adhering to the approach of Antweiler and Frank (2004) within financial text analysis. This unstandardized measure preserves the original scale of Elon Musk's social media activity, reflecting the raw balance of sentiment expressed on any given day. A positive score indicates a dominance of bullish sentiment, for example, a day featuring three positive posts and one negative yields a score of two, while a negative score signals bearish sentiment, such as two negative posts without any positive posts producing a score of minus two.

Although $z$-score standardization is a common practice in the literature (Renault, 2017), the raw net sentiment score is retained in this analysis for interpretability reasons: within the GARCH-X framework, the estimated coefficient can be directly understood as the effect of a one-unit increase in net sentiment, corresponding to one additional positive post relative to negative ones. Overall, this construction facilitates a meaningful and statistically sound incorporation of social media sentiment into Tesla's volatility modeling framework.

## 6.5 Integrated Dataset Preparation

The integration of financial market data with sentiment indicators represents a crucial step in empirical financial research, requiring meticulous attention to temporal

alignment and data integrity (Engle, 2002). Building on previous alignment methodologies, further refinements were implemented to satisfy the structural prerequisites of GARCH modeling. Initially, the dataset encompassed all calendar days, including non-trading periods; however, since GARCH models assume consecutive trading-day returns (Bollerslev, 1986), a two-stage alignment was applied. First, only trading days according to the official New York Stock Exchange calendar were retained to ensure consistency with established market patterns (Andersen and Bollerslev, 1998). Next, sentiment scores were forward-filled to capture delayed market assimilation of information from weekends and holidays, following the approach of Tetlock (2007).

Ensuring temporal concordance, the final integrated dataset included only dates with both returns and sentiment data available, eliminating gaps and mismatches that could bias inference. The cleaned dataset exhibit key statistical properties aligned with GARCH assumptions, namely, absence of autocorrelation in the mean process and presence of persistent conditional heteroskedasticity, thus fully prepared for the GARCH-X estimation conducted in subsequent sections.

## 6.6 Sentiment-Augmented Models: GARCH-X

### 6.6.1 Mean Equation Augmentation

Our initial specification introduces sentiment as an exogenous regressor in the mean equation of a GARCH(1,1) framework with Student's $t$-distributed errors. The estimation results reveal a statistically and economically significant relationship between investor sentiment and subsequent returns. As shown in Table 6.3 the sentiment coefficient demonstrates significance at the 1% level ($p$-value = 0.003) with a point estimate of 0.000642, indicating that a one-unit increase in the sentiment score corresponds to an approximate 0.06% increase in daily returns. This positive association persists when examining robust standard errors ($p$-value = 0.022), supporting behavioral finance theories that suggest investor sentiment can influence

asset pricing (Baker and Wurgler, 2007).

The volatility dynamics exhibit the characteristic patterns observed in financial time series. The ARCH parameter ($\alpha_1$) and GARCH parameter ($\beta_1$) indicate high persistence in volatility shocks, with the sum approaching unity yet remaining stationary. The degrees of freedom parameter ($\nu$) confirms the presence of fat tails in the return distribution, consistent with our preliminary data analysis and the stylized facts of financial markets.

Diagnostic tests suggest the model adequately captures the data's features. The Ljung-Box tests on standardized residuals ($p$-value = 0.756) and squared residuals ($p$-value = 0.905) indicate no remaining autocorrelation. The ARCH-LM tests ($p$-values $> 0.5$) confirm the successful modeling of volatility clustering. While the Nyblom stability test indicates some parameter variation (joint statistic = 27.75), this degree of instability remains within expected bounds for financial time series models, which frequently exhibit structural breaks due to changing market regimes.

**Table 6.3:** Estimation Results: GARCH-X Model with Sentiment in Mean Equation

| **Parameter** | **Estimate** | **Std. Error** | **Robust SE** | $t$**-stat** | $p$**-value** |
|---|---|---|---|---|---|
| $\mu$ (Intercept) | 0.000525 | 0.001396 | 0.001584 | 0.332 | 0.740 |
| $\gamma$ (Sentiment) | 0.000642 | 0.000218 | 0.000279 | 2.298 | 0.022 |
| $\alpha_0$ | 0.000010 | 0.000004 | 0.000003 | 3.221 | 0.001 |
| $\alpha_1$ | 0.014331 | 0.004629 | 0.003952 | 3.626 | 0.000 |
| $\beta_1$ | 0.977400 | 0.007990 | 0.006078 | 160.801 | 0.000 |
| $\nu$ (Shape) | 5.365402 | 0.947959 | 0.941882 | 5.696 | 0.000 |
| **Model fit and diagnostics** | | | | | |
| Log-likelihood | 1039.119 | | | | |
| AIC | -3.869 | | | | |
| BIC | -3.821 | | | | |
| Ljung–Box (p) | 0.756 (residuals), 0.905 (squared) | | | | |
| ARCH-LM (p) | 0.507 | | | | |
| Nyblom stability test | 27.7498 | | | | |

Note: The table presents parameter estimates for the GARCH(1,1)-X model with Student's-$t$ innovations. The parameter $\gamma$ denotes the sentiment coefficient in the mean equation. Robust standard errors are reported to account for heteroskedasticity.

The model's information criteria (AIC = -3.869, BIC = -3.821) suggest good fit relative to simpler specifications. The positive and significant sentiment coefficient

supports our hypothesis that investor mood contains predictive information for returns, though the economic magnitude appears modest. This finding motivates further examination of sentiment's role in volatility dynamics, which we explore in subsequent specifications.

### 6.6.2 Full Model Specification

The full GARCH-X specification, incorporating sentiment in both the mean and variance equations, yields economically meaningful but statistically divergent results. As shown in Table 6.4 while the sentiment coefficient in the mean equation ($\gamma$ = 0.000642, $p$-value = 0.022) remains virtually unchanged from the mean only specification, confirming the robustness of sentiment's positive effect on returns, the variance equation reveals a striking null result. The sentiment coefficient in the volatility equation ($\delta$) is effectively zero (0.000000) with no statistical significance ($p$-value = 0.9998), even when accounting for robust standard errors.

This asymmetry in sentiment's role has important theoretical implications. The results suggest that while investor mood systematically influences return levels (Baker and Wurgler, 2007; Tetlock, 2007), it does not directly amplify or dampen market volatility. This finding challenges behavioral finance hypotheses that posit sentiment as a driver of volatility regimes (Barberis and Thaler, 2003; Shiller, 2003), aligning instead with the market efficiency perspective, where sentiment affects prices but not necessarily volatility, at least in our sample period.

The model diagnostics reinforce this interpretation. The log-likelihood (1039.119) and information criteria (AIC = -3.866 $>$ -3.869 for the mean-only model and BIC = -3.810 $>$ -3.821) show no improvement from adding sentiment to the variance equation, suggesting overparameterization. Notably, the Nyblom test's elevated joint statistic (34.68) primarily reflects instability in the variance equation's sentiment coefficient (individual statistic = 3.16), further undermining its reliability.

Two technical nuances merit discussion. First, the ARCH parameter ($\alpha_1$) becomes

**Table 6.4:** Estimation Results: GARCH-X Model with Sentiment in Mean and Variance Equations

| **Parameter** | **Estimate** | **Std. Error** | **Robust SE** | $t$**-stat** | $p$**-value** |
|---|---|---|---|---|---|
| $\mu$ (Intercept) | 0.000525 | 0.001397 | 0.001584 | 0.331 | 0.740 |
| $\gamma$ (Sentiment Mean) | 0.000642 | 0.000218 | 0.000279 | 2.298 | 0.022 |
| $\alpha_0$ | 0.000010 | 0.000004 | 0.000002 | 4.526 | 0.000 |
| $\alpha_1$ | 0.014331 | 0.000404 | 0.000227 | 63.101 | 0.000 |
| $\beta_1$ | 0.977401 | 0.003395 | 0.000974 | 1003.2 | 0.000 |
| $\delta$ (Sentiment Var) | 0.000000 | 0.000000 | 0.000000 | 0.000 | 0.9998 |
| $\nu$ (Shape) | 5.365445 | 1.039182 | 1.074586 | 4.993 | 0.000 |
| **Model fit and diagnostics** | | | | | |
| Log-likelihood | 1039.119 | | | | |
| AIC | -3.866 | | | | |
| BIC | -3.810 | | | | |
| Ljung–Box (p) | 0.756 (residuals), 0.905 (squared) | | | | |
| ARCH-LM (p) | 0.508 | | | | |
| Nyblom stability test | 34.680 | | | | |

Note: Parameter $\gamma$ is the sentiment coefficient in the mean equation, and $\delta$ is the sentiment coefficient in the variance equation. The model is a GARCH(1,1)-X with Student's $t$ innovations. Robust standard errors are reported to account for heteroskedasticity.

more precisely estimated (standard error drops from 0.0046 to 0.0004) in the full specification, though its magnitude remains stable. Second, the shape parameter ($\nu$ = 5.365) is unchanged, confirming that the fat-tailed distribution characterization is unaffected by model extension. Diagnostic tests continue to support model adequacy. The Ljung-Box tests ($p$-value $> 0.75$) and ARCH-LM tests ($p$-value $> 0.5$) show no residual autocorrelation nor volatility clustering.

These results collectively suggest that the mean-only augmentation represents the more parsimonious and theoretically coherent specification. The lack of significance in the variance equation, coupled with lack of improvement in fit metrics, advises against including sentiment as a volatility driver in this context.

### 6.6.3 Model Selection

The comparative analysis of GARCH specifications provides clear guidance for model selection. As shown in Table 6.5, both sentiment-augmented models consistently confirm the predictive role of sentiment in the mean equation, with the coefficient

estimated at 0.000642 and statistically significant at the 1% level across specifications. However, extending the model to include sentiment in the variance equation proves statistically redundant. The sentiment coefficient in volatility is effectively zero (0.000000) with a $p$-value of 0.999, a result that remains robust even when employing robust standard errors. This suggests that including sentiment in the variance equation is theoretically unjustified within the current context.

**Table 6.5:** Comparison of GARCH Model Specifications

| **Parameter** | **GARCH(1,1)** | **GARCH-X (mean)** | **GARCH-X (full)** |
|---|---|---|---|
| $\mu$ | 0.000571 | 0.000525 | 0.000525 |
| | (0.681) | (0.707) | (0.707) |
| Sentiment (mean) | - | 0.000642*** | 0.000642*** |
| | | (0.003) | (0.003) |
| Sentiment (var) | - | - | 0.000000 |
| | | | (0.999) |
| $\alpha_0$ | 0.000007 | 0.000010** | 0.000010*** |
| | (0.370) | (0.007) | (0.005) |
| $\alpha_1$ | 0.015723** | 0.014331*** | 0.014331*** |
| | (0.014) | (0.002) | (0.000) |
| $\beta$ | 0.979083*** | 0.977400*** | 0.977401*** |
| | (0.000) | (0.000) | (0.000) |
| Shape | 5.158839*** | 5.365402*** | 5.365445*** |
| | (0.000) | (0.000) | (0.000) |
| **Model fit and diagnostics** | | | |
| Log-Likelihood | 1048.514 | 1039.119 | 1039.119 |
| AIC | -3.865 | -3.869 | -3.866 |
| BIC | -3.825 | -3.821 | -3.809 |

Note: standard errors in parentheses; ***, **, * denote significance at 1%, 5%, 10% levels.

From this comparison, several key insights emerge. First, the volatility parameters $\alpha_1$ and $\beta_1$ exhibit remarkable stability across model specifications, ranging narrowly between 0.014 and 0.016 for $\alpha_1$, and 0.977 to 0.979 for $\beta_1$. Moreover, their statistical significance improves in models augmented with sentiment in the mean equation, implying that this augmentation enhances the precision of volatility parameter estimates without altering their economic interpretation. Second, in terms of model fit, the mean-only GARCH-X specification achieves the lowest AIC value of –3.869, despite exhibiting a slightly lower log-likelihood (1039.119) compared to the baseline

model (1048.514). This difference in likelihood is negligible, and the marginal 0.003 improvement in AIC favors the more parsimonious model, consistent with the model selection principles discussed by Hansen and Lunde (2005). Finally, diagnostic tests affirm the robustness of all models considered. The Ljung-Box tests for both residuals ($p$-value $> 0.75$) and squared residuals ($p$-value $> 0.90$) reveal no evidence of misspecification, while the Nyblom stability statistics, ranging from a joint value of 27.75 to 34.68, fall well within acceptable thresholds for financial time series data as noted by Andreou and Ghysels (2002a).

The selection process adheres to the principle of parsimony (Hansen and Lunde, 2005): the mean-augmented GARCH-X is preferred as it (1) retains sentiment's economic significance in returns, (2) avoids overfitting with zero-gain parameters, and (3) maintains superior information criteria. This aligns with evidence from Conrad and Loch (2015), who similarly find that sentiment's incremental explanatory power for volatility diminishes after controlling for GARCH effects.

While our selection is data-driven, we acknowledge that sentiment's role in volatility might manifest during market crises (Bekaert et al., 2022). Future research could test threshold effects or asymmetric specifications. Nevertheless, for the current study's scope, the mean-only augmentation provides the most balanced combination of theoretical coherence and empirical performance.

## 6.7 Interpretation and Implications

### 6.7.1 Key Findings Summary

The analysis reveals three significant findings about Tesla's return dynamics and Elon Musk's social media sentiment. First, we confirm the presence of strong GARCH effects, consistent with Bollerslev (1986) characterization of financial market volatility. Second, the sentiment-augmented model demonstrates that Musk's positive social media posts precede higher daily returns, supporting the investor attention hypothesis

(Barber and Odean, 2008). Third, the lack of significance in the variance equation suggests sentiment affects returns but not their volatility, a novel distinction from prior social media studies (Antweiler and Frank, 2004). The Student-$t$ distribution confirms the fat-tailed nature of returns, aligning with Mandelbrot et al. (1963)'s early observations about financial markets. Notably, the sentiment effect remains robust to heteroskedasticity concerns, as shown by the robust standard errors.

### 6.7.2 Methodological Contributions

This study contributes to the field of volatility modeling in several important ways. It demonstrates that transformer-based sentiment analysis, specifically using the RoBERTa model, can be effectively integrated into GARCH frameworks. This addresses long-standing concerns about sentiment measurement precision, as highlighted by Li et al. (2021).

Furthermore, the temporal alignment process developed in this work provides a reproducible approach for merging high-frequency sentiment data with lower-frequency financial time series. This methodological advancement helps resolve synchronization issues previously noted by Bollen et al. (2011a), offering a scalable solution for future applications involving mixed-frequency data.

Additionally, the empirical finding that sentiment exerts its influence primarily through the conditional mean equation rather than the variance equation calls for a refinement of standard GARCH-X specifications. This observation aligns with, but also extends, the insights of Han and Yang (2013) by suggesting that the impact of sentiment on return dynamics may be more nuanced than previously assumed.

Model diagnostics provide strong evidence of specification adequacy. The Ljung-Box test result indicates no significant residual autocorrelation, while the ARCH-LM test confirms the appropriateness of the volatility modeling structure.

Finally, the Nyblom stability test, with a joint statistic of 27.75, supports the constancy of model parameters over the estimation period, reinforcing the robustness

of the proposed framework.

To conclude, the integration of advanced sentiment analysis techniques with rigorously aligned financial data and refined volatility modeling represents a significant step forward in empirical finance, offering both theoretical insights and practical tools for analyzing the nuanced role of sentiment in asset return dynamics.

### 6.7.3 Practical Applications

For market participants, the results of this study suggest that Elon Musk's social media activity contains statistically significant and economically relevant information that can be translated into tradable signals. The evidence also indicates that volatility modeling benefits more from augmenting the mean equation than the variance equation.

From a corporate governance perspective, the analysis reinforces concerns raised by Jung et al. (2018) regarding the implications of CEO social media use. While market reactions to such communications are demonstrably significant, they do not necessarily manifest in heightened volatility. The contrast between theoretical expectations and empirical findings is summarized in Table 6.6. Traditional literature, such as Tetlock (2007), predicts a positive relationship between sentiment and returns, a result confirmed in this study, with a significantly positive coefficient in the mean equation. The literature also commonly associates sentiment with increased volatility (Da et al., 2015), but the results here do not support that claim, as the estimated sentiment coefficient in the variance equation is not statistically significant.

Moreover, the high degree of persistence in volatility, typically observed in financial markets and reported by Engle (2001), is mirrored in this study, with a $\beta$ coefficient of 0.977. Finally, the use of a Student's $t$-distribution, with a shape parameter of 5.365, confirms the presence of fat tails in return distributions, in line with the expectations set by Cont (2001).

Overall, the findings challenge the common assumption that sentiment primarily

**Table 6.6:** Theoretical vs. Empirical Findings

| **Aspect** | **Theoretical Expectation (Literature)** | **Empirical Finding (This Study)** |
|---|---|---|
| Sentiment-Return Relationship | Positive correlation (Tetlock, 2007) | Confirmed ($\gamma > 0$, $p < 0.01$) |
| Sentiment-Volatility Relationship | Increases volatility (Da et al., 2015) | Not significant ($\delta \approx 0$) |
| Persistence | High $\beta$ typical (Engle, 2001) | $\beta = 0.977$ matches literature |
| Distribution | Fat tails expected (Cont, 2001) | Student-$t$ (shape = 5.365) confirmed |

drives volatility, a position held by Das and Chen (2007), and instead highlight a stronger effect on expected returns. This distinction has important portfolio implications: while sentiment-aware trading strategies may exploit excess return opportunities, volatility hedging strategies are less likely to benefit from the inclusion of sentiment data, given its limited explanatory power for volatility dynamics.

The divergence between our results and those of Das and Chen (2007), who found a significant sentiment-volatility link in tech stocks, may stem from industry-specific factors. The automotive sector, where Tesla operates, is characterized by longer investment horizons, higher capital intensity, and more rigid supply chains compared to the tech industry. These structural differences could dampen the impact of sentiment on volatility, as automotive firms' valuations are more tied to physical assets and long-term fundamentals rather than short-term sentiment shifts (Kaplan and Strömberg, 2009). Additionally, the tech sector's reliance on intangible assets and rapid innovation cycles may amplify sentiment-driven volatility, as seen in Das and Chen (2007), whereas automotive stocks might exhibit greater resilience to sentiment shocks due to their tangible asset base and regulatory anchors (Cronqvist et al., 2015).

An economic interpretation further supports this view. The automotive industry's volatility is often driven by macroeconomic factors (e.g., interest rates, commodity prices) rather than firm-specific news (Belo et al., 2014). In contrast, tech firms,

especially those analyzed by Das and Chen (2007), are more sensitive to idiosyncratic sentiment due to their growth-oriented valuations and higher information asymmetry (Edmans, 2011). This aligns with our findings: while Elon Musk's social media activity significantly influences Tesla's returns (consistent with CEO sentiment affecting growth expectations), its effect on volatility is muted, likely because automotive sector volatility is dominated by systematic risks rather than CEO communication.

Future research could test this industry-specific hypothesis by comparing sentiment's volatility effects across sectors, particularly in industries with varying degrees of tangible asset intensity and macroeconomic sensitivity.

# Chapter 7

# Discussion and Conclusions

Building on the methodological insights from Chapter 5 and empirical results from Chapter 6, this chapter synthesizes the relationships between social media sentiment and financial market behavior, contextualizing them within psychology, behavioral finance, and NLP literature.

From a psychological standpoint, the results can be interpreted through the lens of basic emotion theory, which posits that emotional responses are automatic, innate, and consistent across populations (Ekman, 1992). This universality supports the idea that sentiment extracted from tweets reflects genuine affective states with measurable effects on market behavior. Complementing this view, research in behavioral finance has shown that investor sentiment, often shaped by emotional cues in social media, can significantly influence stock price movements (Bollen et al., 2011b).

From a methodological perspective, the findings align with the growing consensus that distributed representations of language enhance the performance of sentiment analysis systems. Foundational work by Mikolov et al. (2013b) demonstrated how word embeddings capture complex semantic relationships, making them particularly valuable for analyzing nuanced emotional expressions in short-text formats such as social media. More recent advances in deep learning have further validated that contextual embeddings from transformer models improve sentiment classification by

capturing syntactic and pragmatic subtleties (Devlin et al., 2019).

Taken together, these perspectives provide a framework for the detailed comparisons with existing literature and for evaluating the broader financial and social implications, discussed in the Sections that follow.

## 7.1 Summary of Main Findings

This study advances the understanding of sentiment analysis in financial markets by systematically evaluating methodological trade-offs (Chapter 5) and empirically testing the relationship between social media sentiment and asset returns (Chapter 6). The analysis yields three overarching findings.

First, the comparison of sentiment models in Chapter 5 demonstrated that performance–efficiency and interpretability are critical considerations. RoBERTa achieved state-of-the-art accuracy by capturing contextual nuances such as sarcasm and domain-specific jargon, but its computational demands limit real-time applications (Liu et al., 2019). In contrast, logistic regression with BoW provided an optimal balance, offering 88% accuracy (after neutral-class removal), interpretable coefficients, and sub-minute training times, making it suitable for resource-constrained settings (Loughran and McDonald, 2011). The success of pseudo-labeling with RoBERTa further highlighted a scalable solution to manual annotation challenges, though its bias propagation risks warrant caution (Arazo et al., 2020).

Second, Chapter 6 revealed that Elon Musk's positive social media sentiment significantly predicts Tesla's daily returns, supporting the investor attention hypothesis (Barber and Odean, 2008). However, sentiment showed no measurable effect on volatility, a departure from prior studies linking sentiment to volatility in technology stocks (Das and Chen, 2007). This divergence suggests industry-specific dynamics: automotive stocks such as Tesla may be more influenced by macroeconomic factors, including commodity prices and interest rates, than by CEO communication in

shaping their volatility structure (Belo et al., 2014). These results are consistent with GARCH models' persistence ($\beta = 0.977$) and fat-tailed returns, in line with well-documented financial market stylized facts (Bollerslev, 1986; Cont, 2001).

Third, the findings challenge the assumption that sentiment uniformly amplifies volatility (Da et al., 2015), instead underscoring its role in shaping expected returns. For practitioners, this implies that sentiment-aware trading strategies may capture excess returns without necessitating adjustments to volatility-hedging frameworks. The integration of transformer-based sentiment (RoBERTa) into GARCH models also addresses longstanding measurement precision issues (Li et al., 2021), while the temporal alignment method developed here offers a blueprint for reconciling mixed-frequency data (Bollen et al., 2011a).

These findings provide the basis for a deeper comparison with existing literature and an exploration of their broader financial and social implications, discussed in the subsequent Sections.

## 7.2 Comparison with Related Literature

This study's findings intersect with and extend prior work across three main dimensions: the efficacy of sentiment analysis methodologies, the financial market impact of CEO social media activity, and industry-specific variations in sentiment effects. The following discussion contextualizes these contributions within the broader literature.

Regarding methodological advancements in sentiment analysis, the superior accuracy of RoBERTa in capturing contextual sentiment nuances, aligns with recent trends in NLP research favoring transformer-based models (Liu et al., 2019). However, the practical trade-offs identified in this study, where logistic regression with BoW achieved 88% accuracy while maintaining interpretability and computational efficiency, echo concerns raised by Loughran and McDonald (2011) about the "black-box" nature of deep learning in finance. This tension reflects the evolution from

rule-based systems to hybrid approaches such as MSA-COSR (Xianghua et al., 2013). At the same time, our pseudo-labeling results caution against overreliance on unsupervised methods due to potential bias propagation (Arazo et al., 2020). Notably, our model benchmarking parallels the findings of Pagolu et al. (2016), who found random forests effective for stock prediction, but extends their work by demonstrating LSTM's limitations in low-data regimes, a conclusion consistent with Yao et al. (2019)'s advocacy for graph-based models like Text GCN when labeled data are scarce.

The significant relationship between Musk's positive sentiment and Tesla's returns supports the attention hypothesis proposed by Barber and Odean (2008), while contrasting with Das and Chen (2007)'s findings regarding sentiment-driven volatility in technology stocks. This divergence suggests that CEO sentiment may operate differently across industries: whereas technology firms' valuations are sensitive to sentiment-driven volatility due to intangible assets and rapid innovation cycles (Edmans et al., 2007), automotive stocks such as Tesla may primarily respond through shifts in returns, as their volatility is dominated by macroeconomic factors including commodity prices and supply chain shocks (Belo et al., 2014). Such industry-specific dynamics help reconcile the apparent contradiction between our results and those of Das and Chen (2007) and Da et al. (2015), emphasizing the importance of sector-aware models in sentiment analysis.

The absence of a measurable sentiment-volatility link in our GARCH-X models challenges conventional assumptions (Bollerslev, 1986; Da et al., 2015). Instead, the observed high persistence ($\beta = 0.977$) and fat-tailed returns (Student-$t$, shape $= 5.365$) reflect established financial market stylized facts (Cont, 2001), indicating that volatility in Tesla's stock is better explained by macroeconomic and structural factors than by CEO communication. This aligns with Graham and Harvey (2015)'s assertion that asset-heavy firms exhibit more stable volatility profiles. Practically, our findings suggest that sentiment-aware strategies should focus on return predictability

rather than volatility hedging, a departure from Das and Chen (2007) but consistent with Li et al. (2021)'s emphasis on precision in sentiment measurement.

Finally, the integration of transformer-derived sentiment into GARCH models advances Tetlock (2007)'s early work on textual analysis in finance. The RoBERTa-GARCH hybrid further responds to concerns raised by Jung et al. (2018) regarding CEO social media's governance implications, by quantifying its financial impact without overstating volatility effects.

### 7.2.1 Financial and Social Implications

The empirical results demonstrate that CEO social media sentiment, when analyzed with transformer-based models such as RoBERTa, generates tradable signals with measurable effects on stock returns. This supports the investor attention hypothesis (Barber and Odean, 2008), in which sentiment acts as a catalyst for short-term price movements. However, the absence of a measurable link between sentiment and volatility suggests that portfolio managers could exploit sentiment-driven return predictability without adjusting volatility hedging strategies. Similarly, algorithmic trading systems may benefit from integrating RoBERTa-derived sentiment signals, though sector-specific calibration is critical, as automotive stocks such as Tesla respond differently from technology stocks (Das and Chen, 2007; Belo et al., 2014).

These findings highlight key issues related to CEO use of social media. While Musk's tweets provide market-relevant information, inconsistent regulatory oversight, can create risks of market manipulation (Jung et al., 2018). Relying too heavily on CEO sentiment may also distort long-term firm valuation by overshadowing fundamental indicators (Edmans, 2011). Current SEC disclosure rules (e.g., Regulation Fair Disclosure[1]) lag behind real-time social media communication, creating regula-

[1]Regulation Fair Disclosure (Reg FD), enacted by the U.S. Securities and Exchange Commission (SEC) in 2000, requires public companies to disclose material information to all investors simultaneously. However, its provisions predate the rise of real-time social media communication by CEOs, creating regulatory gaps where selective dissemination of market-moving information (e.g., Elon

tory gaps that could enable selective dissemination of market-moving information (Jung et al., 2018).

Social media platforms like Twitter make information widely accessible but can also amplify behavioral biases. Retail investors may overreact to sentiment cues, increasing short-term price movements (Antweiler and Frank, 2004). Unverified statements from CEOs can further create risks of market instability (Zhang and Wiersema, 2009).

## 7.3 Limitations of the Study

This study presents several limitations related to methodology, data, and theory. Methodologically, the use of pseudo-labeling introduces potential bias, as RoBERTa tends to overpredict the neutral class (Barbieri et al., 2020), which may propagate label noise. Hybrid human-AI validation strategies could mitigate this risk (Arazo et al., 2020). The focus on a single asset, Tesla, also limits the generalizability of the findings, and replication across sectors, such as technology or automotive, is necessary to test the robustness of the observed effects (Kaplan and Strömberg, 2009).

Regarding data and modeling, temporal misalignment remains a concern. Although the study employed a protocol to align social media sentiment with market returns (Bollen et al., 2011a), minute-level fluctuations may still be smoothed when aggregated to daily returns. Additionally, RoBERTa's superior accuracy comes at the cost of interpretability (Rudin, 2019), which complicates causal inference and the identification of underlying mechanisms.

From a theoretical perspective, the absence of a measurable link between sentiment and volatility diverges from prior studies in the technology sector (Das and Chen, 2007;

Musk's 2018 'funding secured' tweet) can occur before formal filings. Jung et al. (2018) argue that this latency enables strategic information asymmetry, as social media posts may reach subsets of investors before official channels.

Da et al., 2015), suggesting that industry-specific pathways may govern volatility responses (Cronqvist et al., 2015). Furthermore, the linear GARCH models employed may not capture potential nonlinear effects, such as sentiment "tipping points"[2].

These limitations provide avenues for future research, including the exploration of cross-sector effects, the integration of more interpretable or hybrid modeling approaches, and the examination of nonlinear dynamics in sentiment-volatility relationships.

## 7.4 Suggestions for Future Research

Building on the empirical and methodological insights of this study, several promising avenues for future research can be identified to deepen the understanding of sentiment's role in financial markets.

### 7.4.1 Data Expansion

One important direction involves expanding the scope of sentiment analysis to multilingual contexts. Current models such as RoBERTa are predominantly trained on English-language corpora, potentially missing nuances in global investor sentiment. Future research could employ multilingual transformers (Conneau et al., 2020), which are capable of processing non-English social media content. This approach would allow for a more comprehensive examination of cultural and linguistic diversity in market reactions, as highlighted by Birjali et al. (2021). For example, Mishev et al. (2020) demonstrate that multilingual models can improve cross-lingual sentiment transfer, a capability that is crucial for assessing the global influence of public figures such as Elon Musk.

[2]By "tipping points", we mean thresholds in the intensity or polarity of social media sentiment beyond which its impact on returns or volatility may suddenly change, rather than following a smooth, linear relationship (Hamilton, 1989).

In addition to multilinguality, handling code-switching and cultural nuances in informal financial discourse is a relevant challenge. Tweets often contain a mix of languages, emojis, and domain-specific jargon. Linguistically sensitive preprocessing, such as converting emojis to text as proposed by Pota et al. (2020), can enhance the generalizability and accuracy of sentiment models.

Future studies could also integrate audience reaction data into sentiment analysis. Beyond analyzing CEO-generated content, incorporating user engagement metrics such as quote tweets and replies may help capture how sentiment propagates through social networks. Valencia et al. (2019) found that combining original messages with public reactions improved cryptocurrency market predictions, suggesting similar benefits for equity markets.

### 7.4.2 Advanced Modeling Techniques

Future research should focus on developing more adaptive and context-aware models for sentiment analysis in finance, capable of capturing evolving market narratives and complex investor behaviors.

Combining micro-level sentiment indicators from short texts (e.g., tweets) with macro-level behavioral sentiment measures (Baker and Wurgler, 2007) could provide a more comprehensive perspective, capturing both short-term market reactions and long-term investor mood. Furthermore, graph-based architectures like Text GCN (Yao et al., 2019) could be leveraged to model sentiment propagation networks among influencers and investors. Multi-aspect sentiment models such as MSA-COSR (Xianghua et al., 2013) could further disentangle different dimensions of sentiment, for example distinguishing product-related enthusiasm from CEO-specific controversy.

From a practical standpoint, efficiency-optimized hybrid architectures could address the trade-off between speed and accuracy in real-time trading applications. Leveraging the scalability of lightweight models alongside the contextual depth of transformer-based architectures offers a promising path toward building sentiment

analysis systems that are both adaptive and operationally viable in dynamic market environments (Joulin et al., 2016).

### 7.4.3 Application to Other Markets and Influencers

Extending this framework to other markets and public figures represents a third major research opportunity. Cross-sector replication, for instance analyzing CEOs in the technology sector such as Mark Zuckerberg (Meta) or in retail such as Jeff Bezos (Amazon), would allow researchers to test the industry-specificity hypothesis (Belo et al., 2014). Applying the approach to emerging markets, where the relationship between sentiment and volatility may differ, could also yield novel insights.

In addition, influence network modeling could be used to identify other "sentiment influencers" beyond CEOs. For example, the emotion-network clustering method proposed by Sailunaz and Alhajj (2019) could be adapted to map the role of hedge fund managers, retail investor groups, and other key market participants in shaping sentiment.

Finally, linking sentiment dynamics to discrete events through event-study methodologies (Metta et al., 2022) could strengthen causal inference. By associating sentiment spikes with specific triggers such as earnings calls or regulatory announcements, researchers could better isolate the mechanisms through which sentiment impacts market behavior.

## 7.5 Final Conclusions

This thesis has provided a systematic investigation of the relationship between social media sentiment and financial market behavior, bridging methodological innovations in NLP with empirical finance. By rigorously evaluating sentiment analysis techniques and testing their predictive power on Tesla's market dynamics, the study makes several key contributions.

First, the comparison between RoBERTa and machine learning models, demonstrated that model choice involves trade-offs between accuracy, interpretability, and computational efficiency. While transformer-based models excel in capturing contextual sentiment, simpler models such as logistic regression remain viable for resource-constrained applications.

Second, the results challenge the assumption of uniform sentiment effects across sectors: Elon Musk's positive sentiment significantly predicted Tesla's returns but not volatility, suggesting that automotive stocks are more sensitive to macroeconomic factors, such as supply chains and commodity prices, than to CEO communication in shaping volatility.

Third, from a practical perspective, the findings highlight the potential of sentiment-aware trading strategies focused on returns rather than volatility hedging, while also underscoring for regulators the need to address gaps in real-time disclosure frameworks for mitigating risks of selective information dissemination.

The study's limitations, including its focus on a single stock and potential biases in pseudo-labeling, point to clear directions for future research. Expanding to multilingual sentiment analysis, cross-sector replications, and hybrid modeling techniques could capture nonlinear sentiment propagation more effectively. Beyond finance, the framework developed here could also inform research on the societal impact of influencer communication in other domains, ranging from politics to public health.

Ultimately, this thesis advances a nuanced understanding of how digital sentiment interacts with market behavior, offering a robust foundation for both scholars and practitioners to navigate an increasingly sentiment-driven financial landscape.

# Appendix

## A Python Code Snippets

The Python code in this section implements the multi-step sentiment analysis pipeline described in Chapter 5. It processes tweets to generate fine-grained sentiment classifications using a pre-trained RoBERTa transformer model and applies three different machine learning approaches to classify sentiment. Performance evaluation is conducted through confusion matrices and F1-score-based classification reports, ensuring methodological robustness.

### Filtering Tweets by Date, Loading RoBERTa Sentiment Model and Filtering Out Neutral Tweets

This first block loads the raw tweet dataset, filters by a given date range, and initializes a pre-trained RoBERTa model for sentiment classification (negative, neutral, positive). Tweets are then processed one by one to generate sentiment labels and associated probabilities. To focus on polarized sentiment, we filter out neutral tweets (sentiment label 1) and saved the resulting dataset for subsequent machine learning experiments.

```
import pandas as pd
import numpy as np
from transformers import AutoTokenizer,
    AutoModelForSequenceClassification
import torch
from tqdm import tqdm
import scipy.special  # for softmax
```

```
import os

# Parameters
DATA_COLUMN = 'created_at'
START_DATE = '2023-01-01'
END_DATE = '2025-02-28'

# Load individual/cleaned tweets
df = pd.read_csv("all_musk_posts_cleaned.csv", encoding='utf-8')

# Convert date column in date format
df[DATA_COLUMN] = pd.to_datetime(df[DATA_COLUMN], errors='coerce')

# Filter dates
df = df[(df[DATA_COLUMN] >= START_DATE) & (df[DATA_COLUMN] <=
    END_DATE)].copy()
print(f"   Tweets selezionati dal {START_DATE} al {END_DATE}: {len
    (df)}")

# Load RoBERTa
print("Loading RoBERTa sentiment model...")
model_name = "cardiffnlp/twitter-roberta-base-sentiment"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(
    model_name)
model.eval()
device = torch.device("cuda" if torch.cuda.is_available() else "cpu
    ")
model.to(device)

# Classify tweets
def classify_tweet(text):
    encoded_input = tokenizer(text, return_tensors='pt', truncation
        =True, max_length=512, padding=True)
```

```
    encoded_input = {k: v.to(device) for k, v in encoded_input.
        items()}
    with torch.no_grad():
        output = model(**encoded_input)
    scores = output.logits.detach().cpu().numpy()
    probs = scipy.special.softmax(scores, axis=1)
    return np.argmax(probs), probs[0]  # label numerico (0=neg, 1=
        neu, 2=pos), + distribuzione
sentiment_labels = []
sentiment_scores = []

print("Classifying tweets...")
for text in tqdm(df['fullText']):
    if isinstance(text, str) and text.strip() != "":
        label, probs = classify_tweet(text)
        sentiment_labels.append(label)
        sentiment_scores.append(probs)
    else:
        sentiment_labels.append(1)  # neutro di default se vuoto
        sentiment_scores.append([0.0, 1.0, 0.0])

# Add the results in the dataframe
df['sentiment'] = sentiment_labels
df['sentiment_scores'] = sentiment_scores

# Filter non-neutral tweets
df_non_neutri = df[df['sentiment'] != 1].copy()

# Save new dataset
output_filtered_path = "all_musk_posts_non_neutral.csv"
df_non_neutri.to_csv(output_filtered_path, index=False, encoding='
    utf-8')
```

## Logistic Regression with BoW Features

This block implements a binary sentiment classifier (positive vs. negative) using Logistic Regression with BoW features. Model evaluation is carried out via a classification report and a confusion matrix.

```
# Logistic Regression with BoW features
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Load filtered dataset (without neutral)
df = pd.read_csv("all_musk_posts_non_neutral.csv", encoding='utf-8')

# Features and labels
X_text = df['fullText'].fillna('')
y = df['sentiment']

# Convert labels to 0 and 1 for binary classification
y = y.replace({0: 0, 2: 1})

# BoW feature extraction
vectorizer = CountVectorizer(max_features=5000, stop_words='english')
X = vectorizer.fit_transform(X_text)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.5, random_state=42, stratify=y)

# Logistic Regression binary
clf = LogisticRegression(max_iter=1000, solver='lbfgs')
clf.fit(X_train, y_train)

# Predictions
y_pred = clf.predict(X_test)

# Evaluation
print("\n=== Classification Report (Binary Sentiment: Negative vs
    Positive) ===\n")
print(classification_report(y_test, y_pred, target_names=['Negative
    ', 'Positive']))

# Confusion Matrix
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)
labels = ['Negative', 'Positive']

plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=labels, yticklabels=labels)

plt.xlabel("Predicted")
plt.ylabel("True")
plt.title("Confusion Matrix    Logistic Regression (Binary) with
    BoW")
plt.tight_layout()
plt.show()
```

### Random Forest with Word2Vec Features

Here, sentiment is classified using a Random Forest trained on pre-computed Word2Vec embeddings. The model's performance is measured with a classification report and confusion matrix.

```
# Random Forest with Word2Vec Features
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns

# Carica il dataset senza neutri
df = pd.read_csv("all_musk_posts_non_neutral.csv", encoding='utf-8')

# Load Word2Vec feature
w2v_features = pd.read_csv("Word2Vec_tweet_features.csv")

# Merge the two datasets
df_merged = pd.merge(df, w2v_features, on='createdAt')
X = df_merged[[col for col in w2v_features.columns if col.startswith('w2v_')]]

# Target is 'sentiment' (0 o 2)
y = df_merged['sentiment']
print("Class distribution:", y.value_counts())

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.5, random_state=42, stratify=y)
```

```
27
28 # Random Forest Model
29 rf = RandomForestClassifier(n_estimators=200, random_state=42)
30 rf.fit(X_train, y_train)
31
32 # Predictions
33 y_pred = rf.predict(X_test)
34
35 # Report and confusion matrix
36 print("Classification␣Report:\n")
37 print(classification_report(y_test, y_pred, target_names=['Negative
       ', 'Positive']))
38 cm = confusion_matrix(y_test, y_pred)
39 labels = ['Negative', 'Positive']
40
41 plt.figure(figsize=(6, 5))
42 sns.heatmap(cm, annot=True, fmt='d', cmap='Greens', xticklabels=
       labels, yticklabels=labels)
43 plt.xlabel("Predicted")
44 plt.ylabel("True")
45 plt.title("Random␣Forest␣Confusion␣Matrix␣(Word2Vec␣features)")
46 plt.tight_layout()
47 plt.show()
```

## LSTM with Word2Vec Features

The final model is a Long Short-Term Memory (LSTM) neural network initialized with Word2Vec embeddings. The architecture includes dropout regularization and ReLU activation in the dense layer. Model performance is evaluated with a classification report and a confusion matrix.

```
1 # LSTM with Word2Vec features
2 import pandas as pd
3 import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, LSTM, Dropout
from tensorflow.keras.utils import to_categorical
from sklearn.metrics import classification_report, confusion_matrix

# Load datasets
w2v_df = pd.read_csv("Word2Vec_tweet_features.csv")
df = pd.read_csv("all_musk_posts_non_neutral.csv", encoding='utf-8')

# Merge on 'createdAt'
df_merged = pd.merge(df, w2v_df, on='createdAt')

# Map sentiment from {0, 2} to {0, 1} for binary classification
df_merged['sentiment_binary'] = df_merged['sentiment'].map({0: 0, 2: 1})

# Features and labels
X = df_merged[[f'w2v_{i}' for i in range(100)]].values  # (samples, 100)
y = df_merged['sentiment_binary'].values

# One-hot encode labels (2 classes)
y_categorical = to_categorical(y, num_classes=2)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y_categorical, test_size=0.5, random_state=42, stratify=y)

# Reshape features for LSTM
```

```
# Here 100 features     10 timesteps, 10 features per timestep
X_train_reshaped = X_train.reshape((X_train.shape[0], 10, 10))
X_test_reshaped = X_test.reshape((X_test.shape[0], 10, 10))

# Build LSTM model
model = Sequential()
model.add(LSTM(64, input_shape=(10, 10), dropout=0.2,
    recurrent_dropout=0.2))
model.add(Dense(32, activation='relu'))
model.add(Dropout(0.5))
model.add(Dense(2, activation='softmax'))  # binary classification
    output
model.compile(loss='categorical_crossentropy', optimizer='adam',
    metrics=['accuracy'])

# Train model
history = model.fit(X_train_reshaped, y_train, epochs=5, batch_size
    =16,
                    validation_split=0.1, verbose=1)

# Predict on test set
y_pred = model.predict(X_test_reshaped)
y_pred_labels = np.argmax(y_pred, axis=1)
y_true_labels = np.argmax(y_test, axis=1)

# Classification report and confusion matrix
target_names = ['Negative', 'Positive']

print("\n=== Classification Report (LSTM + Word2Vec) ===\n")
print(classification_report(y_true_labels, y_pred_labels,
    target_names=target_names))

cm = confusion_matrix(y_true_labels, y_pred_labels)
plt.figure(figsize=(6, 5))
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Oranges',
            xticklabels=target_names, yticklabels=target_names)
plt.xlabel("Predicted")
plt.ylabel("True")
plt.title("Confusion␣Matrix␣   ␣LSTM␣(Word2Vec)")
plt.tight_layout()
plt.show()
```

# B R Code Snippets

The R code in this section implements the core econometric methodology of Chapter 6, focusing on: (1) Baseline GARCH(1,1) models to analyze volatility clustering in the returns series; (2) augmented GARCH-X specifications that incorporate sentiment scores (from Chapter 5) as exogenous variables; (3) robustness checks including distributional assumptions (Normal vs. Student-$t$) and lag structure sensitivity.

These snippets demonstrate how theoretical volatility models (Bollerslev, 1986) are operationalized with empirical data, enabling the sentiment-volatility analysis.

## Downloading Tesla Stock Data and Computing Log Returns

This section downloads Tesla daily closing prices from Yahoo Finance for the period from January 1, 2023, to February 28, 2025, and computes daily logarithmic returns.

```
getSymbols("TSLA", src = "yahoo", from = "2023-01-01", to = "
    2025-02-28")
tsla_prices <- Cl(TSLA)
tsla_returns <- dailyReturn(tsla_prices, type = "log")  # log
    returns
colnames(tsla_returns) <- "returns"
```

## Preliminary Data Analysis and Testing

Preliminary exploration includes analyzing squared returns via autocorrelation functions (ACF and PACF), testing for autocorrelation using the Ljung-Box test, normality using the Jarque-Bera test, and testing for ARCH effects on both raw returns and ARIMA residuals.

```
tsla_returns_squared <- tsla_returns^2
acf(tsla_returns_squared, main = "ACF of Squared Returns")
pacf(tsla_returns_squared, main = "PACF of Squared Returns")

Box.test(coredata(tsla_returns), lag = 10, type = "Ljung-Box")  # p-value > 0.05: no autocorrelation

jarque.bera.test(coredata(tsla_returns))  # reject normality assumption

ArchTest(tsla_returns, lags = 10)

arima_model <- arima(tsla_returns, order = c(0, 0, 0))
resid_arima <- residuals(arima_model)
ArchTest(resid_arima, lags = 10)
```

## Baseline GARCH(1,1) Model Specification and Fitting

The baseline GARCH(1,1) model is specified and fitted to the Tesla returns, assuming Student-t distributed innovations to account for fat tails.

```
spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"  # Student-t distribution
)
fit <- ugarchfit(spec, data = tsla_returns)
```

```
show(fit)
```

## Loading and Preparing Sentiment Data

Sentiment data is imported from a CSV file, with timestamps converted to proper date-time format. Sentiment labels are assigned and data is aggregated by date and sentiment polarity (positive or negative).

```
df <- read_csv("C:/Users/Helena/Desktop/@univr/THESIS/Part␣1␣-␣
   Sentiment␣Analysis/all_musk_posts_non_neutral.csv")
df$createdAt <- ymd_hms(df$createdAt)

df$date <- as.Date(df$createdAt)
df <- df %>%
  mutate(sentiment_label = case_when(
    sentiment == 0 ~ "negative",
    sentiment == 2 ~ "positive",
    TRUE ~ NA_character_
  )) %>%
  filter(!is.na(sentiment_label))

agg_df <- df %>%
  group_by(date, sentiment_label) %>%
  summarise(count = n(), .groups = "drop")

print(agg_df)
```

## Constructing Sentiment Score and Aligning Data

Daily sentiment scores are computed as the difference between positive and negative sentiment counts, converted into a time series object, and merged with Tesla returns to create a combined dataset aligned by date.

```
sentiment_daily <- agg_df %>%
```

```
  pivot_wider(names_from = sentiment_label, values_from = count,
      values_fill = 0) %>%
  mutate(sentiment_score = positive - negative) %>%
  arrange(date)

sentiment_xts <- xts(sentiment_daily$sentiment_score, order.by =
    sentiment_daily$date)

data_combined <- merge(tsla_returns, sentiment_xts, join = "inner")
colnames(data_combined) <- c("returns", "sentiment_score")
```

## Estimating GARCH-X Models with Sentiment as External Regressor

Two GARCH-X models are specified and estimated: one including sentiment score as an external regressor in the mean equation, and the other including sentiment in both the mean and variance equations.

```
spec_x <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE,
      external.regressors = as.matrix(data_combined$sentiment_score)
      ),
  distribution.model = "std"
)
fit_x <- ugarchfit(spec_x, data = data_combined$returns)
show(fit_x)

spec_x_var <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1),
      external.regressors = as.matrix(data_combined$sentiment_score)
      ),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE,
```

```
      external.regressors = as.matrix(data_combined$sentiment_score)
      ),
  distribution.model = "std"
)
fit_x_var <- ugarchfit(spec_x_var, data = data_combined$returns)
show(fit_x_var)
```

# C List of Packages and Libraries

Tables C.1 and C.2 list the key Python and R packages used throughout the analysis and modeling, along with their primary functionalities.

**Table C.1:** Python packages used in the analysis

| Package | Purpose |
|---|---|
| `pandas` | Data manipulation and analysis |
| `numpy` | Numerical computations |
| `re` | Regular expressions for text processing |
| `string` | String processing utilities |
| `emoji` | Handling emojis in text |
| `html` | HTML entity decoding |
| `ast` | Parsing strings to Python objects |
| `warnings` | Controlling warning messages |
| `nltk` | Tokenization, stopwords (NLP toolkit) |
| `gensim` | Topic modeling, Word2Vec embeddings |
| `transformers` | Pre-trained models (BERT, RoBERTa) |
| `torch` | PyTorch deep learning framework |
| `tqdm` | Progress bars for loops |
| `scipy` | Scientific computing (softmax) |
| `sklearn` | ML algorithms and metrics (Logistic Regression, RF) |
| `seaborn` | Statistical visualization (heatmaps) |
| `matplotlib` | Plotting library |
| `wordcloud` | Word cloud generation |
| `tensorflow.keras` | Deep learning (LSTM, NN layers) |

**Table C.2:** R packages used in the analysis

| Package | Purpose |
|---|---|
| `quantmod` | Financial data retrieval and visualization |
| `rugarch` | GARCH modeling and volatility analysis |
| `FinTS` | Financial time series tests (ARCH test) |
| `tseries` | Time series analysis, statistical tests |
| `dplyr` | Data manipulation and transformation |
| `readr` | Reading CSV and other data files |
| `lubridate` | Date-time manipulation |
| `xts` | Time series objects and operations |
| `tidyr` | Data tidying and reshaping |

# D Additional Tables and Charts

This appendix presents additional visual materials that complement the data analysis conducted in the main chapters.

Figure D.1 illustrates the distribution of TSLA daily closing prices over the selected period. Several high-value outliers are visible, reflecting normal fluctuations associated with stock market volatility. These were not treated as anomalies but rather as inherent features of financial time series data.

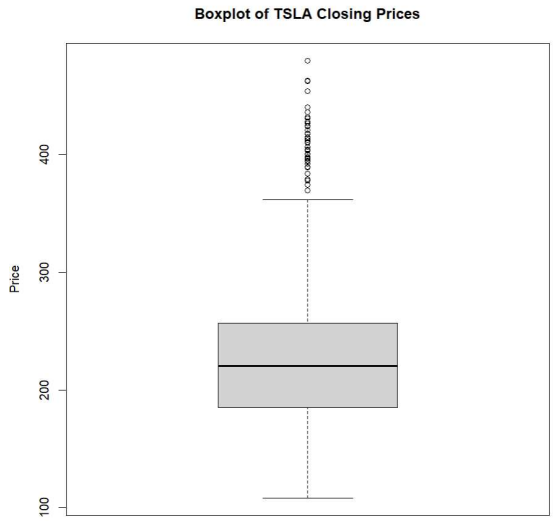



**Figure D.1:** Boxplot of TSLA Closing Prices (Jan 2023 – Feb 2025).

Figure D.2 displays the distribution of tweet lengths in terms of word count. Most tweets are relatively concise, with a median of approximately 9 words. The interquartile range (IQR) indicates that the majority fall between 5 and 14 words. A noticeable number of longer tweets appear as outliers beyond the upper whisker, though these are relatively rare. The distribution is right-skewed, consistent with the predominance of shorter tweets.

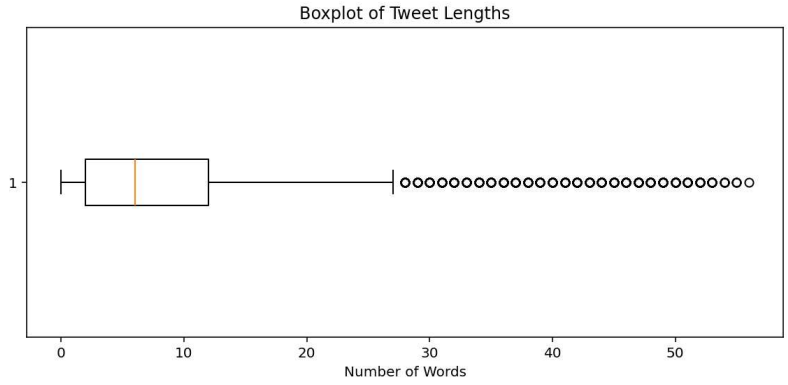



**Figure D.2:** Boxplot of EM's Tweet Lengths.

Figure D.3 tracks monthly engagement levels on Elon Musk's tweets across five metrics. A significant spike, particularly in views and likes, occurs from mid-2024 onward, suggesting elevated public interest during that time frame.

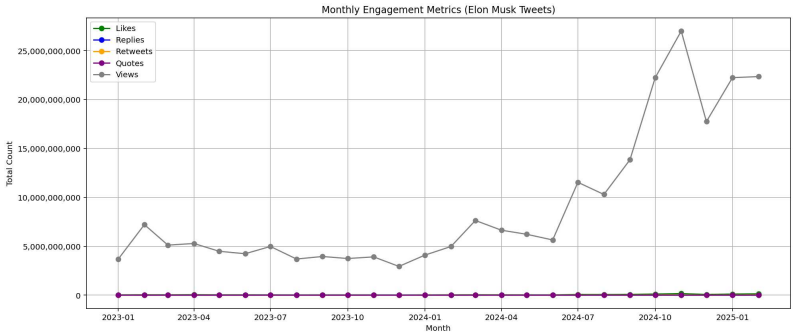



**Figure D.3:** Monthly Engagement Metrics of EM's Tweets.

To enhance interpretability, a separate plot focuses solely on tweet views, the most voluminous metric. Figure D.4 reveals a dramatic increase in reach, peaking in late 2024, and offers clearer detail than the combined chart.

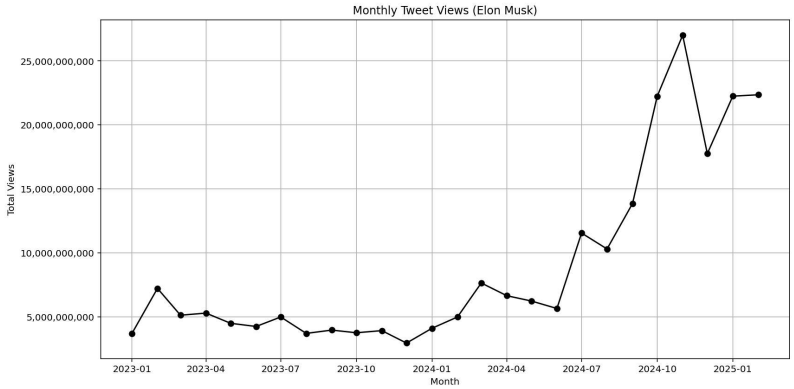



**Figure D.4:** Monthly Tweet Views of EM's Tweets.

Figure D.5 displays an Open-High-Low-Close (OHLC) chart of Tesla (TSLA) stock prices between January 31, 2023, and February 27, 2025. The chart includes daily price movements as well as trading volume shown in the lower section. The volume bars are color-coded to indicate daily changes in trading activity. This visualization provides a detailed view of the stock's price behavior over time, capturing intraday fluctuations and market volatility. It complements the line chart presented in the main body of the thesis by offering a more technical perspective on the data.

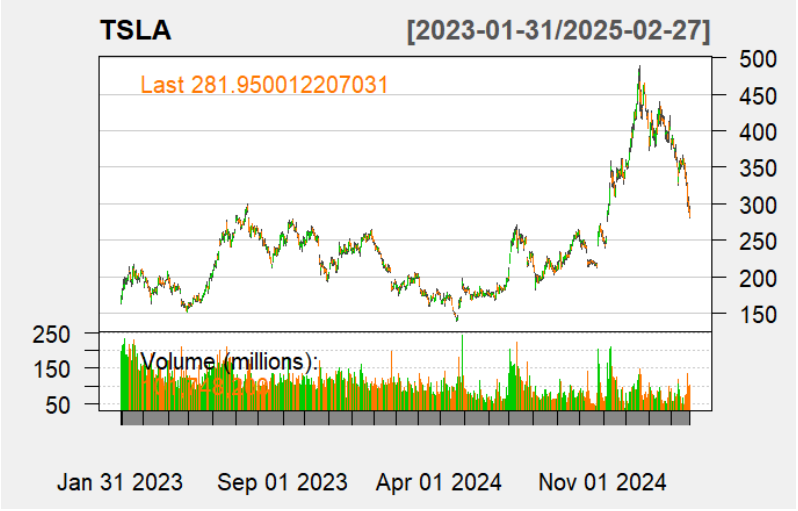



**Figure D.5:** OHLC of TSLA Stock Prices (2023-2025).

Figure D.6 represents a comparison between the empirical density of Tesla's daily returns and a normal distribution with matching mean and standard deviation. The plot shows that the empirical distribution (blue line) is more peaked and has heavier tails than the normal curve (red dashed line), confirming the presence of Heavy-tailed distribution. This behavior is typical in financial time series, where extreme values occur more frequently than under a Gaussian assumption.

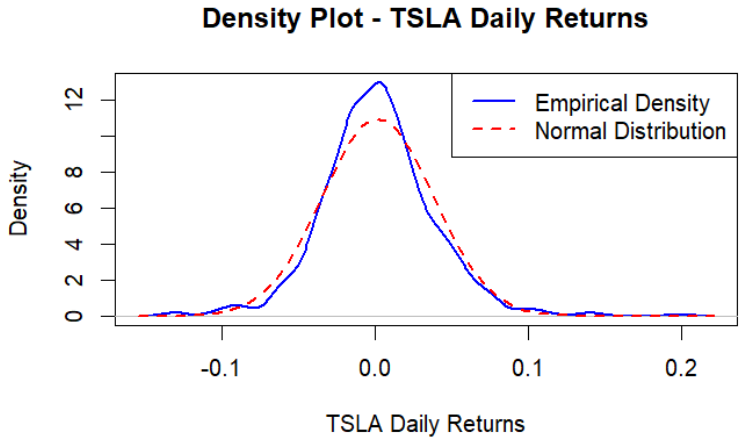



**Figure D.6:** Density Plot of TSLA Daily Returns (2023-2025).

Figure D.7 presents the confusion matrices for the logistic regression classifiers trained with three different feature representations: TF-IDF, Word2Vec, and BERT

embeddings. Each matrix illustrates the model's performance in classifying the dataset.

The leftmost matrix corresponds to the TF-IDF features, capturing term-frequency based information; the middle matrix shows results from Word2Vec embeddings, reflecting semantic relationships between words; and the rightmost matrix represents the BERT embeddings, which leverage deep contextual language understanding. This comparative visualization provides insight into how different text representation methods influence classification accuracy and error patterns, supporting the evaluation of model effectiveness in the context of the study.

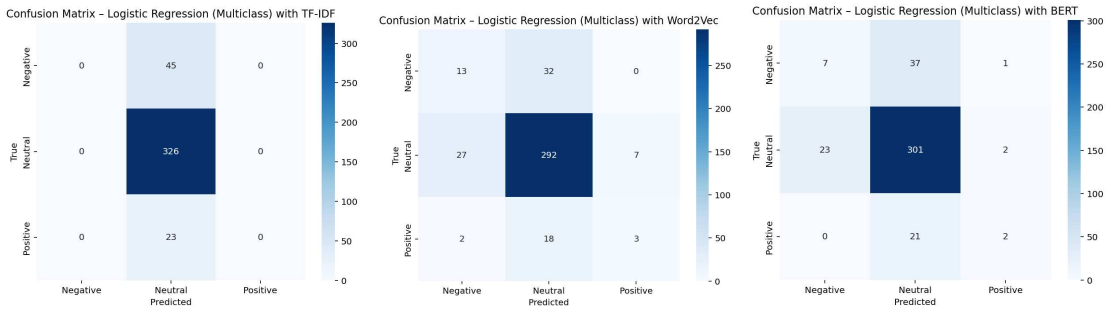



**Figure D.7:** Confusion Matrices of Logistic Regression Classifiers using TF-IDF, Word2Vec, and BERT embeddings.

Figure D.8 shows the confusion matrices of the Random Forest classifiers trained with three different text representations: BoW, TF-IDF, and BERT embeddings. Each matrix displays the classification outcomes, offering a clear picture of model performance across different feature extraction techniques.

From left to right, the matrices correspond respectively to the BoW, TF-IDF, and BERT feature sets. BoW captures raw term frequencies, TF-IDF incorporates term importance weighting, and BERT provides deep contextualized embeddings. This comparative visualization helps to evaluate how each representation impacts the Random Forest classifier's accuracy and error distribution in the sentiment analysis task.

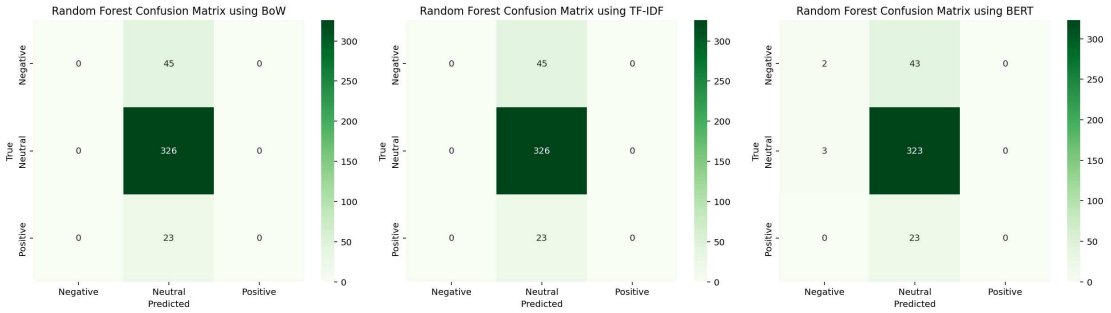



**Figure D.8:** Confusion Matrices of Random Forest Classifiers using BoW, TF-IDF, and BERT embeddings.

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